5.2 INTEGRITY OF REACTOR COOLANT PRESSURE BOUNDARY

This section discusses the measures employed to provide and maintain the integrity of the reactor coolant pressure boundary (RCPB) for the plant design lifetime[a]. Section 50.2 of 10 CFR 50 defines the RCPB as extending to the outermost containment isolation valve in system piping which penetrates the containment and is connected to the reactor coolant system (RCS). This section is limited to a description of the components of the RCS, as defined in section 5.1, unless otherwise noted. Components which are part of the RCPB (as defined in 10 CFR 50) but are not described in this section are described in the following sections:

A. Section 6.3 - RCPB components which are part of the emergency core cooling system.

B. Subsection 9.3.4 - RCPB components which are part of the chemical and volume control system.

C. Subsection 3.9.N.1 - Design loading, stress limits, and analyses applied to the RCS and American Society of Mechanical Engineers (ASME) Code Class 1 components.

D. Subsection 3.9.N.3 - Design loadings, stress limits, and analyses applied to ASME Code Class 2 and 3 components.

The abbreviation RCS, as used in this section, is as defined in section 5.1. When the term RCPB is used in this section, its definition is that of Section 50.2 of 10 CFR 50.

5.2.1 COMPLIANCE WITH CODES AND CODE CASES

5.2.1.1 Compliance with 10 CFR 50.55a

RCS components are designed and fabricated in accordance with 10 CFR 50, Section 50.55a, Codes and Standards. The addenda of the ASME code applied in the design of each component are listed in table 5.2.1-1.

5.2.1.2 Applicable Code Cases

Regulatory Guides 1.84 and 1.85 are discussed in section 1.9. The following discussion addresses only unapproved or conditionally approved code cases (per Regulatory Guides 1.84 and 1.85) used on Class 1 components.

Code Case 1528 (SA-508, Class 2a) material has been used in the manufacture of the VEGP steam generators and pressurizers. Regulatory Guide 1.85 presently reflects a conditional Nuclear Regulatory Commission (NRC) approval of Code Case 1528. Westinghouse has conducted a test program which demonstrates the adequacy of Code Case 1528 material. The

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

results of the test program are documented in reference 1. Reference 1 and a request for approval (reference 2) of the use of Code Case 1528 have been submitted to the NRC.

5.2.1.3 References

1. Letter NS-CE-1228, dated October 4, 1976, C. Eicheldinger (Westinghouse) to J. F. Stolz (NCR).

2. Letter NS-CE-173, dated March 17, 1978, C. Eicheldinger (Westinghouse) to J. F. Stolz (NRC).

5.2.2 OVERPRESSURE PROTECTION

Reactor coolant system (RCS) overpressure protection is provided by the pressurizer safety valves, steam generator safety valves, and the reactor protection system and associated equipment. Combinations of these systems ensure compliance with the overpressure requirements of the American Society of Mechanical Engineers (ASME), Boiler and Pressure Vessel Code, Section III, Paragraphs NB-7300 and NC-7300, for pressurized water reactor systems.

The only portions of an auxiliary system connected to the RCS that are utilized for overpressure protection of the RCS are the liquid relief valves of the residual heat removal system (RHRS). These valves protect the RCS at low temperatures when the RHRS is in operation.

5.2.2.1 Design Bases

Overpressure protection is provided for the RCS by the pressurizer safety valves. This protection is afforded for the following events which envelope those credible events that could lead to overpressure of the RCS if adequate overpressure protection is not provided:

- Loss of electrical load and/or turbine trip.
- Uncontrolled rod withdrawal at power.
- Loss of reactor coolant flow.
- Loss of normal feedwater.
- Loss of offsite power to the station auxiliaries.

The sizing of the pressurizer safety valves is based on analysis of a complete loss of steamflow to the turbine with the reactor operating at 102 percent of engineered safeguards design power. In this analysis, feedwater flow is also assumed to be lost, and no credit is taken for operation of the pressurizer power-operated relief valves (PORV), pressurizer level control system, pressurizer spray system, rod control system, steam dump system, or steam line PORV. The reactor is maintained at full power (no credit for direct reactor trip on turbine trip), and steam relief through the steam generator safety valves is considered. The total pressurizer safety valve capacity is required to be at least as large as the maximum surge rate into the pressurizer during this transient.

This sizing procedure results in a safety valve capacity well in excess of the capacity required to prevent exceeding 110 percent of system design pressure for the events listed above.

Overpressure protection for the steam system is provided by steam generator safety valves. The steam system safety valve capacity is based on providing enough relief to remove the engineered safeguards design steamflow. This must be done while limiting the maximum steam system pressure to less than 110 percent of the steam generator shell side design pressure.

Blowdown and heat dissipation systems of the nuclear steam supply system (NSSS) connected to the discharge of these pressure relieving devices are discussed in subsection 5.4.11.

Steam generator blowdown systems are discussed in subsection 10.4.8.

5.2.2.2 Design Evaluation

A description of the pressurizer safety valves performance characteristics along with the design description of the incidents, assumptions made, method of analysis, and conclusions are discussed in chapter 15.

The relief capacities of the pressurizer and steam generator safety valves are determined from the postulated overpressure transient conditions in conjunction with the action of the reactor protection system. An evaluation of the functional design of the overpressure protection system and an analysis of the capability of the system to perform its function for a typical plant are presented in reference 1. The report describes in detail the types and number of pressure relief devices employed, relief device description, locations in the systems, reliability history, and the details of the methods used for relief device sizing based on typical worst-case transient conditions and analysis data for each transient condition. An overpressure protection report specifically for the VEGP is prepared in accordance with Article NB-7300 of Section III of the ASME Code. The description of the analytical model used in the analysis of the overpressure protection system and the basis for its validity are discussed in reference 2.

The capacities of the pressurizer safety and relief valves are discussed in subsection 5.4.13. The setpoints and reactor trip signals which occur during overpressure transients are discussed in subsection 5.4.10.

5.2.2.3 Piping and Instrumentation Diagrams

Overpressure protection for the RCS is provided by the pressurizer safety and relief valves shown in drawing 1X4DB112. These valves discharge to the pressurizer relief tank through a common manifold.

The steam system safety valves are discussed in section 10.3 and are shown in drawings 1X4DB159-1, 1X4DB159-2, and 1X4DB159-3.

5.2.2.4 Equipment and Component Description

The operation, significant design parameters, number and types of operating cycles, and environmental conditions of the pressurizer safety valves are discussed in section 3.11.N and subsections 3.9.N.1 and 5.4.13.

Section 10.3 contains a discussion of the equipment and components of the steam system overpressure protection features.

5.2.2.5 Mounting of Pressure Relief Devices

The design and installation of the pressure relief devices for the RCS are described in subsection 5.4.11. The design basis for the assumed loads for the primary and secondary side pressure relief devices are described in section 3.9.N. Subsection 10.3.2 provides a discussion of the main steam safety valves and the power-operated atmospheric steam relief valves.

5.2.2.6 Applicable Codes and Classification

The requirements of the ASME Boiler and Pressure Vessel Code, Section III, Paragraphs NB-7300 (Overpressure Protection Report) and NC-7300 (Overpressure Protection Analysis), are met.

Piping, valves, and associated equipment used for overpressure protection are classified in accordance with American National Standards Institute (ANSI) N18.2a-1975, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants. These safety class designations are delineated in table 3.2.2-1 and shown in drawings 1X4DB111, 2X4DB111, 1X4DB112, 2X4DB112, and 1X4DB113.

5.2.2.7 Material Specifications

Refer to subsection 5.2.3 for a description of material specifications.

5.2.2.8 Process Instrumentation

Each pressurizer safety valve discharge line incorporates a control board temperature indicator and alarm to notify the operator of steam discharge due to either leakage or actual valve operation. For a further discussion on process instrumentation associated with the system, refer to chapter 7.

5.2.2.9 System Reliability

The reliability of the pressure relieving devices is discussed in section 4 of reference 1.

5.2.2.10 RCS Pressure Control During Low-Temperature Operation[a]

An important aspect of RCS overpressure protection at low temperatures is the use of administrative controls which are discussed in some detail in paragraph 5.2.2.10.2. Although specific alarms do not exist to invoke specific administrative procedures, annunciation is provided to alert the operator to arm the cold overpressure mitigation system. Operating procedures maximize the use of a pressurizer steam bubble, since a steam bubble reduces the maximum pressure reached for some transients, and slows the rate of pressure increase for others, and aids the operator in controlling RCS pressure during low temperature operation.

[a] The cold overpressure protection system setpoint calculation was evaluated as a time-limited aging analysis (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in paragraph 19.4.6.4.

When the RCS is at temperatures below approximately 350°F, it is opened to the RHRS for the purposes of removing residual heat from the core, providing a path for letdown to the purification subsystem, and controlling the RCS pressure when the plant is operating in a water solid mode. The RHRS is provided with self-actuated water relief valves to prevent overpressure in this relatively low design pressure system caused either within the system itself or from transients transmitted from the RCS. The RHRS relief valves mitigate pressure transients originated in the RCS to maximum pressure values determined by the relief valve set pressure.

The low design pressure RHRS is normally isolated from the high design pressure RCS during reactor power operation at temperatures above approximately 350°F by two isolation valves in series. Therefore, the RHRS can be inadvertently isolated from the RCS by these same isolation valves. The PORVs and associated logic provide overpressure mitigation for those transients which might occur if the RHRS isolation valves were inadvertently closed. The PORV logic is manually armed at the system setpoint.

Two pressurizer PORVs are each supplied with actuation logic. The logic for each PORV continuously monitors RCS temperature and pressure, converts an auctioneered RCS temperature to the Appendix G allowable pressure, and then compares the allowable pressure to the actual RCS pressure. As the actual RCS pressure approaches the allowable pressure, a main control board alarm is annunciated. If the RCS pressure continues to increase, an actuation signal is transmitted to a PORV and the valve opens to mitigate the transient.

As described in subsection 5.4.13, the VEGP PORVs are safety related. They were designed in accordance with the ASME Code and are qualified via the Westinghouse pump and valve operability program which is described in paragraph 3.10.N.2.2.

The hardware and logic associated with this function will operate following an operating basis earthquake. Offsite power is not required for the system to function. The actuation logic in the system is testable. However, the PORVs and RHR relief valves are not exercised with the reactor at power. They are capable of being tested as required by the ASME Code and the VEGP Technical Specifications.

5.2.2.10.1 Transient Evaluation

Potential overpressurization transients to the RCS, while at relatively low temperatures, can be caused by either of two types of events to the RCS; i.e., mass input or heat input. Both types result in more rapid pressure changes when the RCS is water solid.

Anticipated mass and heat input transients are evaluated to demonstrate conformance with Appendix G. The most limiting heat input transient is an inadvertant reactor coolant pump startup in a loop where the steam generator secondary temperature is 50°F higher than the primary temperature in any loop. The most limiting mass input transient is a charging-letdown mismatch where two emergency core cooling system (ECCS) centrifugal charging pumps and one normal centrifugal charging pump are charging water into the reactor coolant system with the letdown path isolated.

It should be noted that the following transient is also addressed. With the plant in a cooled down and depressurized condition in which the cold overpressure protection system is required to be operable and with charging and letdown established and RHRS open to RCS, a dc vital bus fails. This failure causes normal letdown to isolate and also results in the loss of one of the two PORVs. However, RHRS relief valves mitigate the transient.

5.2.2.10.2 Administrative Controls

During plant operation the following precautions are observed:

A. At least one RHR inlet line from the reactor coolant loop is not isolated unless there is a steam bubble in the pressurizer. This precaution ensures that there is a relief path from the reactor coolant loop to a RHR suction line relief valve when the RCS is at low temperature and is water solid.

B. Whenever the plant is water solid and the reactor coolant pressure is being maintained by the low pressure letdown control valve, letdown should include flow from the operating RHR loop through the RHRS cleanup to the letdown heat exchanger valve.

C. One RCP should normally be running anytime RCS temperature is changed by more than 10°F in 1 h. Additionally, RCPs should not be started if steam generator secondary water temperature is greater than 10°F above the RCS temperature.

D. During a typical plant cooldown, operable steam generators should be connected to the steam header to ensure a uniform cooldown of the reactor coolant loops.

E. To preclude inadvertent emergency core cooling system (ECCS) actuation during heatup and cooldown, blocking of the high pressurizer pressure, and low steam line pressure safety injection, signal actuation logic at 1970 psig is required. These manual blocking features are further discussed in paragraph 7.3.1.2.2.6.

   During further cooldown, closure and power lockout of the accumulator isolation valves is performed at 1000 psig and power lockout to the safety injection pumps is performed at approximately 220°F in the RCS.

F. Periodic ECCS pump performance testing requires the testing of the pumps during normal power operation or at hot shutdown conditions. This precludes any potential for developing a cold overpressurization transient.

   Should cold shutdown testing of the pumps be required, the test is done when the vessel is open to the atmosphere, again precluding overpressurization potential.

   If cold shutdown testing with the vessel closed is necessary, the procedures require only one pump to be tested with ECCS discharge valve closure and RHRS alignment to both isolate potential ECCS pump input and to provide backup benefit of the RHRS relief valves.

   The SI signal circuitry testing, if performed during cold shutdown, also requires RHRS alignment and safety injection pumps power lockout to preclude developing cold overpressurization transients.

G. A steam bubble will be maintained in the pressurizer when the RCS temperature is greater than 220°F.

5.2.2.11 Consequences of a Postulated Loss of a dc Bus Coupled with a Single Failure Disabling a PORV Allowing a Cold Pressurization Event

This discussion addresses the following postulated event:

With the plant in a cooled down and depressurized condition in which the cold overpressure protection system is required to be operable and with charging and letdown established, a dc vital bus fails. This failure causes normal letdown to isolate and also results in the loss of one of the two PORVs.

In addition to the dc bus failure, an additional random failure of the second PORV is postulated to occur. This sequence of events places the plant in a condition in which letdown is isolated, the automatic cold overpressure protection system is inoperable, and charging flow is filling the pressurizer, increasing system pressure towards the Appendix G limits.

To begin this discussion the limitations placed on plant operation by the VEGP Technical Specifications are addressed.

A. With RCS temperature below 200°F (i.e., cold shutdown) one RHR pump is required to be in operation. This requirement ensures that at least one RHR suction relief valve is available for overpressure protection of the RCS. This valve will relieve the flow from two ECCS centrifugal charging pumps and one normal centrifugal charging pump at the valve lift setting pressure.

B. Whenever the cold overpressure protection system is required to be operable, only the two ECCS centrifugal charging pumps and the normal centrifugal charging pumps are allowed to be operable. The safety injection pumps will be inoperable. This ensures that the maximum charging letdown mismatch will be that stated in 5.2.2.10.1.

C. Considering these requirements, anytime RHR is in operation and the RCS is in a condition requiring the cold overpressure protection to be operable, there is no overpressure event as a result of the prescribed event. Assuming the event as described[(a)] did occur, the RHR relief valve would prevent RCS pressure from reaching the Appendix G limit by relieving all charging flow.

Typically at least one RHR train is in operation, or at a minimum one RHR loop suction valve is open, providing an open path from the RCS to a RHR suction relief valve, whenever RCS temperature is below 350°F. For this reason an overpressure event resulting from the prescribed event is very unlikely; however, the discussion is extended to the infrequent case where the RHRS is isolated from the RCS, and the cold overpressure protection system is required to be operable.

To gain a better understanding of the results of the event, it is necessary to address the functions of some of the chemical and volume control system control valves. As stated earlier, the letdown valves fail closed on loss of dc power isolating letdown. Between the charging pumps and the normal charging isolation valves are two normally throttled valves which receive their power from the process and control racks powered by the essential ac instrument buses. These valves are unaffected by a dc bus failure and continue to work normally during the event.

One of these valves is the charging flow control valve (1-FCV-121), which automatically regulates flow to maintain a prescribed pressurizer level. Assuming this valve continues to function normally, as pressurizer level rises, charging flow is reduced until the charging flow is limited to that required for seal injection (32 gal/min) plus a minimal amount (15 gal/min) required for regenerative heat exchanger cooling. At this flowrate, ample time is provided (as discussed below) to allow appropriate operator action. If valve control is in manual, the valve position remains unchanged. The other valve is the charging flow backpressure regulator

[(a)] As additional information, during RHR operation letdown is typically taken from the discharge of the RHR pumps and is not isolated by the dc bus failure.

(1-HCV-182), which is manually positioned to regulate flow to the seal. This valve remains in its initial position. The effect of these two valves is to limit charging flow to its value at the beginning of the event. Assuming maximum letdown at the initiation of the event, total flow (charging plus seal injection) to the RCS is limited to approximately 130 gal/min.

An additional consideration is that, with the plant in the hot shutdown condition and RHR isolated from the RCS, normal operation is to have a steam bubble in the pressurizer of approximately 1350 ft^3. At a maximum charging rate of 130 gal/min, it would take in excess of 30 min to reach the Appendix G limit at 200°F, the temperature corresponding to the coldest RCS temperature at which RHR is permitted to be isolated. As an extreme case, with a bubble of only half the normal size, the corresponding time available for appropriate action would be in excess of 15 min.

To summarize:

A. The postulated event is unlikely to occur since the dc buses have a battery as an emergency power supply, and should the dc bus fail, it must be coupled with the additional failure of the second PORV for overpressurization.

B. In the unlikely event that the prescribed event did occur, RHR would normally be online and capable of mitigating any potential overpressure resulting from two ECCS centrifugal charging pumps and one normal centrifugal charging pump.

C. In the highly unlikely event that the prescribed event should occur when RHR is isolated from the RCS, the operator would have sufficient time to mitigate the event.

D. The Appendix G curves are excessively conservative for their intended purpose of ensuring vessel integrity during cold shutdown.

No further action is necessary to address this postulated event, and the existing plant design and operational techniques result in successful event mitigation.

5.2.2.12 Testing and Inspection

Testing and inspection of the overpressure protection components are discussed in paragraph 5.4.13.4 and chapter 14. Testing capabilities of the overpressurization protection system are consistent with testing principles for systems' electronics in paragraph 7.2.2.2.3. Operational surveillance procedures will demonstrate the operability of the overpressure protection system when system operation is required. Inservice inspection is performed in accordance with Section XI of the ASME Code. Relief requests, augmented examinations, and alternate techniques will be utilized as appropriate.

5.2.2.13 References

1. Cooper, L., Miselis, V., and Starek, R. M., "Overpressure Protection for Westinghouse Pressurized Water Reactors," WCAP-7769, Revision 1, June 1972 (also letter NS-CE-622 dated April 16, 1975, C. Eicheldinger (Westinghouse) to D. B. Vassallo (NRC), Additional Information on WCAP-7769, Revision 1).

2. Burnett, T. W. T., et al., "LOFTRAN Code Description," WCAP-7907, October 1972.

5.2.3 REACTOR COOLANT PRESSURE BOUNDARY (RCPB) MATERIALS

5.2.3.1 Material Specifications

Typical material specifications used for the principal pressure-retaining applications in Class 1 primary components and for Class 1 and 2 auxiliary components in systems required for reactor shutdown and for emergency core cooling are listed in table 5.2.3-1. Typical material specifications used for the reactor internals required for emergency core cooling, for any mode of normal operation or under postulated accident conditions, and for core structural load bearing members are listed in table 5.2.3-2.

Tables 5.2.3-1 and 5.2.3-2 may not be totally inclusive of the material specifications used in the listed applications; however, the listed specifications are representative. Identification of actual materials is available in VEGP quality assurance records.

The materials utilized conform to the applicable American Society of Mechanical Engineers (ASME) code rules.

The welding materials used for joining the ferritic base materials of the RCPB conform to or are equivalent to ASME Material Specifications SFA 5.1, 5.2, 5.5, 5.17, 5.18, and 5.20. They are qualified to the requirements of the ASME Code, Section III.

The welding materials used for joining the austenitic stainless steel base materials of the RCPB conform to ASME Material Specifications SFA 5.4 and 5.9. They are qualified to the requirements of the ASME Code, Section III.

The welding materials used for joining nickel-chromium-iron alloy in similar base material combination and in dissimilar ferritic or austenitic base material combination conform to ASME Material Specifications SFA 5.11 and 5.14. They are qualified to the requirements of the ASME Code, Section III.

5.2.3.2 Compatibility with Reactor Coolant

5.2.3.2.1 Chemistry of Reactor Coolant[a]

The reactor coolant system (RCS) chemistry specifications are given in table 5.2.3-3.

The RCS water chemistry is selected to minimize corrosion. Routinely scheduled analyses of the coolant chemical composition are performed to verify that the reactor coolant chemistry meets the specifications.

The chemical and volume control system (CVCS) provides a means for adding chemicals to the RCS which perform the following functions:

- Control the pH of the coolant during prestartup testing and subsequent operation.
- Scavenge oxygen from the coolant during heatup.
- Control radiolysis reactions involving hydrogen, oxygen, and nitrogen during all power operations subsequent to startup.

[a] The Water Chemistry Control Program is credited as a license renewal aging management program (see subsection 19.2.28).

The normal limits for chemical additives and reactor coolant impurities for power operation are shown in table 5.2.3-3.

The pH control chemical utilized is lithium hydroxide monohydrate, enriched in the lithium-7 isotope to 99.9 percent. This chemical is chosen for its compatibility with the materials and water chemistry of borated water/stainless steel/zirconium/ Inconel systems. In addition, lithium-7 is produced in solution from the neutron irradiation of the dissolved boron in the coolant. The lithium-7 hydroxide is introduced into the RCS via the charging flow. The solution is prepared in the laboratory and transferred to the chemical additive tank. Reactor makeup water is then used to flush the solution to the suction header of the charging pumps. The concentration of lithium-7 hydroxide in the RCS is maintained in the range specified for pH control. If the concentration exceeds this range, the cation bed demineralizer is employed in the letdown line in series operation with the mixed bed demineralizer.

During reactor startup from the cold condition, hydrazine is employed as an oxygen scavenging agent. The hydrazine solution is introduced into the RCS in the same manner as described above for the pH control agent.

The reactor coolant is treated with dissolved hydrogen to control the net decomposition of water by radiolysis in the core region. The hydrogen also reacts with oxygen and nitrogen introduced into the RCS as impurities under the impetus of core radiation. Sufficient partial pressure of hydrogen is maintained in the volume control tank so that the specified equilibrium concentration of hydrogen is maintained in the reactor coolant. A self-contained pressure control valve maintains a minimum pressure in the vapor space of the volume control tank.

This can be adjusted to provide the correct equilibrium hydrogen concentration.

Boron, in the chemical form of boric acid, is added to the RCS for long-term reactivity control of the core.

A soluble zinc compound may be added to the reactor coolant as a means to reduce radiation fields within the primary system. The zinc used may be either natural zinc or zinc depleted of ^{64}Zn. When used, the target system zinc concentration is normally maintained to a concentration no greater than 40 ppb.

Suspended solid (corrosion product particulates) and other impurity concentrations are maintained below specified limits by controlling the chemical quality of makeup water and chemical additives and by purification of the reactor coolant through the CVCS.

5.2.3.2.2 Compatibility of Construction Materials with Reactor Coolant

All of the ferritic low-alloy and carbon steels which are used in principal pressure-retaining applications have corrosion-resistant cladding on all surfaces that are exposed to the reactor coolant. The corrosion resistance of the cladding material is at least equivalent to the corrosion resistance of types 304 and 316 austenitic stainless steel alloys or nickel-chromium-iron alloy, martensitic stainless steel, and precipitation-hardened stainless steel. These corrosion-resistant cladding materials may be subjected to the ASME code required postweld heat treatment for ferritic base materials.

Ferritic low-alloy and carbon steel nozzles have safe ends of either stainless steel wrought materials, stainless steel weld metal analysis A-7 (designated A-8 in the 1974 edition of the ASME code), or nickel-chromium-iron alloy weld metal F-Number 43. The latter buttering material requires further safe ending with austenitic stainless steel base material after completion of the postweld heat treatment when the nozzle is larger than a 4-in. nominal inside diameter and/or the wall thickness is greater than 0.531 in.

All of the austenitic stainless steel and nickel-chromium-iron alloy base materials with primary pressure-retaining applications are used in the solution annealed condition. These heat treatments are as required by the material specifications.

During subsequent fabrications, these materials are not heated above 800°F other than locally by welding operations. The solution-annealed surge line material is subsequently formed by hot bending followed by a resolution annealing heat treatment.

Components employing stainless steel sensitized in the manner expected during component fabrication and installation operate satisfactorily under normal plant chemistry conditions in pressurized water reactor (PWR) systems because chlorides, fluorides, and oxygen are controlled to very low levels.

5.2.3.2.3 Compatibility with External Insulation and Environmental Atmosphere

In general, all of the materials listed in table 5.2.3-1 which are used in principal pressure-retaining applications and are subject to elevated temperature during system operation are in contact with thermal insulation that covers their outer surfaces.

The thermal insulation used on the RCPB is either reflective stainless steel type or made of compounded materials which yield low leachable chloride and/or fluoride concentrations. The compounded materials in the form of blocks, boards, cloths, tapes, adhesives, cements, etc., are silicated to provide protection of austenitic stainless steels against stress corrosion which may result from accidental wetting of the insulation by spillage, minor leakage, or other contamination from the environmental atmosphere. Section 1.9 indicates the degree of conformance with Regulatory Guide 1.36, Nonmetallic Thermal Insulation for Austenitic Stainless Steel.

In the event of coolant leakage, the ferritic materials will show increased general corrosion rates. Where minor leakage is anticipated from service experience, such as valve packing, pump seals, etc., only materials that are compatible with the coolant are used. These are as shown in table 5.2.3-1. Ferritic materials exposed to coolant leakage can be readily observed as part of the inservice visual and/or nondestructive inspection program to ensure the integrity of the component for subsequent service.

5.2.3.3 Fabrication and Processing of Ferritic Materials

5.2.3.3.1 Fracture Toughness

The fracture toughness properties of the RCPB components meet the requirements of the ASME Code, Section III, Paragraphs NB, NC, and ND-2300, as appropriate.

The fracture toughness properties of the reactor vessel materials are discussed in section 5.3.

Limiting steam generator and pressurizer reference temperature for a nil ductility transition (RT_{NDT}) temperatures are guaranteed at 60°F for the base materials and the weldments.

These materials meet the 50-ft-lb absorbed energy and 35-mils lateral expansion requirements of the ASME Code, Section III, at 120°F. The actual results of these tests are provided in the ASME material data reports which are supplied for each component and submitted to the owner at the time of shipment of the component.

Calibration of temperature instruments and Charpy impact test machines are performed to meet the requirements of the ASME Code, Section III, Paragraph NB-2360.

Westinghouse has conducted a test program to determine the fracture toughness of low-alloy ferritic materials with specified minimum yield strengths greater than 50,000 psi to demonstrate compliance with Appendix G of the ASME Code, Section III. In this program, fracture toughness properties were determined and shown to be adequate for base metal plates and forgings, weld metal, and heat-affected zone metal for higher strength ferritic materials used for components of the RCPB. The results of the program are documented in reference 1, which has been submitted to the Nuclear Regulatory Commission (NRC) for review.

5.2.3.3.2 Control of Welding

All welding is conducted utilizing procedures qualified according to the rules of Sections III and IX of the ASME Code. Control of welding variables, as well as examination and testing, during procedure qualification and production welding is performed in accordance with ASME Code requirements.

Westinghouse practices for storage and handling of welding electrodes and fluxes comply with ASME Code, Section III, Paragraph NB-2400.

Section 1.9 indicates the degree of conformance of the ferritic materials components of the RCPB with Regulatory Guides 1.34, Control of Electroslag Weld Properties, 1.43, Control of Stainless Steel Weld Cladding of Low-Alloy Steel Components, 1.50, Control of Preheat Temperature for Welding of Low-Alloy Steel, and 1.71, Welder Qualification for Areas of Limited Accessibility.

5.2.3.4 Fabrication and Processing of Austenitic Stainless Steel

Paragraphs 5.2.3.4.1 through 5.2.3.4.5 address Regulatory Guide 1.44, Control of the Use of Sensitized Stainless Steel, and present the methods and controls utilized by Westinghouse to avoid sensitization and prevent intergranular attack (IGA) of austenitic stainless steel components. Also, section 1.9 indicates the degree of conformance with Regulatory Guide 1.44.

5.2.3.4.1 Cleaning and Contamination Protection Procedures

It is required that all austenitic stainless steel materials used in the fabrication, installation, and testing of nuclear steam supply components and systems be handled, protected, stored, and cleaned according to recognized and accepted methods which are designed to minimize contamination which could lead to stress corrosion cracking. The rules covering these controls are stipulated in Westinghouse process specifications. As applicable, these process specifications supplement the equipment specifications and purchase order requirements of every individual austenitic stainless steel component or system which Westinghouse procures for the VEGP nuclear steam supply systems (NSSSs), regardless of the ASME Code classification. Westinghouse process specifications are also given to Bechtel, the architect-engineer, and to Georgia Power Company for recommended use within their scope of supply and activity.

The process specifications that define these requirements and that follow the guidance of the American National Standards Institute (ANSI) N-45 committee specifications include the following:

Number	Process Specification
82560HM	Requirements for Pressure Sensitive Tapes for Use on Austenitic Stainless Steels.
83336KA	Requirements for Thermal Insulation Used on Austenitic Stainless Steel Piping and Equipment.
83860LA	Requirements for Marking of Reactor Plant Components and Piping.
8435OHA	Site Receiving Inspection and Storage Requirements for Systems, Material, and Equipment.
8435INL	Determination of Surface Chloride and Fluoride on Austenitic Stainless Steel Materials.
853I0QA	Packaging and Preparing Nuclear Components for Shipment and Storage.
292722	Cleaning and Packaging Requirements of Equipment for Use in the NSSS.
597756	Pressurized Water Reactor Auxiliary Tanks Cleaning Procedures.
597760	Cleanliness Requirements During Storage Construction, Erection, and Startup Activities of Nuclear Power System.

Section 1.9 indicates the degree of conformance of the austenitic stainless steel components of the RCPB with Regulatory Guide 1.37, Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants.

5.2.3.4.2 Solution Heat Treatment Requirements

The austenitic stainless steels listed in tables 5.2.3-1 and 5.2.3-2 are utilized in the final heat-treated condition required by the respective ASME Code, Section II materials specification for the particular type or grade of alloy.

5.2.3.4.3 Material Testing Program

Westinghouse practice is that austenitic stainless steel materials of product forms with simple shapes need not be corrosion tested provided that the solution heat treatment is followed by water quenching. Simple shapes are defined as all plates, sheets, bars, pipe, and tubes, as well as forgings, fittings, and other shaped products that do not have inaccessible cavities or chambers that would preclude rapid cooling when water quenched. When testing is required, the tests are performed in accordance with ASTM A 262, Practice A or E, as amended by Westinghouse Process Specification 84201MW.

5.2.3.4.4 Prevention of Intergranular Attack of Unstabilized Austenitic Stainless Steels

Unstabilized austenitic stainless steels are subject to IGA provided that three conditions are present simultaneously. These are:

- An aggressive environment; e.g., an acidic aqueous medium containing chlorides or oxygen.
- A sensitized steel.
- A high temperature.

If any one of the three conditions described above is not present, IGA will not occur. Since high temperatures cannot be avoided in all components in the NSSS, reliance is placed on the elimination of the other two conditions to prevent IGA on wrought stainless steel components.

This is accomplished by:

- Control of primary water chemistry to ensure a benign environment.
- Utilization of materials in the final heat-treated condition and the prohibition of subsequent heat treatments in the 800°F and 1500°F temperature range.
- Control of welding processes and procedures to avoid heat-affected zone sensitization.
- Confirmation that the welding procedures used for the manufacture of components in the primary pressure boundary and the reactor internals do not result in the sensitization of heat-affected zones.

Further information on each of these steps is provided in the following paragraphs.

The water chemistry in the RCS is controlled to prevent the intrusion of aggressive species. In particular, the maximum permissible oxygen and chloride concentrations are 0.1 ppm and 0.15 ppm, respectively. Table 5.2.3-3 lists the recommended reactor coolant water chemistry specifications. The precautions taken to prevent the intrusion of chlorides into the system during fabrication, shipping, and storage are stipulated in the appropriate process specifications. The use of hydrogen overpressure precludes the presence of oxygen during operation. The effectiveness of these controls has been demonstrated by both laboratory tests and operating experience. The long-term exposure of severely sensitized stainless steels to reactor coolant environments in early Westinghouse PWRs has not resulted in any sign of IGA. Reference 2 describes the laboratory experimental findings and reactor operating experience. The additional years of operations since the issuing of reference 2 have provided further confirmation of the earlier conclusions that severely sensitized stainless steels do not undergo any IGA in Westinghouse PWR coolant environments.

Although there is no evidence that PWR coolant water attacks sensitized stainless steels, Westinghouse considers it good metallurgical practice to avoid the use of sensitized stainless steels in the NSSS components. Accordingly, measures are taken to prohibit the purchase of sensitized stainless steels and to prevent sensitization during component fabrication. Wrought austenitic stainless steel stock is used for components that are part of:

- The RCPB.
- Systems required for reactor shutdown.
- Systems required for emergency core cooling.

- Reactor vessel internals relied upon to permit adequate core cooling for normal operation or under postulated accident conditions.

The wrought austenitic stainless steel stock is utilized in one of the following conditions:

- Solution annealed and water quenched.
- Solution annealed and cooled through the sensitization temperature range within less than approximately 5 min.

It is generally accepted that these practices will prevent sensitization. Westinghouse has verified this by performing corrosion tests on wrought material as it was received.

The heat-affected zones of welded components must, of necessity, be heated into the sensitization temperature range, 800°F to 1500°F. However, severe sensitization (i.e., continuous grain boundary precipitates of chromium carbide, with adjacent chromium depletion) can be avoided by controlling welding parameters and welding processes. The heat input and associated cooling rate through the carbide precipitation range are of primary importance. Westinghouse has demonstrated this by corrosion testing a number of weldments.

Heat input is calculated according to the formula:

$$H = \frac{(E)(I)(60)}{S}$$

where:

H = joules/in.

E = volts.

I = amperes.

S = travel speed (in./min).

Of 25 production and qualification weldments tested, representing all major welding processes and a variety of components and incorporating base metal thicknesses from 0.10 to 4.0 in., only portions of two were severely sensitized. Of these, one involved a heat input of 120,000 J, and the other involved a heavy socket weld in relatively thin-walled material. In both cases, sensitization was caused primarily by high-heat inputs relative to the section thickness. In only the socket weld did the sensitized condition exist at the surface, where the material is exposed to the environment. The component has been redesigned, and a material change has been made to eliminate this condition.

The heat input in all austenitic pressure boundary weldments has been controlled by:

- Prohibiting the use of block welding.
- Limiting the maximum interpass temperature to 350°F.
- Exercising approval rights on all welding procedures.

5.2.3.4.5 Retesting Unstabilized Austenitic Stainless Steels Exposed to Sensitization Temperatures

As described in the previous section, it is not normal Westinghouse practice to expose unstabilized austenitic stainless steels to the sensitization range of 800°F to 1500°F during fabrication into components. If during the course of fabrication, the steel is inadvertently exposed to the sensitization temperature range, 800°F to 1500°F, the material may be tested in

accordance with ASTM A 262, as amended by Westinghouse Process Specification 84201MW, to verify that it is not susceptible to IGA, except that testing is not required for:

A. Cast metal or weld metal with a ferrite content of 5 percent or more.

B. Material with a carbon content of 0.03 percent or less that is subjected to temperatures in the range of 800°F to 1500°F for less than 1 h.

C. Material exposed to special processing, provided the processing is properly controlled to develop a uniform product and provided that adequate documentation exists of service experience and/or test data to demonstrate that the processing will not result in increased susceptibility to intergranular stress corrosion.

If it is not verified that such material is not susceptible to IGA, the material is resolution annealed and water quenched or rejected.

5.2.3.4.6 Control of Welding

The following paragraphs address Regulatory Guide 1.31, Control of Ferrite Content in Stainless Steel Weld Metal, and present the methods used, and the verification of these methods, for austenitic stainless steel welding.

The welding of austenitic stainless steel is controlled to mitigate the occurrence of microfissuring or hot cracking in the weld. Although published data and experience have not confirmed that fissuring is detrimental to the quality of the weld, it is recognized that such fissuring is undesirable in a general sense. Also, it has been well documented in the technical literature that the presence of delta ferrite is one of the mechanisms for reducing the susceptibility of stainless steel welds to hot cracking. However, there is insufficient data to specify a minimum delta ferrite level below which the material will be prone to hot cracking. It is assumed that such a minimum lies somewhere between 0- and 3-percent delta ferrite.

The scope of these controls discussed herein encompasses welding processes used to join stainless steel parts in components designed, fabricated, or stamped in accordance with the ASME Code, Section III, Class 1 and 2, and core support components. Delta ferrite control is appropriate for the above welding requirements, except where no filler metal is used or where for other reasons such control is not applicable. These exceptions include electron beam welding, autogenous gas shielded tungsten arc welding, explosive welding, and welding using fully austenitic welding materials.

The fabrication and installation specifications require welding procedure and welder qualification in accordance with Section III and include the delta ferrite determinations for the austenitic stainless steel welding materials that are used for welding qualification testing and for production processing. Specifically, the undiluted weld deposits of the "starting" welding materials are required to contain a minimum of 5-percent delta ferrite (the equivalent ferrite number may be substituted for percent delta ferrite) as determined by chemical analysis and calculation using the appropriate weld metal constitution diagrams in Section III. When new welding procedure qualification tests are evaluated for these applications, including repair welding of raw materials, they are performed in accordance with the requirements of Sections III and IX.

The results of all the destructive and nondestructive tests are reported in the procedure qualification record in addition to the information required by Section III.

The starting welding materials used for fabrication and installation welds of austenitic stainless steel materials and components meet the requirements of Section III. The austenitic stainless steel welding material conforms to ASME weld metal analysis A-7 (designated A-8 in the 1974 edition of the ASME Code), type 308 or 308L for all applications. Bare weld filler metal, including consumable inserts, used in inert gas welding processes conform to ASME SFA 5.9, and are procured to contain not less than 5-percent delta ferrite according to Section III. Weld filler metal materials used in flux shielded welding processes conform to ASME SFA 5.4 or 5.9 and are procured in a wire-flux combination to be capable of providing not less than 5-percent delta ferrite in the deposit according to Section III. Welding materials are tested using the welding energy inputs to be employed in production welding.

Combinations of approved heat and lots of starting welding materials are used for all welding processes. The welding quality assurance program includes identification and control of welding material by lots and heats as appropriate. All of the weld processing is monitored according to approved inspection programs which include review of starting materials, qualification records, and welding parameters. Welding systems are also subject to:

- Quality assurance audit including calibration of gauges and instruments.
- Identification of starting and completed materials.
- Welder and procedure qualifications.
- Availability and use of approved welding and heat-treating procedures.
- Documentary evidence of compliance with materials, welding parameters, and inspection requirements.

Fabrication and installation welds are inspected using nondestructive examination methods according to Section III rules.

To ensure the reliability of these controls, Westinghouse has completed a delta ferrite verification program, described in reference 3. This program has been approved as a valid approach to verify the Westinghouse hypothesis and is considered an acceptable alternative for conformance with the NRC Interim Position on Regulatory Guide 1.31. The Regulatory Staff's acceptance letter and topical report evaluation were received on December 30, 1974. The program results, which do support the hypothesis presented in reference 3, are summarized in reference 4.

Section 1.9 indicates the degree of conformance of the austenitic stainless steel components of the RCPB with Regulatory Guides 1.34, Control of Electroslag Properties, and 1.71, Welder Qualification for Areas of Limited Accessibility.

5.2.3.5 References

1. Logsdon, W. A., Begley, J. A., and Gottshall, C. L., "Dynamic Fracture Toughness of ASME SA-508 Class 2a and ASME SA-533 Grade A Class 2 Base and Heat-Affected Zone Material and Applicable Weld Metals," WCAP-9292, March 1978.

2. Golik, M. A., "Sensitized Stainless Steel in Westinghouse PWR Nuclear Steam Supply Systems," WCAP-7477-L (Proprietary), March 1970, and WCAP-7735 (Nonproprietary), August 1971.

3. Enrietto, J. F., "Control of Delta Ferrite in Austenitic Stainless Steel Weldments," WCAP-8324-A, June 1975.

4. Enrietto, J. F., "Delta Ferrite in Production Austenitic Stainless Steel Weldments," WCAP-8693, January 1976.

5.2.4 INSERVICE INSPECTION AND TESTING OF REACTOR COOLANT PRESSURE BOUNDARY[a]

Inservice inspection and testing of Class 1 pressure-retaining components such as vessels, piping, pumps, valves, bolting, and supports within the reactor coolant pressure boundary shall be performed in accordance with Section XI of the American Society of Mechanical Engineers (ASME) Code including any applicable addenda in accordance with 10 CFR 50.55a(g)(4)(ii) (specific edition and any applicable addenda of the code will be delineated in each program), with certain exceptions whenever specific written relief is granted by the Nuclear Regulatory Commission (NRC) in accordance with 10 CFR 50.55a(a)3 and 10 CFR 50.55a(g)(6)(i). The inservice testing of pumps and valves in accordance with the requirements of Articles IWP and IWV of the code is discussed in subsection 3.9.6. Class 2 and 3 components examinations are addressed in section 6.6.

The preservice inspection program requirements for each unit were completed prior to the commercial operation date for each of the respective units. The preservice inspection program for Unit 1 complied with the ASME Code, Section XI, 1980 Edition including addenda through Winter 1980, except that reactor pressure vessel examinations were performed using the 1980 Edition including addenda through Winter 1981. The preservice inspection program for Unit 2 complied with the ASME Code, Section XI, 1983 Edition including addenda through Summer 1983, except that reactor pressure vessel examinations were performed using the 1980 Edition including addenda through Winter 1981. Certain preservice inspection requirements of the ASME Code, Section XI were determined to be impractical and relief requests were granted by the NRC, pursuant to 10 CFR 50.55a(g)(i). The relief requests were supported by information pursuant to 10 CFR 50.55a (a) (3). The inservice inspection program and inservice test program were submitted to the NRC prior to commercial operation. These programs comply with applicable inservice inspection provisions of 10 CFR 50.55a(g) and the NRC guidelines attached as an appendix to section 121.0 of review questions entitled "Guidance for Preparing Preservice and Inservice Inspection Programs and Relief Requests Pursuant to 10 CFR 50.55a(g)." Where compliance with code requirements is not practical, relief requests have been submitted to the NRC for review and approval. The inservice programs detail the areas subject to examination and method, extent, and frequency of examinations. Additionally, component supports and snubber testing requirements are included in the inspection programs.

5.2.4.1 System Boundary Subject to Inspection

In addition to the reactor pressure vessel, all Class 1 components such as vessels, piping, pumps, valves, bolting, and supports shall be inspected to the extent practical, in accordance with Article IWB of ASME Code, Section XI. Class 1 pressure-retaining components and their specific boundaries are identified in the inspection plan documents.

[a] The Inservice Inspection Program is credited as a license renewal aging management program (see subsection 19.2.13).

5.2.4.2 Arrangement and Accessibility

The physical arrangement of components was designed to allow personnel and equipment access to the extent practical to perform the required inservice examinations. Removable insulation and shielding was provided on those piping systems requiring volumetric and surface examination. Temporary or permanent working platforms, scaffolding, and ladders are provided to facilitate access to piping welds.

An inservice inspection design review was undertaken to identify exceptions to the access requirements of the code with subsequent design modifications and/or inspection technique development to ensure code compliance to the extent practical. Additional exceptions may be identified and reported to the NRC after plant operation, as specified in 10 CFR 50.55a(g)(5)(iv).

Space has been provided to handle and store insulation, structural members, shielding, and other materials related to the inspection. Suitable hoists and other handling equipment, lighting, and sources of power for inspection equipment were installed at appropriate locations. The reactor pressure vessel (RPV) inspections are performed primarily from the vessel internal surfaces. Other areas of the RPV such as the closure head are accessible from the outer surfaces of the vessel for inspection. Closure studs, nuts, and washers are removed to a dry location for direct inspection.

5.2.4.3 Examination Techniques and Procedures

The visual, surface, and volumetric examination techniques, procedures, and special techniques are in accordance with the requirements of subarticle IWA-2200 and table IWB-2500-1 of the ASME Code, Section XI except where compliance with code requirements is not practical and relief has been requested from the NRC. Liquid penetrant methods and/or magnetic particle methods are used for surface examinations. Radiography and/or ultrasonic techniques, whether manual or remote, are used for volumetric examinations. A special vessel inspection tool is used to inspect the RPV welds from the vessel internal surfaces. Welds located in the reactor vessel beltline region are examined to meet the requirements of Regulatory Guide 1.150 to the extent practicable. Other RPV welds are examined to meet the requirements of Regulatory Guide 1.150, with the exception of the near surface examination, to the extent practicable. Other examination techniques may be used provided that the results are demonstrated to be equivalent or superior to the above techniques.

5.2.4.4 Inspection Intervals

Inspection intervals are as defined in subarticles IWA-2400 and IWB-2400 of ASME Code, Section XI. The interval may be extended by as much as 1 year to permit inspections to be concurrent with plant outages. It is intended that inservice examinations be performed during normal plant outages such as refueling shutdowns or maintenance shutdowns occurring during the inspection interval.

5.2.4.5 Examination Categories and Requirements

The examination categories and requirements are in accordance with subarticle IWB-2500 and table IWB-2500-1 of ASME Code, Section XI. The preservice examinations complied with IWB-2200.

5.2.4.6 Evaluation of Examination Results

Examination results are evaluated in accordance with IWB-3000, with flaw indications in accordance with IWB-3400 and table IWB-3410. Repair procedures are in accordance with IWB-4000 of ASME Code, Section XI.

5.2.4.7 System Leakage and Hydrostatic Pressure Tests

System pressure tests comply with IWA-5000 and IWB-5000 of ASME Code, Section XI.

5.2.5 DETECTION OF LEAKAGE THROUGH REACTOR COOLANT PRESSURE BOUNDARY

The reactor coolant pressure boundary (RCPB) leakage detection systems monitor leaks from the reactor coolant and associated systems. These systems provide information which permit the plant operators to take corrective action if a leak is evaluated as detrimental to the safety of the facility.

5.2.5.1 Design Bases

The leak detection systems are designed in accordance with the requirements of 10 CFR 50 and the general design criterion 30 to provide a means of detecting and, to the extent practical, identifying the source of the reactor coolant leakage. The systems conform with Regulatory Guide 1.45. Main systems that monitor the environmental condition of the containment include the sump level monitoring system, the airborne particulate radioactivity monitoring systems, and the containment fan cooler condensate measuring system. In addition to the above systems, the humidity, temperature, pressure, and radiogas monitors provide indirect indication of leakage to the containment.

Associated systems and components connected to the reactor coolant system have intersystem leakage monitoring devices.

These leakage detection systems are qualified for all seismic events not requiring a shutdown. The airborne radioactivity monitoring system is qualified for a safe shutdown earthquake (SSE).

5.2.5.1.1 Leakage Classification

RCPB leakage is classified as either identified or unidentified leakage. Identified leakage includes: leakage into closed systems, such as pump seal or valve packing leaks that are captured, flow metered, and conducted to a sump or collecting tank; or leakage into the containment atmosphere from sources that are both specifically located and known either not to interfere with the operation of unidentified leakage monitoring systems or not to be from a flaw in the RCPB; or leakage into auxiliary systems and secondary systems. Unidentified leakage is all other leakage.

5.2.5.1.2 Limits for Reactor Coolant Leakage

Limits for reactor coolant leakage are identified in the Technical Specifications.

5.2.5.2 Identified Intersystem Leakage Detection

Unidentified leakage into closed primary systems is directed to the reactor coolant drain tank or pressurizer relief tank. Identified leakage, such as pump seal or valve packing leakage, is directed to the reactor coolant drain tank where it is monitored by tank pressure, temperature, level, and flow instrumentation on the reactor coolant drain tank discharge lines.

Identified leakage, such as leakage past the pressurizer safety valves or power-operated relief valves (PORVs), is directed to the pressurizer relief tank. This leakage is monitored by temperature instrumentation in the piping system and tank pressure, temperature, and level instrumentation. Leakage collected in the pressurizer relief tank is directed to the reactor coolant drain tank for subsequent treatment and discharge.

An important identified leakage path for reactor coolant into other systems is through the steam generator tubes into the secondary side of the steam generator. Identified leakage to the steam generators is detected by means of the steam generator sample liquid or condenser air ejector radiation monitors. Two additional primary-to-secondary leak detection systems are also provided: a noble gas detector and a system utilizing N16 as the detection medium. The N16 detector is installed in the turbine building main steam pipe chase, between the two main steam pipes. The N16 leak monitor is independent of the primary loop fission and corrosion product radioactivity, so when the tube leak rates increase without considerable changes in the secondary side radioactivity levels, the system can still detect small leaks. The noble gas detector is located in the condenser steam jet air ejector discharge header immediately prior to the filtration unit. For details of these radiation monitors, see subsection 11.5.2.

Auxiliary systems connected to the RCPB incorporate design and administrative provisions that serve to limit leakage. These provisions include isolation valves designed for low seat leakage, periodic testing of RCPB check valves (paragraph 6.3.4.2), and inservice inspection (subsection 5.2.4 and section 6.6). Leakage is detected by increasing auxiliary system level, temperature, and pressure indications or lifting of relief valves accompanied by increasing values of monitored parameters in the relief valve discharge path. These systems are isolated from the RCS by normally closed valves and/or check valves.

5.2.5.2.1 Description and Operation of Identified Leak Detection System

A. Residual Heat Removal System (RHRS) (Suction Side)

The RHRS is isolated from the RCS on the suction side by motor-operated valves HV-8701A/B and HV-8702A/B. Leakage past these valves is detected by lifting of relief valves PSV-8708A or PSV-8708B, accompanied by increasing pressurizer relief tank level, pressure, and temperature indications and alarms on the main control board.

B. Safety Injection System (SIS)/Accumulators

The accumulators are isolated from the RCS by check valves 1204-U6-083 through -086 and 1204-U6-079 through -082. Leakage past these valves and into the accumulator subsystem is detected by redundant control room accumulator pressure and level indications and alarms.

C. SIS/RHR Discharge Subsystem

The RHR pump portion of the SIS is isolated from the RCS by check valves 1204-U6-083 through -086, 1204-U6-147 through -150, 1204-U6-125 and -126, 1204-U6-128 and -129, and normally closed motor-operated valve HV-8840.

Leakage past these valves will eventually pressurize the RHR discharge header and the pump suction header through the normally open pump miniflow isolation valves FV-610 or FV-611. A continued increase in RHR pump discharge pressure will be indicated in the control room and ultimately result in lifting relief valves PSV-8708A and PSV-8708B in the suction header.

D. SIS/Safety Injection Pump Subsystem

The safety injection pump portion of the SIS is isolated from the RCS by check valves 1204-U4-083 through -086, 1204-U4-143 through -146, 1204-U6-124 through -127, 1204-U4-120 through -123, and normally closed motor- operated valves HV-8802A/B. Leakage past these valves will pressurize the safety injection pump discharge header, resulting in control room indication of increasing pressure and, eventually, lifting of relief valve PSV-8851 or PSV-8853A/B.

E. SIS/Centrifugal Charging Pump Subsystem

The charging pump subsystem of the SIS is isolated from the RCS by check valves 1204-U4-026 through -029, 1204-U6-013, and motor-operated valves HV-8801A/B. Leakage past valves HV-8801A/B is not possible, since the valve inlets are pressurized by the operating charging pump.

F. Head Gasket Monitoring Connections

The reactor vessel flange and head are sealed by two metallic O-rings. These gaskets are of the hollow self-energizing type in which pressure of the fluid being sealed enters the interior of the gasket. The O-rings are fastened to the closure head by a mechanical connection to facilitate removal.

Seal leakage is detected by means of two leak-off connections: one between the inner and outer ring, and one outside the outer O-ring. A manual isolation valve is installed just outside the missile barrier of each leak-off line. Downstream of these valves the lines are headered before being routed to the reactor coolant drain tank in the waste processing system. An air-operated isolation valve, actuated from the control board, is installed in the common line. During normal plant operation, the leak-off piping is aligned such that leakage across the inner O-ring passes through valves 1201-U4-088 and HV-8032 into the drain tank. A surface mounted, resistance temperature detector, installed on the bottom of the common pipe, signals leakage at an alarm setpoint. A blind flanged branch line containing isolation valve 1201-U4-089 is provided to confirm and establish the magnitude of the leakage.

Once inner O-ring leakage is discovered, valve 1201-U4-087 should be opened and valve 1201-U4-088 closed so that possible leakage across the second O-ring would be monitored.

In addition, during plant refueling operations both the inner and outer reactor vessel flange leak-off valves are closed. This prevents possible gas leakage from the reactor coolant drain tank to the containment atmosphere. Refer to drawings 1X4DB111 and 2X4DB111 for the flow diagram representation.

The reactor vessel is the only flanged vessel within the RCPB that is provided with leak-off collection provisions.

Leakage past the reactor vessel head gaskets results in temperature indication and alarm in the control room.

G. Component and Auxiliary Component Cooling Water Systems

Leakage from the RCS to the component cooling water (CCW) and auxiliary component cooling water (ACCW) systems, which service all RCPB associated components that require cooling, is detected by the CCW and ACCW radioactivity monitoring system (subsection 11.5.2) and/or increasing surge tank level. Components serviced by these auxiliary cooling systems include: reactor coolant pump thermal barriers, RHR heat exchangers, letdown line heat exchangers, reactor coolant pump seal water heat exchangers.

5.2.5.3 Unidentified Leakage Detection

Normally, unidentified leakage from the RCS is very low. The RCS is an all-welded system, with the exception of the connections on the pressurizer safety valves, reactor vessel head, pressurizer and steam generator manways, and reactor vessel head vent, which are flanged.

In general, valves in the RCS that are 2 in. and under are of the packless type. All valves larger than 2 in. have dual packing with a leak-off connection to the reactor coolant drain tank between the two packings or a reduced packing configuration with the valve stem leakoff line capped.

Primary indications of unidentified coolant leakage to the containment are provided by air particulate radioactivity monitors, gaseous radioactivity monitors, fan cooler condensate flow monitors, and containment sump level monitors.

In normal operation, these primary monitors show a background level that is indicative of the normal level of unidentified leakage inside the containment. Variations in airborne radioactivity or specific humidity above the normal level signify an increase in unidentified leakage rates and signal to the plant operators that corrective action may be required. Similarly, increases in containment sump level signify an increase in unidentified leakage.

RCS unidentified leakage may also be indicated by increasing charging pump flowrate compared with normal RCS inventory changes and by unscheduled increases in reactor makeup water usage.

Reactor coolant inventory monitoring provides an indication of system leakage. Net level changes in the pressurizer and volume control tank are indicative of system leakage, since the chemical and volume control system is a closed loop connected to the RCS. Monitoring net makeup to the chemical and volume control system, as well as net collected leakage, provides an important method of obtaining information for use in establishing a water inventory balance. An abnormal increase in makeup water requirements or a significant change in the water inventory balance can be indicative of increased system leakage.

The sensitivity and response time of the detection equipment for unidentified leakage is such that a leakage rate, or its equivalent, of 1 gal/min can be detected in approximately 1 h.

The above methods are supplemented by visual and ultrasonic inspections of the RCPB during plant shutdown periods, in accordance with the inservice inspection program (subsection 5.2.4).

5.2.5.3.1 Description and Operation of Main Unidentified Leak Detection Systems

Systems employed for detecting leakage to the containment from unidentified sources are:

- Containment airborne particulate radioactivity monitor.
- Containment gaseous radioactivity monitor.

- Containment air cooler condensate flow monitor.
- Containment sump level monitor.

Additionally, humidity, temperature, and pressure monitoring of the containment atmosphere are used for alarms and indirect indication of leakage to the containment.

A. Containment Airborne Particulate Radioactivity Monitoring System

An air sample is drawn outside the containment into a closed system by a sample pump and is then consecutively passed through a particulate filter with detector and a gaseous monitor chamber with detector. The filter collects 99 percent of the particulate matter greater than 1 mm in size. The sample transport system includes:

- A pump to obtain the air sample.
- A flow control valve to provide flow adjustment.
- A flow meter to indicate the flowrate.
- A flow alarm assembly to provide high- and low-flow alarm signals.

The particulate filter is continuously monitored by a scintillation crystal with a photomultiplier tube that provides an output signal proportional to the activity collected on the filter. The particulate monitor has a minimum detectable concentration of 10^{-11} $\mu Ci/cm^3$and a range of 10^{-11} to 10^{-6} $\mu Ci/cm^3$. More details concerning the particulate monitors can be found in subsection 11.5.2.

Particulate activity can be correlated with the coolant fission and corrosion product activities. Any increase of more than two standard deviations above the count rate for background would indicate a possible leak. The total particulate activity concentration above background, due to an abnormal leak and natural decay, increases almost linearly with time for the first several hours after the beginning of a leak. As shown in figure 5.2.5-1, with 0.01-percent failed fuel, containment background airborne particulate radioactivity equivalent to 10^{-3} percent/day, and a partition factor equal to 0.001, a 1-gal/min leak would be detected in approximately 1 h. Larger leaks would be detected in proportionately shorter times (exclusive of sample transport time, which remains constant). The detection capabilities and response times are shown in figure 5.2.5-1.

The activity is indicated on displays and electronically recorded. High-activity alarm indications are displayed on the radiation monitoring cabinets. Local alarms provide operational status of supporting equipment such as pumps, motors, and flow and pressure controllers.

The leakage flowrate can be determined by performing a water inventory balance of the reactor coolant system when the count rate indicates a possible leak as explained above.

B. Containment Gaseous Radioactivity Monitoring System

The containment gaseous radioactivity monitor determines gaseous radioactivity in the containment by monitoring continuous air samples from the containment atmosphere. After passing through the gas monitor, the sample is returned via the closed system to the containment atmosphere.

Each sample is continuously mixed in a fixed, shielded volume where its activity is monitored. The monitor has a range of 10^{-7} to 10^{-2} μCi/cm^3 and a minimum detectable concentration of 5 x 10^{-7} μCi/cm^3.

The containment gaseous radioactivity monitors are described in subsection 11.5.2.

Gaseous radioactivity can be correlated with the gaseous activity of the reactor coolant. Any increase more than two standard deviations above the count rate for background would indicate a possible leak. The total gaseous activity level above background increases almost linearly for the first several hours after the beginning of the leak. As specified in figure 5.2.5-1, with 0.01-percent failed fuel, containment background airborne gaseous radioactivity equivalent to 1 percent/day, and a partition factor equal to 1, a 1-gal/min leak would be detected in approximately 1 h. Larger leaks would be detected in proportionately shorter times (exclusive of the sample transport time, which remains constant). The detection capabilities and response times are shown in figure 5.2.5-1.

The detector outputs are transmitted to the radiation monitoring system cabinets in the control room, where the activity is indicated by displays and electronically recorded. High-activity alarm indications are displayed on the control board annunciator in addition to the radiation monitoring system cabinets. Local alarms annunciate the operational status of the supporting equipment.

The leakage flowrate can be determined by performing a water inventory balance of the reactor coolant system when the count rate indicates a possible leak as explained above.

The containment purge system radioactivity monitors (subsection 11.5.2) serve as backup to the containment air particulate and gaseous airborne radioactivity monitoring system while the purge is in operation.

The containment purge monitors function in the same manner as the containment air particulate and gaseous radioactivity monitors, except that the purge monitors sample from the containment purge exhaust line.

C. Containment Air Cooler Condensate Monitoring System

The condensate monitoring system permits measurements of the liquid runoff from the containment cooler units. It consists of a containment cooler drain collection header, a vertical standpipe, valving, and standpipe level instrumentation. The condensation from the containment coolers flows via the collection header to the vertical standpipe. A differential pressure transmitter provides standpipe level signals. The system provides measurements of low leakages by monitoring standpipe level increase versus time.

The condensate flowrate is a function of containment humidity, nuclear service cooling water (NSCW) temperature, and containment purge rate. The water vapor dispersed by a 1-gal/min leak is usually greater than the water vapor brought in with the outside air. Air brought in from the outside is heated to 60°F before it enters the containment.

After the air enters the containment, it mixes with the containment atmosphere and is heated to between 100°F and 120°F while the relative humidity drops. The most important factor in condensing the water vapor is the temperature of the NSCW which is supplied to the containment coolers. This water is assumed to vary in temperature between 35°F and 95°F.

Drainage flowrate from the units due to normal condensation is calculated for the ambient (background) atmospheric conditions present within the containment. With the initiation of an additional or abnormal leak, the containment atmosphere humidity and condensation runoff rate both begin to increase, the water level rises in the vertical pipe, and the high condensate flow alarm is actuated. Level changes of as little as 0.25 in. in the cooler condensate standpipes can be detected.

Figure 5.2.5-1 shows the detection capabilities of the system for various conditions. Normal background leakage increases containment humidity to the point where the condensation rate will increase, which improves the detection capabilities of this system. As shown in figure 5.2.5-1, a sensitivity of 1 gal/min in approximately 1 h can be achieved when supplying cold NSCW to the containment coolers or with the initial background leakage.

The rate of leakage can be determined when the precise NSCW, outside air, and containment air temperatures, and the outside relative humidity are known.

D. Containment Sump Level Monitoring System

Since a leak in the primary system would result in reactor coolant flowing into the containment normal or reactor cavity sumps, leakage would be indicated by a level increase in the sump. Indication of increasing sump level is transmitted from the sump to the control room level indicator by means of a sump level transmitter. The system provides measurements of low leakages by monitoring level increase versus time.

The detection capabilities of the containment normal sump and reactor cavity sump are shown in figure 5.2.5-1, assuming that the water from the leak is collected in the sump.

The actual reactor coolant leakage rate can be established from the increase above the normal rate of change of sump level. A check of other instrumentation would be required to eliminate possible leakage from nonradioactive systems as a cause of an increase in sump level. The leakage rate can also be determined from the frequency of sump pump operation.

Under normal conditions, the containment normal sump pumps operate infrequently and reactor cavity sump pump operates very infrequently. Gross leakage can be surmised from unusual frequency of pump operation. Sump level and pump running indication are provided in the control room to alert the operators.

5.2.5.3.2 Additional Unidentified Leakage Detection Methods

Other methods available for detecting leakage are:

A. Charging Pump Operation

During normal operation, one of the charging pumps is in operation. If a gross increase in reactor coolant leakage occurs, the flowrate of the charging pump would increase, indicating leakage from the RCS. This leakage must be sufficient to cause a decrease in pressurizer or volume control tank level that is within the sensitivity range of the level indicators. The flowrate of the charging

pump would automatically increase to try to maintain pressurizer level. Charging pump discharge flow indication is provided in the control room.

The leakage rate can be determined by the amount that the charging pump flowrate increases above the letdown flowrate to maintain constant pressurizer level. Any significant increase in the charging flowrate is a possible indication of a leak.

B. Containment Humidity Monitoring System

The containment humidity system, utilizing temperature-compensated humidity detectors, is provided to determine the water-vapor content of the containment atmosphere. An increase in the humidity of the containment atmosphere indicates release of water within the containment. The range of the containment humidity measuring system is 5- to 99-percent relative humidity at 80°F with a temperature range of 40 to 120°F. The accuracy of the humidity detectors is ±3 percent.

The response of the containment humidity under various outside air conditions and no leakage falls within the extremes shown in figure 5.2.5-1. The humidity monitor supplements the condensate monitor. It is most sensitive under conditions when there is no condensation.

A rapid increase of humidity over the background level by more than 10 percent can be taken as a probable indication of a leak.

The leakrate can be determined when the outside air temperature and humidity and the containment atmosphere temperature are known.

C. Liquid Inventory

The operators can surmise gross leakage from changes in the reactor coolant inventory. Noticeable decreases in the pressurizer level not associated with known changes in operation are investigated. Likewise, makeup water usage information which is available from the plant computer is checked frequently for unusual makeup rates not due to plant operations.

5.2.5.4 Safety Evaluation

The leak detection system has no safety design basis; however, the containment atmosphere radioparticulate and radiogas monitors are qualified for an SSE per the recommendation of Regulatory Guide 1.45.

5.2.5.5 Tests and Inspections

Periodic testing of leakage detection systems is conducted to verify the operability and sensitivity of detector equipment. These tests include installation calibrations and alignments, periodic channel calibrations, frictional tests, and channel checks. A description of calibration and maintenance procedures and frequencies for the containment radioactivity monitoring system is presented in subsection 11.5.2.

The humidity detector and condensate measuring system are also periodically tested to ensure proper operation and verify sensitivity.

Inservice inspection criteria, the equipment used, procedures involved, the frequency of testing, inspection, surveillance, and examination of the structural and leaktight integrity of RCPB components are described in detail in subsection 5.2.4.

5.2.5.6 Instrumentation Applications

The following indications are provided in the control room to allow operating personnel to monitor for leakage:

A. Containment air particulate monitor - air particulate activity.

B. Containment gaseous activity monitor - gaseous activity.

C. Containment cooler condensate monitoring system - standpipe level.

D. Containment humidity measuring system - containment humidity.

E. Containment normal sump level and reactor cavity sump level.

F. Gross leakage detection methods - charging pump flowrate, letdown flowrate, pressurizer level, and reactor coolant temperatures are available for the charging pump flow method. Containment sump levels and pump operation are available for the sump pump operation method. Total makeup waterflow is available from the plant computer for liquid inventory.

TABLE 5.2.1-1

APPLICABLE CODE ADDENDA FOR RCS COMPONENTS

Reactor vessel	ASME III, 1971 Edition through Summer 1972 Addenda
Steam generator	ASME III, 1971 Edition through Summer 1972 Addenda
Pressurizer	ASME III, 1971 Edition through Summer 1972 Addenda
Control rod drive mechanism (CRDM) housing	ASME III, 1974 Edition through Summer 1975 Addenda
CRDM head adapter	ASME III, 1971 Edition through Summer 1972 Addenda
Reactor coolant pump	ASME III, 1971 Edition through 1972 Winter Addenda
Reactor coolant pipe	ASME III, 1974 Edition through Winter 1975 Addenda
Surge line	ASME III, 1974 Edition through Winter 1975 Addenda
Valves	
Pressurizer safety	ASME III, 1971 Edition through Winter 1972 Addenda
Motor operated	ASME III, 1974 Edition through Summer 1974 Addenda
Manual (3 in. and larger)	ASME III, 1974 Edition through Summer 1974 Addenda
Control	ASME III, 1971 Edition through Summer 1972 Addenda

TABLE 5.2.3-1 (SHEET 1 OF 5)

PRIMARY AND AUXILIARY COMPONENTS MATERIAL SPECIFICATIONS

Reactor Vessel Components	
Shell and head plates (other than core region)	SA-533, Grade A, B, or C, Class 1 or 2 (vacuum treated)
Shell plates (core region)	SA-533, Grade A or B, Class 1 (vacuum treated)
Shell, flange, and nozzle forgings, nozzle safe ends	SA-508, Class 2 or 3; SA-182; Grade F304 or F316
Control rod drive mechanism (CRDM) and/or ECCS appurtenances, upper head	SB-166 or SB-167 and SA-182 Grade F304
Instrumentation tube appurtenances, lower head	SB-166 or SB-167 and SA-182, Grade F304, F304L, or F316
Closure studs, nuts, washers, inserts, and adaptors	SA-540, Class 3, Grade B23 or B24
Core support pads	SB-166 with carbon less than 0.10 percent
Monitor tubes and vent pipe	SA-312 or SA-376, Grade TP304 or TP316 or SB-166, SB-167, or SA-182, Grade F316
Vessel supports, seal ledge, and head lifting lugs	SA-516, Grade 70 (quenched and tempered) or SA-533, Grade A, B, or C, Class 1 or 2 (Vessel supports may be of weld metal buildup of equivalent strength of the nozzle material.)
Cladding and buttering	Stainless Steel Weld Metal Analysis A-8 and Ni-Cr-Fe Weld Metal F-Number 43

TABLE 5.2.3-1 (SHEET 2 OF 5)

Steam Generator Components	
Pressure plates	SA-533, Grade A, B, or C, Class 1 or 2
Pressure forgings (including nozzles and tube sheet)	SA-508, Class 1, 2, 2a, or 3
Nozzle safe ends	Stainless Steel Weld Metal Analysis A-8
Channel heads	SA-533, Grade A, B, or C, Class 1 or 2 or SA-216, Grade WCC
Tubes	SB-163 (Ni-Cr-Fe annealed)
Cladding and buttering	Stainless Steel Weld Metal Analysis A-8 and Ni-Cr-Fe Weld Metal F-Number 43
Closure studs/nuts	SA-193, Grade B7/SA-194 Gr 7
Pressurizer Components	
Pressure plates	SA-533, Grade A, Class 2
Pressure forgings	SA-508, Class 2 or 2a
Nozzle safe ends	SA-182, Grade F316L
Cladding and buttering	Stainless Steel Weld Metal Analysis A-8 and Ni-Cr-Fe Weld Metal F-Number 43
Closure studs/nuts	SA-193, Grade B7/SA-194 Gr 7
Reactor Coolant Pump	
Pressure forgings	SA-182, Grade F304, F316, F347, or F348
Pressure casting	SA-351, Grade CF8, CF8A, or CF8M
Tube and pipe	SA-213; SA-376 or SA-312, Seamless, Grade TP304 or TP316

TABLE 5.2.3-1 (SHEET 3 OF 5)

Pressure plates	SA-240, Type 304 or 316
Bar material	SA-479, Type 304 or 316
Closure bolting	SA-193, SA-320, SA-540, or SA-453, Grade 660
Flywheel	SA-533, Grade B, Class 1
Reactor Coolant Piping	
Reactor coolant pipe	SA-351, Grade CF8A Centrifugal Casting
Reactor coolant fittings, branch nozzles	SA-351, Grade CF8A and SA-182, (Code Case 1423-2) Grade 316N
Surge line	SA-376, Grade TP304, TP316, or F304N
Auxiliary piping	SA-312 and SA-376 Grades TP304 and TP316 to ANSI B36.10 or B36.19
Socket weld fittings	ANSI B16.11
Piping flanges	ANSI B16.5
Full-Length CRDM	
Latch housing	SA-182, Grade F304 or SA-351, Grade CF8
Rod travel housing	SA-182, Grade F304 or SA-336 Class 304
Cap	SA-479, Type 304
Welding materials	Stainless Steel Weld Metal Analysis A-8
Valves	
Bodies	SA-182, Grade F316 or SA-351, Grade CF8 or CF8M

TABLE 5.2.3-1 (SHEET 4 OF 5)

Bonnets	SA-182, Grade F316 or SA-351, Grade CF8 or CF8M
Discs	SA-182, Grade F316 or SA-564, Grade 630 or SA-351, Grade CF8 or CF8M
Stems	SA-182, Grade F316 or SA-564, Grade 630
Pressure retaining bolting	SA-453, Grade 660
Pressure retaining nuts	SA-453, Grade 660 or SA-194, Grade 6
Auxiliary Heat Exchangers	
Heads	SA-240, Type 304
nozzle necks	SA-182, Grade F304, SA-240 and 312, Type 304
Tubes	SA-213, Grade TP304; SA-249 Type 304
Tube sheets	SA-182, Grade F304; SA-240, Type 304 and SA-516 GR 70 clad with Stainless Steel Weld Metal Analysis A-8
Shells	SA-240 and SA-312, Grade TP304
Auxiliary Pressure Vessels, Tanks, Filters, etc.	
Shells and heads	SA-240, Type 304 or SA-351 Grade CFA8, or SA264 (consisting of SA-537, Class 1 with Stainless Steel Weld Metal Analysis A-8 Cladding)
Flanges and nozzles	SA-182, Grade F304 and SA-105 or SA-350, Grade LF2 and LF3 with Stainless Steel Weld Metal Analysis A-8 Cladding
Piping	SA-312 Type 304 and SA-240, Grade TP304 or TP316

TABLE 5.2.3-1 (SHEET 5 OF 5)

Pipe fittings	SA-403, Grade WP304 Seamless
Closure bolting and nuts	SA-193, Grade B7 and SA-194, Grade 2H or Grade 7
Auxiliary Pumps	
Pump casing and heads	SA-351, Grade CF8 or CF8M; SA-182, Grade F304 or F316
Flanges and nozzles	SA-182, Grade F304 or F316; SA-403 Grade WP316L Seamless
Piping	SA-312, Grade TP304 or TP316 Seamless
Stuffing or packing box cover	SA-351, Grade CF8 or CF8M; SA-240 Type 304 or 304L or 316
Pipe fittings	SA-403, Grade WP316L Seamless
Closure bolting and nuts	SA-193, Grade B6, B7, or B8M; SA-194, Grade 2H or 8M; SA-453 Grade 660, and Nuts, SA-194 Grade 2H, 6 and 8M

TABLE 5.2.3-2

REACTOR VESSEL INTERNALS MATERIAL SPECIFICATIONS

Forgings	SA-182, Grade F304
Plates	SA-240, Type 304
Pipes	SA-312, Grade TP304 Seamless or SA-376, Grade TP304
Tubes	SA-213, Grade TP304 or ASTM A-511 Grade MT304, Code Case1618
Bars	SA-479, Type 304
Castings	SA-351, Grade CF8
Bolting	SA-193, Class 2 (65-90 YS/90 MTS) Code Case 1618 Inconel-750; SA-637, Grade 688, Type 2
Nuts	SA-193, Grade B8
Locking devices	SA-479, Type 304

TABLE 5.2.3-3 (SHEET 1 OF 2)

RECOMMENDED[a] REACTOR COOLANT WATER CHEMISTRY SPECIFICATION

Electrical conductivity	Determined by the concentration of boric acid and alkali present. Expected range is <1 to 40 μmhos/cm at 25°C.
Solution pH	Determined by the concentration of boric acid and alkali present. Expected values range between 4.2 (high boric acid concentration) and 10.5 (low boric acid concentration) at 25°C. Values will be 5.0 or greater at normal operating temperatures.
Oxygen[b]	0.10 ppm, maximum
Chloride[c]	0.15 ppm, maximum
Fluoride[c]	0.15 ppm, maximum
Hydrogen[d]	25 to 50 cm^3 (STP)/kg H_2O
Suspended solids[e]	0.2 ppm, maximum
pH control agent (Li_7OH)	Lithium is coordinated with boron in accordance with the principles of the EPRI PWR Primary Water Chemistry Guidelines, Volume 1.
Boric acid	Variable from 0 to 4000 ppm as boron
Zinc[g]	0.04 ppm, maximum
Silica[f]	1.0 ppm, maximum
Aluminum[f]	0.05 ppm, maximum
Calcium[f] + magnesium	0.05 ppm, maximum
Magnesium[f]	0.025 ppm, maximum

TABLE 5.2.3-3 (SHEET 2 OF 2)

a. Refer to the Technical Requirements Manual for required reactor coolant chemistry limits.

b. Oxygen concentration must be controlled to less than 0.1 ppm in the reactor coolant by scavenging with hydrazine prior to plant operation above 250°F. During power operation with the specified hydrogen concentration maintained in the coolant, the residual oxygen concentration will not exceed 0.005 ppm.

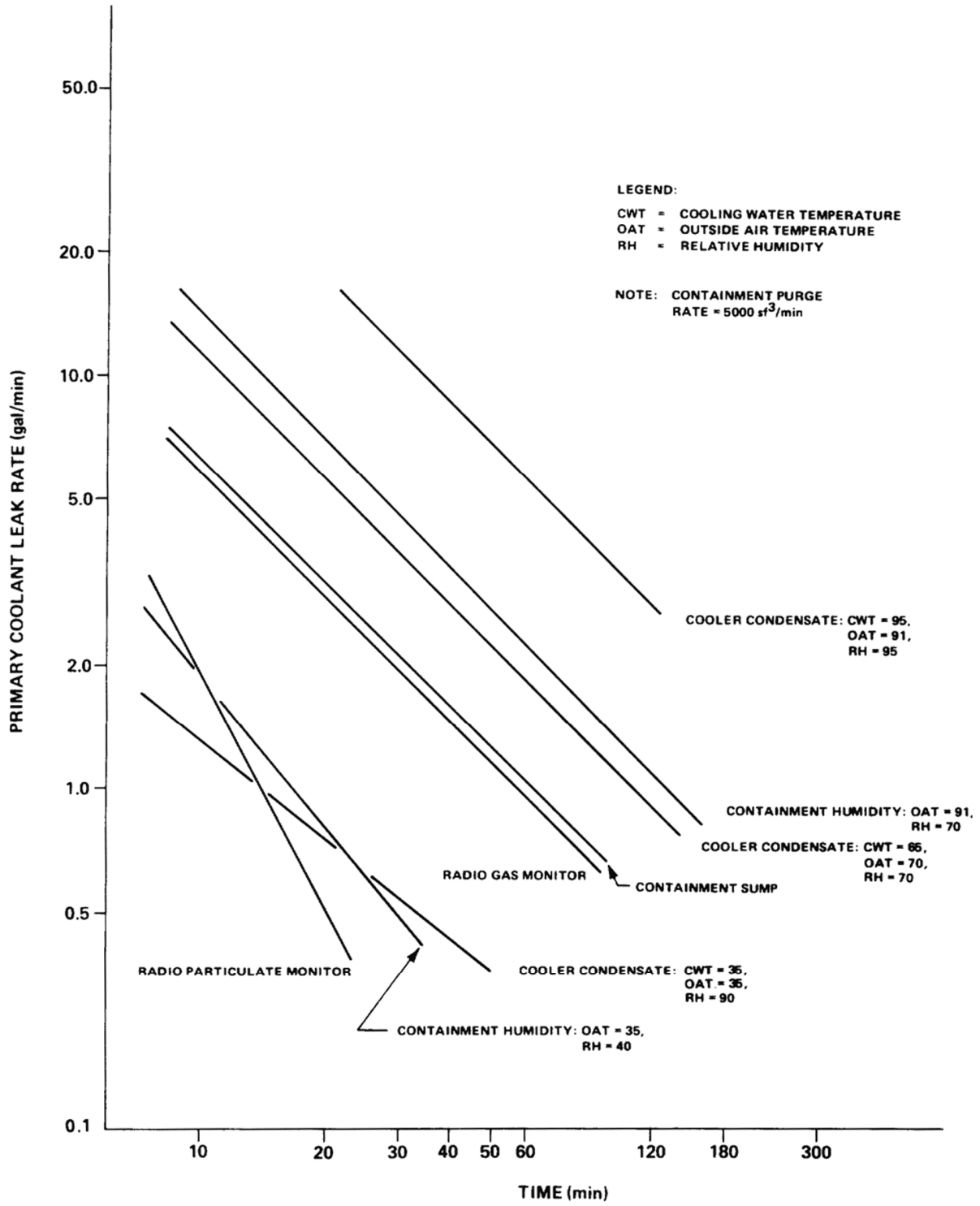
c. Halogen concentrations must be maintained below the specified values at all times when fuel is in the reactor vessel regardless of system temperature.

d. Hydrogen must be maintained in the reactor coolant for all plant operations with nuclear power above 1 MW.

e. Solids concentration determined by filtration through filter having 0.45-μm pore size.

f. These limits are included in the table of reactor coolant specifications as recommended standards for monitoring coolant purity. Establishing coolant purity within the limits shown for these species is judged desirable with the current data base to minimize fuel clad crud deposition which affects the corrosion resistance and heat transfer of the clad.

g. Specification is applicable during power operation when zinc is being injected. Zinc target concentrations are maintained at the lower of 0.04 ppm or that specified in the reload safety analysis.



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PRIMARY COOLANT LEAK DETECTION RESPONSE TIME

FIGURE 5.2.5–1

5.3 REACTOR VESSEL

5.3.1 REACTOR VESSEL MATERIALS

5.3.1.1 Material Specifications

Material specifications are in accordance with the American Society of Mechanical Engineers (ASME) Code requirements and are given in subsection 5.2.3. All ferritic reactor vessel materials comply with the fracture toughness requirements of Section 50.55a and Appendices G and H of 10 CFR 50.

The ferritic materials of the reactor vessel beltline are restricted to the following maximum limits of copper and phosphorus to reduce sensitivity to irradiation embrittlement in service:

Element	Base Metal (percent)	As Deposited Weld Metal (percent)
Copper	0.10 (ladle)	0.10
	0.12 (check)	
Phosphorus	0.012 (ladle)	0.020
	0.017 (check)	

5.3.1.2 Special Processes Used for Manufacturing and Fabrication

A. The vessel is Safety Class 1. Design and fabrication of the reactor vessel is carried out in strict accordance with ASME Code, Section III, Class 1 requirements. The vessel head, flanges, and nozzles are manufactured as forgings. The cylindrical portion of the vessel is made of several shells, each consisting of formed plates joined by full penetration longitudinal and girth weld seams. The hemispherical heads are made from dished plates. The reactor vessel parts are joined by welding, using the single or multiple wire submerged arc and the shielded metal arc processes.

B. The use of severely sensitized stainless steel as a pressure boundary material has been prohibited and has been eliminated by either a select choice of material or by programming the method of assembly.

C. The control rod drive mechanism (CRDM) head adapter threads and surfaces of the guide studs are chrome plated to prevent possible galling of the mated parts.

D. At all locations in the reactor vessel where stainless steel and Inconel are joined, the final joining beads are Inconel weld metal in order to prevent cracking.

E. The location of full penetration weld seams in the upper closure head and vessel bottom head are restricted to areas that permit accessibility during inservice inspection.

F. The stainless steel clad surfaces are sampled to ensure that composition requirements are met.

G. Freedom from underclad cracking is ensured by special evaluation of the procedure qualification for cladding applied on low-alloy steel (SA-508, Class 2).[a]

H. Minimum preheat requirements have been established for pressure boundary welds using low-alloy material. The preheat is maintained until either an intermediate postweld heat treatment or a full postweld heat treatment is completed or until the completion of welding.

I. A field weld is made, after the reactor vessel has been set, to install the permanent reactor vessel cavity seal ring. This stainless steel filler weld joins the seal ring to the reactor vessel seal ledge. A minimum preheat is specified for this weld in compliance with the ASME Code requirements.

5.3.1.3 Special Methods for Nondestructive Examination

The nondestructive examination (NDE) of the reactor vessel and its appurtenances is conducted in accordance with ASME Code, Section III requirements; also, numerous examinations are performed in addition to ASME Code, Section III requirements. The NDE of the vessel is discussed in the following paragraphs, and the reactor vessel quality assurance program is given in table 5.3.1-1.

5.3.1.3.1 Ultrasonic Examination

A. In addition to the required ASME Code straight beam ultrasonic examination, angle beam inspection over 100 percent of one major surface of plate material is performed during fabrication to detect discontinuities that may be undetected by the straight beam examination.

B. In addition to the ASME Code, Section III NDE, all full penetration ferritic pressure boundary welds in the reactor vessel are ultrasonically examined during fabrication. This test is performed upon completion of the welding and intermediate heat treatment but prior to the final postweld heat treatment.

C. After hydrotesting, all full penetration ferritic pressure boundary welds in the reactor vessel, as well as the nozzle to safe end welds, are ultrasonically examined. These inspections are also performed in addition to the ASME Code, Section III NDE.

5.3.1.3.2 Penetrant Examinations

The partial penetration welds for the CRDM head adapters and the bottom instrumentation tubes are inspected by dye penetrant after the root pass, in addition to code requirements. Core support block attachment welds are inspected by dye penetrant after the first layer of weld metal and after each 1/2 in. of weld metal. All clad surfaces and other vessel and head internal surfaces are inspected by dye penetrant after the hydrostatic test.

[a] Underclad cracking of the reactor pressure vessel was evaluated as a time-limited aging analysis (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in paragraph 19.4.6.5.

5.3.1.3.3 Magnetic Particle Examination

The magnetic particle examination requirements below are in addition to the magnetic particle examination requirements of Section III of the ASME Code.

All magnetic particle examinations of materials and welds are performed in accordance with the following:

- Prior to the final postweld heat treatment, only by the prod, coil, or direct contact method.

- After the final postweld heat treatment, only by the yoke method.

The following surfaces and welds are examined by magnetic particle methods. The acceptance standards are in accordance with Section III of the ASME Code.

A. Surface Examinations

1. Magnetic particle examination of all exterior vessel and head surfaces after the hydrostatic test.

2. Magnetic particle examination of all exterior closure stud surfaces and all nut surfaces after final machining or rolling. Continuous circular and longitudinal magnetization is used.

3. Magnetic particle examination of all inside diameter surfaces of carbon and low alloy steel products that have their properties enhanced by accelerated cooling. This inspection is performed after forming and machining and prior to cladding.

B. Weld Examination

Magnetic particle examination of the welds attaching the closure head lifting lugs and refueling seal ledge to the reactor vessel after the first layer and each 1/2 in. of weld metal is deposited. All pressure boundary welds are examined after back-chipping or back-grinding operations.

5.3.1.4 Special Controls for Ferritic and Austenitic Stainless Steels

Welding of ferrite steels and austenitic stainless steels is discussed in subsection 5.2.3. Subsection 5.2.3 includes discussions which indicate the degree of conformance with Regulatory Guide 1.44. Section 1.9 discusses the degree of conformance with Regulatory Guides 1.43, 1.50, 1.71, and 1.99.

5.3.1.5 Fracture Toughness[a]

Assurance of adequate fracture toughness of ferritic materials in the reactor vessel (ASME Code, Section III, Class 1 component) is provided by compliance with the requirements for fracture toughness testing included in NB-2300 to Section III of the ASME Code and Appendix G of 10 CFR 50.

The initial Charpy V-notch minimum upper shelf fracture energy levels for the reactor vessel beltline (including welds) are 75 ft-lb, as required by Appendix G of 10 CFR 50. The vessel

[a] Reactor vessel neutron embrittlement was evaluated as a TLAA for license renewal in accordance with 10 CFR 54.21 (see subsection 19.4.1).

fracture toughness data for Units 1 and 2 are given in tables 5.3.1-2 and 5.3.1-3, respectively. The end-of-life RT_{NDT} and upper shelf energy projections estimated using Regulatory Guide 1.99 for the end-of-life neutron fluence at the 1/4 T and ID reactor vessel locations for Units 1 and 2 are given in tables 5.3.3-2 and 5.3.3-3.

5.3.1.6 Material Surveillance[a]

The reactor vessel material irradiation surveillance specimens shall be removed and examined to determine changes in material properties as required by 10 CFR Part 50, Appendix H, in accordance with the schedule in FSAR tables 5.3.1-8 and 5.3.1-9. The results of these examinations shall be used to update figures in the Pressure Temperature Limits Report (PTLR).(16) In the surveillance program, the evaluation of radiation damage is based on preirradiation testing of Charpy V-notch and tensile specimens and post irradiation testing of Charpy V-notch, tensile, and 1/2-T compact tension (CT) fracture mechanics test specimens. The program is directed toward evaluation of the effect of radiation on the fracture toughness of reactor vessel steels based on the transition temperature approach and the fracture mechanics approach. The program conforms to American Society of Testing Materials (ASTM) E-185-82, Conducting Surveillance Tests for Light-Water-Cooled Nuclear Reactor Vessels, and 10 CFR 50, Appendix H.

[HISTORICAL] The reactor vessel surveillance program uses six specimen capsules. The capsules are located in guide baskets welded to the outside of the neutron shield pads and positioned directly opposite the center portion of the core. The capsules can be removed when the vessel head is removed and can be replaced when the internals are removed. The six capsules contain reactor vessel steel specimens, oriented both parallel and normal (longitudinal and transverse) to the principal rolling direction of the limiting base material located in the core region of the reactor vessel and associated weld metal and weld heat-affected zone metal. The six capsules contain 54 tensile specimens, 360 Charpy V-notch specimens (which include weld metal and weld heat-affected zone material), and 72 CT specimens. Archive material sufficient for two additional capsules is retained at Westinghouse. The surveillance program withdrawal schedule, lead factor, test samples, and materials in the reactor vessel are given in tables 5.3.1-7 and 5.3.1-8. [HISTORICAL]

Removal of these specimen capsules was completed for Unit 1 at refueling outage 1R14 and for Unit 2 at refueling outage 2R14. Refer to paragraph 5.3.1.6.1.4 for a description of the external neutron monitoring system also known as external vessel neutron dosimetry system (EVNDS) used following removal of the last specimen capsules.

Dosimeters, as described below, are placed in filler blocks drilled to contain them. The dosimeters permit evaluation of the flux seen by the specimens and the vessel wall. In addition, thermal monitors made of low melting point alloys are included to monitor the maximum temperature of the specimens. The specimens are enclosed in a tight-fitting stainless steel sheath to prevent corrosion and ensure good thermal conductivity. The complete capsule is helium leak tested. As part of the surveillance program, a report of the residual elements in weight percent to the nearest 0.01 percent is made for surveillance material and as deposited weld metal. Each of the six capsules contains the following specimens:

[a] The Reactor Vessel Surveillance Program is credited as a license renewal aging management program (see subsection 19.2.25).

Material	Number of Charpys	Number of Tensiles	Number of CTs
Limiting base material[a]	15	3	4
Limiting base material[b]	15	3	4
Weld metal[(c)]	15	3	4
Heat-affected zone	15	-	-

The following dosimeters and thermal monitors are included in each of the six capsules:

A. Dosimeters

1. Iron.
2. Copper.
3. Nickel.
4. Cobalt-aluminum (0.15-percent cobalt).
5. Cobalt-aluminum (cadmium shielded).
6. Uranium-238 (cadmium shielded).
7. Neptunium-237 (cadmium shielded).

B. Thermal Monitors

1. 97.5-percent lead, 2.5-percent silver, (579°F melting point).
2. 97.5-percent lead, 1.75-percent silver, 0.75-percent tin (590°F melting point).

The fast neutron exposure of the specimens occurs at a faster rate than that experienced by the vessel wall, with the specimens being located between the core and the vessel. Since these specimens experience accelerated exposure and are actual samples from the materials used in the vessel, the transition temperature shift measurements are representative of the vessel at a later time in life. Data from CT fracture toughness specimens are expected to provide additional information for use in determining allowable stresses for irradiated material.

Correlations between the calculations and measurements of the irradiated samples in the capsules, assuming the same neutron spectrum at the samples and the vessel inner wall, are described in paragraph 5.3.1.6.1. The anticipated degree to which the specimens perturb the fast neutron flux and energy distribution is considered in the evaluation of the surveillance specimen data. Verification and possible readjustment of the calculated wall exposure is made by the use of data on all capsules withdrawn. The schedule for removal of the capsules for postirradiation testing conforms with ASTM E-185-82 and Appendix H of 10 CFR 50.

5.3.1.6.1 Measurement of Integrated Fast Neutron (E > 1.0 MeV) Flux at the Irradiation Samples

The use of passive neutron sensors such as those included in the internal surveillance capsule dosimetry sets does not yield a direct measure of the energy dependent neutron flux level at the

[a] Specimens oriented in the major rolling or working direction.
[b] Specimens oriented normal to the major rolling or working direction.

[(c)] Weld metal to be selected in accordance with ASTM E-185-82.

measurement location. Rather, the activation or fission process is a measure of the integrated effect that the time- and energy-dependent neutron flux has on the target material over the course of the irradiation period. An accurate assessment of the average flux level and, hence, time integrated exposure (fluence) experienced by the sensors may be developed from the measurements only if the sensor characteristics and the parameters of the irradiation are well known. In particular, the following variables are of interest:

- The measured specific activity of each sensor.
- The physical characteristics of each sensor.
- The operating history of the reactor.
- The energy response of each sensor.
- The neutron energy spectrum at the sensor location.

This section describes the procedures used to determine sensor specific activities, to develop reaction rates for individual sensors from the measured specific activities and the operating history of the reactor, and to derive key fast neutron exposure parameters from the measured reaction rates.

5.3.1.6.1.1 Determination of Sensor Reaction Rates. The specific activity of each of the radiometric sensors is determined using established ASTM procedures. Following sample preparation and weighing, the specific activity of each sensor is determined by means of a high purity germanium gamma spectrometer. In the case of the surveillance capsule multiple foil sensor sets, these analyses are performed by direct counting of each of the individual wires or, as in the case of U-238 and Np-237 fission monitors, by direct counting preceded by dissolution and chemical separation of cesium from the sensor.

The irradiation history of the reactor over its operating lifetime is determined from plant power generation records. In particular, operating data are extracted on a monthly basis from reactor startup to the end of the capsule irradiation period. For the sensor sets utilized in the surveillance capsule irradiations, the half-lives of the product isotopes are long enough that a monthly histogram describing reactor operation has proven to be an adequate representation for use in radioactive decay corrections for the reactions of interest in the exposure evaluations.

Having the measured specific activities, the operating history of the reactor, and the physical characteristics of the sensors, reaction rates referenced to full power operation are determined from the following equation:

$$R = \frac{A}{N_0 F Y \sum_j \frac{P_j}{P_{ref}} C_j \left[1 - e^{-\lambda t_j}\right] e^{-\lambda t_d}}$$

where:

A	=	measured specific activity provided in terms of disintegrations per second per gram of target material (dps/grn).
R	=	reaction rate averaged over the irradiation period and referenced to operation at a core power level of P_{ref} expressed in terms of reactions per second per nucleus of target

isotope (rps/nucleus).

N_0 = number of target element atoms per gram of sensor.

F = weight fraction of the target isotope in the sensor material.

Y = number of product atoms produced per reaction.

P_j = average core power level during irradiation period j (MW).

P_{ref} = maximum or reference core power level of the reactor (MW).

C_j = calculated ratio of $\varphi(E > 1.0\ MeV)$ during irradiation period j to the time weighted average $\varphi(E > 1.0\ MeV)$ over the entire irradiation period.

λ = decay constant of the product isotope (sec^{-1}).

t_j = length of irradiation period j (sec).

t_d = decay time following irradiation period j (sec).

and the summation is carried out over the total number of monthly intervals comprising the total irradiation period.

In the above equation, the ratio P_j/P_{ref} accounts for month-by-month variation of power level within a given fuel cycle. The ratio C_j is calculated for each fuel cycle and accounts for the change in sensor reaction rates caused by variations in flux level due to changes in core power spatial distributions from fuel cycle to fuel cycle. Since the neutron flux at the measurement locations within the surveillance capsules is dominated by neutrons produced in the peripheral fuel assemblies, the change in the relative power in these assemblies from fuel cycle to fuel cycle can have a significant impact on the activation of neutron sensors. For a single-cycle irradiation, $C_j = 1.0$. However, for multiple-cycle irradiations, particularly those employing low-leakage fuel management, the additional C_j correction must be utilized in order to provide accurate determinations of the decay-corrected reaction rates for the dosimeter sets contained in the surveillance capsules.

5.3.1.6.1.2 Corrections to Reaction Rate Data. Prior to using the measured reaction rates in the least squares adjustment procedure discussed in paragraph 5.3.1.6.1.3, additional corrections are made to the U-238 measurements to account for the presence of U-235 impurities in the sensors as well as to adjust for the build-in of plutonium isotopes over the course of the irradiation.

In addition to the corrections made for the presence of U-235 in the U-238 fission sensors, corrections are also made to both the U-238 and Np-237 sensor reaction rates to account for gamma ray induced fission reactions occurring over the course of the irradiation.

5.3.1.6.1.3 Least Squares Adjustment Procedure. Least squares adjustment methods provide the capability of combining the measurement data with the neutron transport calculation resulting in a Best Estimate neutron energy spectrum with associated uncertainties. Best Estimates for key exposure parameters such as neutron fluence (E > 1.0 MeV) or iron atom displacements (dpa) along with their uncertainties are then easily obtained from the adjusted spectrum. The use of measurements in combination with the analytical results reduces the uncertainty in the calculated spectrum and acts to remove biases that may be present in the analytical technique.

In general, the least squares methods, as applied to pressure vessel fluence evaluations, act to reconcile the measured sensor reaction rate data, dosimetry reaction cross-sections, and the calculated neutron energy spectrum within their respective uncertainties. For example,

$$R_i \pm \delta_{Ri} = \sum_g \left(\sigma_{ig} \pm \delta_{\sigma_{ig}}\right)\left(\phi_g \pm \delta_{\phi g}\right)$$

relates a set of measured reaction rates, R_i, to a single neutron spectrum, Φg, through the multigroup dosimeter reaction cross-section, σ_{ig}, each with an uncertainty δ.

The use of least squares adjustment methods in light water reactor (LWR) dosimetry evaluations is not new. ASTM has addressed the use of adjustment codes in ASTM Standard E944, "Application of Neutron Spectrum Adjustment Methods in Reactor Surveillance," and many industry workshops have been held to discuss the various applications. For example, the ASTM-EURATOM Symposia on Reactor Dosimetry holds workshops on neutron spectrum unfolding and adjustment techniques at each of its biannual conferences.

The primary objective of the least squares evaluation is to produce unbiased estimates of the neutron exposure parameters at the location of the measurement. The analytical method alone may be deficient because it inherently contains uncertainty due to the input assumptions to the calculation. Typically these assumptions include parameters such as the temperature of the water in the peripheral fuel assemblies, bypass region and downcomer regions, component dimensions, and peripheral core source. Industry consensus indicates that the use of the calculation alone results in overall uncertainties in the neutron exposure parameters in the range of 15-20% (1σ).

By combining the calculated results with available measurements, the uncertainties associated with the key neutron exposure parameters can be reduced. Specifically ASTM Standard E 944 states, "The algorithims of the adjustment codes tend to decrease the variances of the adjusted data compared to the corresponding input values. The least squares adjustment codes yield estimates for the output data with minimum variances, that is, the "best estimates." This is the primary reason for using these adjustment procedures." ASTM E 944 provides a comprehensive listing of available adjustment codes.

The FERRET least squares adjustment code[(1)] was initially developed at the Hanford Engineering Development Laboratory (HEDL) and has had extensive use in both the Liquid Metal Fast Breeder (LMFBR) program and the NRC Sponsored Light Water Reactor Dosimetry Improvement Program (LWR-PV-SDIP). As a result of participation in several cooperative efforts associated with the LWR-PV-SDIP, the FERRET approach was adopted by Westinghouse in the mid 1980's as the preferred approach for the evaluation of LWR surveillance dosimetry. The least squares methodology was judged superior to the previously employed spectrum averaged cross-section approach that is totally dependent on the accuracy of the shape of the calculated neutron spectrum at the measurement locations.

The FERRET code is employed to combine the results of plant specific neutron transport calculations and multiple-foil reaction-rate measurements to determine best estimate values of exposure parameters in terms of both neutron fluence greater than 1.0 MeV, (Φ(E > 1.0 MeV),

and iron atom displacements, (dpa), along with associated uncertainties in the measurement locations.

The application of the least squares methodology requires the following input:

- The calculated neutron energy spectrum and associated uncertainties at the measurement location.
- The measured reaction rate and associated uncertainty for each sensor contained in the multiple foil set.
- The energy dependent dosimetry reaction cross-sections and associated uncertainties for each sensor contained in the multiple foil sensor set.

For a given application, the calculated neutron spectrum is obtained from the results of plant-specific neutron transport calculations applicable to the irradiation period experienced by the dosimetry sensor set. This calculation is performed using the benchmarked transport calculational methodology described in paragraph 5.3.1.6.2. The sensor reaction rates are derived from the measured specific activities obtained from the counting laboratory using the specific irradiation history of the sensor set to perform the radioactive decay corrections. The dosimetry reaction cross-sections and uncertainties are obtained from the SNLRML dosimetry cross-section library(2). The SNLRML library is an evaluated dosimetry reaction cross-section compilation recommended for use in LWR evaluations by ASTM Standard E1018, "Application of ASTM Evaluated Cross-Section Data File, Matrix E 706 (IIB)." There are no additional data or data libraries built into the FERRET code system. All of the required input is supplied externally at the time of the analysis.

The uncertainties associated with the measured reaction rates, dosimetry cross-sections, and calculated neutron spectrum are input to the least squares procedure in the form of variances and covariances. The assignment of the input uncertainties also follows the guidance provided in ASTM Standard E 944.

5.3.1.6.1.4 External Neutron Monitoring System. The external neutron monitoring system also known as the external vessel neutron dosimetry system (EVNDS) provides for continuing neutron fluence measurement after sufficient specimen material exposure has been achieved and even after the last of the six internal surveillance capsules has been removed from the reactor vessel. It enables verification of fast neutron exposure distributions within the reactor vessel wall beltline region and establishes a mechanism to enable long term monitoring of this portion of the reactor vessel as required per 10 CFR 50 Appendix H. These fluence data can also support potential license renewal activities.

The EVNDS is located external to the reactor vessel, allowing for ease of dosimetry removal and replacement. It is installed in the annular air gap between the reactor vessel insulation and the primary concrete shield wall. The EVNDS is a passive system consisting of six aluminum dosimeter capsules containing radiometric monitors and four stainless steel gradient chains, which are bead chains connecting and supporting the dosimeter capsules. The bead chains are in turn supported by an arrangement of stainless steel hardware-tubular brackets on a support bar suspended by chains from bracket plate assembly, which is welded to the ventilation port liner plate under a banana cover. The bead chains are mechanically secured to the concrete floor below the reactor vessel. The system is shown on drawings 1X6AB03-00020, 1X6AB03-00021, and 1X6AB03-00022 for Unit 1 and is shown on drawings 2X6AB03-00022, 2X6AB03-00023, and 2X6AB03-00024 for Unit 2.

The EVNDS measures fluence for approximately 1/8 of the vessel wall circumference, positioned relative to well known reactor features. Neutron transport calculations then determine the fluence for the entire vessel beltline wall. The system assists in the evaluation of radiation damage to the reactor vessel beltline region by measuring the fluence to this region, which can be used to predict the shift in the reference nil ductility transition temperature (RT_{NDT}). When used in conjunction with previously removed dosimetry from the internal surveillance capsules and with the results of neutron transport calculations, the external vessel neutron measurements allow the projection of embrittlement gradients through the reactor vessel wall with minimum uncertainty. Minimizing the uncertainty in the neutron exposure projections will help to assure that the reactor can be operated in the least restrictive mode possible with respect to:

- 10 CFR 50 Appendix G pressure / temperature limit curves for normal heatup and cooldown of the reactor coolant system,
- Emergency Response Guideline (ERG) pressure / temperature limit curves, and
- Pressurized thermal shock (PTS) RT_{NDT} screening criteria.

Comprehensive sensor sets are employed at discrete locations within the reactor cavity to characterize the neutron energy spectrum variations axially and azimuthally over the beltline region of the reactor vessel. In addition, the stainless steel gradient chains are used in conjunction with the encapsulated dosimeters to complete the mapping of the neutron environment between the discrete locations chosen for spectrum determinations.

The first replacement of irradiation dosimetry is at refueling outage 1R15 (Sept. 2009). The first replacement of irradiation dosimetry is at refueling outage 2R14 (March 2010). An irradiation interval of five fuel cycles between replacements is typical.

5.3.1.6.2 Calculation of Integrated Fast (E > 1.0 MeV) Exposure at the Irradiation Samples and Reactor Vessel Wall

Discrete ordinates transport calculations are performed on a fuel cycle-specific basis to determine the neutron and gamma ray environment within the reactor geometry. The specific methods applied have been benchmarked according to the guidelines of Regulatory Guide 1.190, "Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence," U. S. Nuclear Regulatory Commission, Office of Nuclear Regulatory Research, March 2001, and have been approved by the NRC staff for general application to pressurized water reactor (PWR) analysis. A description of the transport methodology along with the SER documenting NRC staff approval of the method and computer codes are provided in Reference 13.

In the application of this methodology to the fast neutron exposure evaluations for the surveillance capsules and reactor vessel, a series of two-dimensional plant-specific transport calculations are carried out and then synthesized to generate a three-dimensional neutron flux distribution, $\Phi(r,\theta,z)$ throughout the geometry of interest using the procedures outlined in Regulatory Guide 1.190. These three-dimensional mappings of the neutron environment are completed for each operating fuel cycle and then integrated to determine the neutron fluence experienced by the surveillance test specimens and the pressure vessel wall.

In the approved analysis methodology, the transport calculations are completed using the DORT discrete ordinates code Version 3.1[3] and the BUGLE-96 cross-section library[10]. The BUGLE-96 library provides a 67 group coupled neutron-gamma ray cross-section data set

produced specifically for LWR application. In these analyses, anisotropic scattering is treated with a P_5 legendre expansion and the angular discretization is modeled with an S_{16} order of angular quadrature.

Energy- and space-dependent core power distributions, as well as system operating temperatures, are treated on a fuel cycle-specific basis. The spatial variation of the neutron source is obtained from a burnup-weighted average of the respective power distributions from individual fuel cycles including pinwise gradients for all fuel assemblies located on the periphery of the core. The energy distribution of the source is determined on a fuel assembly-specific basis and includes the effects of fissioning in both uranium and plutonium isotopes.

The results of the transport calculations are validated on a plant-specific basis by comparison with the results of surveillance capsule dosimetry developed using the procedures described in paragraph 5.3.1.6.1. These comparisons are used to demonstrate that the plant-specific application is consistent with the uncertainty evaluations provided in Reference 13 and to establish that the 20% uncertainty criterion listed in Regulatory Guide 1.190 is met. These comparisons are not used to modify or bias the results of the transport calculations.

5.3.1.6.2.1 Reference Forward Calculation. The forward transport calculation for the reactor is carried out in r,θ geometry using the DORT two-dimensional discrete ordinates code[3] and the BUGLE-96 cross-section library[10]. The BUGLE-96 library is a 47 neutron group, ENDFB-VI based, data set produced specifically for LWR applications. In these analyses, anisotropic scattering is treated with a P_3 expansion of the scattering cross-sections and the angular discretization is modeled with an S_8 order of angular quadrature. The reference forward calculation is normalized to a core midplane power density characteristic of operation at the stretch rating for the reactor.

The spatial core power distribution utilized in the reference forward calculation is derived from statistical studies of long-term operation of Westinghouse 4-loop plants. Inherent in the development of this reference core power distribution is the use of an out-in fuel management strategy; i.e., fresh fuel on the core periphery. Furthermore, a 2σ uncertainty derived from the statistical evaluation of plant-to-plant and cycle-to-cycle variations in peripheral power is used for the peripheral fuel assemblies.

Due to the use of this bounding spatial power distribution, the results from the reference forward calculation establish conservative exposure projections for reactors of this design operating at the stretch rating. Since it is unlikely that actual reactor operation would result in the implementation of a power distribution at the nominal +2σ level for a large number of fuel cycles and, further, because of the widespread implementation of low-leakage fuel management strategies, the fuel cycle-specific calculations for this reactor will result in exposure rates well below these conservative predictions.

5.3.1.6.2.2 Cycle-Specific Adjoint Calculations. All adjoint analyses are also carried out using an S_8 order of angular quadrature and the P_3 cross-section approximation from the BUGLE-96 library. Adjoint source locations are chosen at several key azimuths on the pressure vessel inner radius. In addition, adjoint calculations were carried out for sources positioned at the geometric center of all surveillance capsules. Again, these calculations are run in r,θ geometry to provide neutron source distribution importance functions for the exposure parameter of interest; in this case, $\varphi(E > 1.0\ MeV)$.

The importance functions generated from these individual adjoint analyses provide the basis for all absolute exposure projections and comparison with measurement. These importance functions, when combined with cycle-specific neutron source distributions, yield absolute predictions of neutron exposure at the locations of interest for each of the operating fuel cycles and establish the means to perform similar predictions and dosimetry evaluations for all subsequent fuel cycles.

Having the importance functions and appropriate core source distributions, the response of interest can be calculated as:

$$\varphi(R_o,\theta_o) = \int_r \int_\theta \int_E I(r,\theta,E)\, S(r,\theta,E) r\, dr\, d\theta\, dE$$

where: $\varphi(R_0,\theta_0)$ = Neutron flux (E > 1.0 MeV) at radius R_0 and azimuthal angle θ_0.

$I(r,\theta,E)$ = Adjoint importance function at radius r, azimuthal angle θ, and neutron source energy E.

$S(r,\theta,E)$ = Neutron source strength at core location r,θ and energy E.

It is important to note that the cycle-specific neutron source distributions, S(r,θ,E), utilized with the adjoint importance functions, I(r,θ,E), permit the use not only of fuel cycle-specific spatial variations of fission rates within the reactor core, but also allow for the inclusion of the effects of the differing neutron yield per fission and the variation in fission spectrum introduced by the build-in of plutonium isotopes, as the burnup of individual fuel assemblies increases.

5.3.1.7 Reactor Vessel Fasteners

The reactor vessel closure studs, nuts, and washers are designed and fabricated in accordance with the requirements of the ASME Code, Section III. The closure studs are fabricated of SA-540, Class 3, Grade B24. The closure stud material meets the fracture toughness requirements of the ASME Code, Section III, and 10 CFR 50, Appendix G. Conformance with Regulatory Guide 1.65, Materials and Inspections for Reactor Vessel Closure Studs, is discussed in section 1.9. Nondestructive examinations are performed in accordance with the ASME Code, Section III. Bolting material properties for Units 1 and 2 are given in tables 5.3.1-4 and 5.3.1-5, respectively.

Refueling procedures require that the reactor vessel studs, nuts, and washers are lifted part of the way out of their respective holes and a stud support collar be put in place prior to the lift of the integrated head assembly during preparation for refueling. In this way the studs are lifted with and stored on the head. An alternative method of the procedures is that the reactor vessel studs, nuts, and washers may be removed from the reactor closure and placed in storage racks during preparation for refueling. In this method, the storage racks are removed from the refueling cavity and stored at convenient locations on the containment operating deck prior to removal of the reactor closure head and refueling cavity flooding. In either case, the reactor closure studs are not exposed to the borated refueling cavity water. Additional protection against the possibility of incurring corrosion effects is ensured by the use of a manganese base phosphate surfacing treatment.

The stud holes in the reactor flange are sealed with special plugs before removing the reactor closure, thus preventing leakage of the borated refueling water into the stud holes.

5.3.1.8 References

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14. RSIC Computer Code Collection PSR-145, “FERRET: Least-Squares Solution to Nuclear Data and Reactor Physics Problems,” Radiation Shielding Information Center, Oak Ridge National Laboratory, January 1980.
15. Westinghouse LTR-REA-06-136, “Alvin W. Vogtle Units 1 and 2 Ex-Vessel Neutron Dosimetry System Description and Safety Evaluation Factors,” (AX6AB03-00019).
16. NRC Letter to C. K. McCoy, “Vogtle Electric Generating Plant, Units 1 and 2 - Acceptance for Referencing of Pressure Temperature Limits Report,” February 12, 1996.

5.3.2 PRESSURE-TEMPERATURE LIMITS

5.3.2.1 Limit Curves

Startup and shutdown operating limitations are based on the properties of the reactor pressure vessel beltline materials. Actual material property test data are used. The methods outlined in Appendix G Section XI of the American Society of Mechanical Engineers (ASME) Code are employed for the shell regions in the analysis of protection against nonductile failure. The initial operating curves are calculated, assuming a period of reactor operation such that the beltline material will be limiting. The heatup and cooldown curves are given in the Pressure and Temperature Limits Report as required by the Technical Specifications. Beltline material properties degrade with radiation exposure, and this degradation is measured in terms of the adjusted reference nil ductility temperature, which includes a reference nil ductility temperature shift (ΔRT_{NDT}). The reference temperature, RT_{NDT}, for materials in the reactor vessel closure flange region and the beltline regions are shown in tables 5.3.1-2 and 5.3.1-3. Tables 5.3.2-2 through 5.3.2-5 give the properties for the vessel beltline materials and data for the C_v curve (energy vs. temperature).

Predicted ΔRT_{NDT} values are derived using guidance provided in Regulatory Guide 1.99 Revision 2. For a selected time of operation, this shift is assigned a sufficient magnitude so that no unirradiated ferritic materials in other components of the reactor coolant system (RCS) will be limiting in the analysis.

The operating curves including pressure-temperature limitations shown in the Pressure and Temperature Limits Report are calculated in accordance with 10 CFR 50, Appendices G and H, and ASME Code, Section XI, Appendix G requirements, Code Case N-640 and WCAP-16142, “Reactor Vessel Closure Head/Vessel Flange Requirements Evaluation for Vogtle Units 1 and 2, Revision 1.”

The results of the material surveillance program described in paragraph 5.3.1.6 will be used to verify that the ΔRT_{NDT} predicted from the effects of the fluence, copper content curve is appropriate if ΔRT_{NDT} determined from the surveillance program is greater than the predicted ΔRT_{NDT}. Temperature limits for inservice leak and hydrotests will be calculated in accordance with ASME Code, Section XI, Appendix G Conformance with Regulatory Guide 1.99, Revision 2, is discussed in section 1.9.

5.3.2.2 Operating Procedures

The transient conditions that are considered in the design of the reactor vessel are presented in paragraph 3.9.1.1. These transients are representative of the operating conditions that should prudently be considered to occur during plant operation. The transients selected form a conservative basis for evaluation of the RCS to ensure the integrity of the RCS equipment.

Those transients listed as upset condition transients are given in table 3.9.N.1-1. None of these transients will result in pressure-temperature changes which exceed the heatup and cooldown limitations, as described in the Pressure and Temperature Limits Report.

5.3.3 REACTOR VESSEL INTEGRITY

5.3.3.1 Design

The reactor vessel is cylindrical with a welded hemispherical bottom head and a removable, bolted, flanged, and gasketed hemispherical upper head. The reactor vessel flange and head are sealed by two hollow metallic O-rings. Seal leakage is detected by means of two leakoff connections, one between the inner and outer ring and one outside the outer O-ring. The vessel contains the core, core support structures, control rods, and other parts directly associated with the core. The reactor vessel closure head contains head adapters. These head adapters are tubular members, attached by partial penetration welds to the underside of the closure head. The upper end of these adapters contains acme threads for the assembly of control rod drive mechanisms (CRDMs) or instrumentation adapters. The seal arrangement at the upper end of these adapters consists of a welded flexible canopy seal. Mechanical assemblies may be used to fix or prevent leaks in the canopy seal weld. Inlet and outlet nozzles are located symmetrically around the vessel. Outlet nozzles are arranged on the vessel to facilitate optimum layout of the reactor coolant system equipment. The inlet nozzles are tapered from the coolant loop vessel interfaces to the vessel inside wall to reduce loop pressure drop.

The bottom head of the vessel contains penetration nozzles for connection and entry of the nuclear incore instrumentation. Each nozzle consists of a tubular member made of either an Inconel or an Inconel-stainless steel composite tube. Each tube is attached to the inside of the bottom head by a partial penetration weld.

Internal surfaces of the vessel which are in contact with primary coolant are weld overlay with 0.125-in. minimum of stainless steel or Inconel.

The reactor vessel is designed and fabricated in accordance with the requirements of the American Society of Mechanical Engineers (ASME) Code, Section III. Principal design parameters of the reactor vessel are given in table 5.3.3-1. The reactor vessel is shown in figure 5.3.3-1.

There are no special design features which would prohibit the in situ annealing of the vessel. If the unlikely need for an annealing operation was required to restore the properties of the vessel material opposite the reactor core because of neutron irradiation damage, a metal temperature greater than 650 °F for a maximum period of 168 h would be applied.(4) Various modes of heating may be used, depending on the temperature required.

The reactor vessel materials surveillance program is adequate to accommodate the annealing of the reactor vessel. Sufficient specimens are available to evaluate the effects of the annealing treatment.

Cyclic loads are introduced by normal power changes, reactor trips, and startup and shutdown operations. These design base cycles are selected for fatigue evaluation and constitute a conservative design envelope for the projected plant life[a]. Vessel analysis results in a usage factor that is less than 1.

The design specifications require analysis to prove that the vessel is in compliance with the fatigue and stress limits of the ASME Code, Section III. The loadings and transients specified for the analysis are based on the most severe conditions expected during service. The heatup

[a] Metal fatigue is evaluated as a TLAA for license renewal (see subsection 19.4.2).

and cooldown rates are 100°F/h for normal operations and under abnormal or emergency conditions. This rate is reflected in the vessel design specifications.

5.3.3.2 Materials of Construction

The materials used in the fabrication of the reactor vessel are discussed in subsection 5.2.3.

5.3.3.3 Fabrication Methods

The VEGP reactor vessel manufacturer is Combustion Engineering Corporation.

The fabrication methods used in the construction of the reactor vessel are discussed in paragraph 5.3.1.2.

5.3.3.4 Inspection Requirements

The nondestructive examinations performed on the reactor vessel are described in paragraph 5.3.1.3.

5.3.3.5 Shipment and Installation

The reactor vessel is shipped in a horizontal position on a shipping sled with a vessel-lifting truss assembly. All vessel openings are sealed to prevent the entrance of moisture, and an adequate quantity of desiccant bags is placed inside the vessel. These are usually placed in a wire mesh basket attached to the vessel cover. All carbon steel surfaces, except for the vessel support surfaces and the top surface of the external seal ring, are painted with a heat-resistant paint before shipment.

The closure head is also shipped with a shipping cover and skid. An enclosure attached to the ventilation shroud support ring protects the control rod mechanism housings. All head openings are sealed to prevent the entrance of moisture, and an adequate quantity of desiccant bags is placed inside the head. These are placed in a wire mesh basket attached to the head cover. All carbon steel surfaces are painted with heat- resistant paint before shipment. A lifting frame is provided for handling the vessel head.

5.3.3.6 Operating Conditions

Operating limitations for the reactor vessel are presented in the Pressure Temperature Limit Report (PTLR).

In addition to the analysis of primary components discussed in paragraph 3.9.1.4, the reactor vessel is further qualified to ensure against unstable crack growth under faulted conditions. Actuation of the emergency core cooling system (ECCS) following a loss-of-coolant accident produces relatively high thermal stresses in regions of the reactor vessel which come into contact with ECCS water. Primary consideration is given to these areas, including the reactor vessel beltline region and the reactor vessel primary coolant nozzle, to ensure the integrity of the reactor vessel under this severe postulated transient. The Westinghouse Owners Group evaluated TMI Action Item II.K.2.13, and the item is satisfied upon submittal of RT_{NDT} values which are below the pressurized thermal shock (PTS) rule screening values. Additionally, new

calculations were performed based on the requirements of Generic Letter 92-01 and the results are given in tables 5.3.3-2 and 5.3.3-3.

For the beltline region, significant developments have recently occurred in order to address PTS events. On the basis of recent deterministic and probabilistic studies, taking U.S. pressurized water reactor operating experience into account, the Nuclear Regulatory Commission staff concluded that conservatively calculated screening criterion values of RT_{NDT} less than 270° for plate material and axial welds, and less than 300° for circumferential welds, present an acceptably low risk of vessel failure from PTS events. These values were chosen as the screening criterion in the PTS rule for 10 CFR 50.34 (new plants) and 10 CFR 50.61 (operating plants).(2) The conservative methods chosen by the NRC staff for the calculation of RT_{PTS} for the purpose of comparison with the screening criterion is presented in paragraph (b)(2) of 10 CFR 50.61. Details of the analysis method and the basis for the PTS rule can be found in SECY-82-465.(3)

The reactor vessel beltline materials are specified in subsection 5.3.1. The fluence of 4.76 x 10^{19} n/cm^2 which is the design basis fluence at the vessel inner radius, at 48 EFPY, at the peak location, was used for calculating all of the RT_{PTS} values. Based on the latest capsule data for each unit, the expected fluence at the vessel inner radius after 56.3 EFPY will be significantly less than the 4.76 x 10^{19} n/cm^2 design basis fluence[a]. RT_{PTS} is RT_{NDT}, the reference nilductility transition temperature as calculated by the method chosen by the NRC staff as presented in paragraph (b)(2) of 10 CFR 50.61, and the PTS rule. The PTS rule states that this method of calculating RT should be used in reporting values used to be compared to the above screening criterion set in the PTS rule. The screening criteria will not be exceeded using the method of calculation prescribed by the PTS rule for the vessel design lifetime. The material properties, initial RT_{NDT}, and end-of-life RT_{PTS} values are in tables 5.3.3-2 and 5.3.3-3. The materials identified in tables 5.3.3-2 and 5.3.3-3 are those materials that are exposed to high fluence levels at the beltline region of the reactor vessel and are, therefore, the subject of the PTS rule. These materials, therefore, are a subset of the materials identified in subsection 5.3.1.

The principles and procedures of linear elastic fracture mechanics (LEFM) are used to evaluate thermal effects in the regions of interest. The LEFM approach to the design against failure is basically a stress intensity consideration in which criteria are established for fracture instability in the presence of a crack. Consequently, a basic assumption employed in LEFM is that a crack or crack-like defect exists in the structure. The essence of the approach is to relate the stress field developed in the vicinity of the crack tip to the applied stress on the structure, the material properties, and the size of defect necessary to cause failure.

The elastic stress field at the crack tip in any cracked body can be described by a single parameter designated as the stress intensity factor, K. The magnitude of the stress intensity factor K is a function of the geometry of the body containing the crack, the size and location of the crack, and the magnitude and distribution of the stress.

The criterion for failure in the presence of a crack is that failure will occur whenever the stress intensity factor exceeds some critical value. For the opening mode of loading (stresses perpendicular to the major plane of the crack), the stress intensity factor is designated as K_I and the critical stress intensity factor is designated K_{IC}. Commonly called the fracture toughness, K_{IC} is an inherent material property which is a function of temperature. Any combination of applied load, structural configuration, crack geometry, and size which yields a stress intensity factor greater than K_{IC} for the material will result in crack instability.

The criterion of the applicability of LEFM is based on plasticity considerations at the postulated crack tip. Strict applicability (as defined by American Society of Testing Materials (ASTM)) of

[a] Reactor vessel embrittlement is evaluated as a TLAA for license renewal (see subsection 19.4.1).

LEFM to large structures where plane strain conditions prevail requires that the plastic zone developed at the tip of the crack does not exceed approximately 2 percent of the crack depth. In the present analysis, the plastic zone at the tip of the postulated crack can reach 20 percent of the crack depth. However, LEFM has been successfully used to provide conservative brittle fracture prevention evaluations, even in cases where strict applicability of the theory is not permitted due to excessive plasticity. Recently, experimental results from the Heavy Section Steel Technology program intermediate pressure vessel tests have shown that LEFM can be applied conservatively as long as the pressure component of the stress does not exceed the yield strength of the material. The addition of the elastically calculated thermal stresses, which results in total stresses in excess of the yield strength, does not affect the conservatism of the results, provided that these thermal stresses are included in the evaluation of the stress intensity factors. Therefore, for faulted conditions analyses, LEFM is considered applicable for the evaluation of the vessel inlet nozzle and beltline region.

In addition, it has been well established that the crack propagation of existing flaws in a structure subjected to cyclic loading can be defined in terms of fracture mechanics parameters. Thus, the principles of LEFM are also applicable to fatigue growth of a postulated flaw at the vessel inlet nozzle and beltline region.

Additional details on this method of analysis of reactor vessels under severe thermal transients are given in reference 1.

5.3.3.7 Inservice Surveillance

The internal and external surfaces of the reactor vessel are accessible for periodic inspection. Visual and/or nondestructive techniques are used. During refueling, the vessel cladding is capable of being inspected in certain areas between the closure flange and the primary coolant inlet nozzles, and, if deemed necessary, the core barrel is capable of being removed, making the entire inside vessel surface accessible.

The closure head is examined visually during each refueling. Optical devices permit a selective inspection of the cladding, CRDM nozzles, and the gasket seating surface. The knuckle transition piece, which is the area of highest stress of the closure head, is accessible on the outer surface for visual inspection, dye penetrant or magnetic particle testing, and ultrasonic testing. The closure studs and nuts can be inspected periodically using visual, magnetic particle, and ultrasonic techniques.

The closure studs, nuts, washers, and the vessel flange seal surface, as well as the full-penetration welds in the following areas of the installed reactor vessel, are available for nondestructive examination:

A. Vessel shell, from the inside and outside surfaces.

B. Primary coolant nozzles, from the inside and outside surfaces.[(a)]

C. Closure head, from the inside and outside surfaces; bottom head, from the inside and outside surfaces.

D. Field welds between the reactor vessel nozzle safe ends and the main coolant piping, from the inside and outside surfaces.

The design considerations which have been incorporated into the system design to permit the above inspection are as follows:

[(a)] Only partial outside diameter coverage is provided.

A. All reactor internals are completely removable. The tools and storage space required to permit these inspections are provided.

B. The closure head is stored dry on the reactor operating deck during refueling to facilitate direct visual inspection.

C. All reactor vessel studs, nuts, and washers can be removed to dry storage during refueling.

D. Access is provided to the reactor vessel nozzle safe ends. The insulation covering the nozzle-to-pipe welds may be removed.

The reactor vessel presents access problems because of the radiation levels and remote underwater accessibility to this component. Because of these limitations on access to the reactor vessel, several steps have been incorporated into the design and manufacturing procedures in preparation for the periodic nondestructive tests which are required by the ASME inservice inspection code. These are as follows:

A. Shop ultrasonic examinations are performed on all internally clad surfaces to an acceptance and repair standard to ensure an adequate cladding bond to allow later ultrasonic testing of the base metal from inside surface. The size of cladding bond defect allowed is 1/4 in. by 3/4 in. with the greater direction parallel to the weld in the region bounded by 2T (T = wall thickness) on both sides of each full-penetration pressure boundary weld. Unbounded areas exceeding 0.442 $in.^2$ (3/4-in. diameter) in all other regions are rejected.

B. The design of the reactor vessel shell is an uncluttered cylindrical surface to permit future positioning of the test equipment without obstruction.

C. The weld-deposited clad surface on both sides of the welds to be inspected is specifically prepared to ensure meaningful ultrasonic examinations.

D. During fabrication, all full-penetration ferritic pressure boundary welds are ultrasonically examined in addition to code examinations.

E. After the shop hydrostatic testing, all full-penetration ferritic pressure boundary welds (with the exception of the closure head welds), as well as the nozzles to safe end welds, are ultrasonically examined from both the inside and outside diameters in addition to ASME Code, Section III requirements. The closure head ferritic pressure boundary welds are examined from the outside diameter only.

The vessel design and construction enables inspection in accordance with the ASME Code, Section XI. The reactor vessel inservice inspection requirements are detailed in the VEGP inservice inspection program.

The Reactor Vessel Closure Head Stud Program is credited as a license renewal aging management program (see subsection 19.2.23).

5.3.3.8 References

1. Buchalet, C., Bamford, W. H., and Chirigos, J. N., "Method for Fracture Mechanics Analysis of Nuclear Reactor Vessels Under Severe Thermal Transients," WCAP-8510, December 1975.

2. PTS Rule, Federal Register Vol. 50, No. 141, July 23, 1985, 10 CFR 50.34.

3. NRC Policy Issue, "Pressurized Thermal Shock," SECY-82-465, November 23, 1982.

4. EPRI NP 2712, "Feasibility of and Methodology for Thermal Annealing an Embrittled Reactor Vessel," November 1982.

TABLE 5.3.1-1 (SHEET 1 OF 2)
REACTOR VESSEL QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]
Forgings				
Flanges		Yes		Yes
Studs and nuts		Yes		Yes
CRDM head adapter flange		Yes	Yes	
CRDM head adapter tube		Yes	Yes	
Instrumentation tube		Yes	Yes	
Main nozzles		Yes		Yes
Nozzle safe ends		Yes	Yes	
Plates		Yes		Yes
Weldments				
CRDM head adapter to closure head connection			Yes	
Instrumentation tube to bottom head connection			Yes	
Main nozzle	Yes	Yes		Yes
Cladding		Yes	Yes	
Nozzle to safe ends	Yes	Yes	Yes	
CRDM head adapter flange to CRDM head adapter tube	Yes		Yes	
All full-penetration ferritic pressure boundary welds accessible after hydrotest		Yes		Yes
Full-penetration nonferritic pressure boundary welds accessible after hydrotest				
a. Nozzle to safe ends		Yes	Yes	
b. CRDM head adapter flange to CRDM head adapter tube			Yes	
Seal ledge				Yes
Head lift lugs				Yes
Core pad welds			Yes	

TABLE 5.3.1-1 (SHEET 2 OF 2)

a. RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.

NOTE:

Base metal weld repairs as a result of UT, MT, RT, and/or PT indications are cleared by the same NDE technique/procedure by which the indications were found. The repairs meet all Section III requirements.

In addition, UT examination in accordance with the inprocess/posthydro UT requirements is performed on base metal repairs in the core region and base metal repairs in the inservice inspection zone (1/2 T).

TABLE 5.3.1-2[a]

VEGP UNIT 1 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

Component	Code No.	Material Spec. No.	Cu (%)	Ni (%)	P (%)	NDT (°F)	RT_{NDT} (°F)	Use NMWD[b] (ft-lb)
Closure head dome	B8807-1	A533B Cl. 1	0.16	0.67	0.008	-50	15	88
Closure head torus[c]	B8808-1	A533B Cl. 1	0.14	0.56	0.010	-30	8	85
Closure head flange[c]	B8801-1	A508 Cl. 2	-	0.70	0.011	20	20	132
Vessel flange[c]	B8802-1	A508 Cl. 2	-	0.71	0.014	0	0	119
Inlet nozzle	B8809-1	A508 Cl. 2	-	0.86	0.011	-20	-20	107
Inlet nozzle	B8809-2	A508 Cl. 2	-	0.84	0.014	-10	-10	95
Inlet nozzle	B8809-3	A508 Cl. 2	-	0.82	0.013	-10	-10	117
Inlet nozzle	B8809-4	A508 Cl. 2	-	0.87	0.014	-20	-20	105
Outlet nozzle	B8810-1	A508 Cl. 2	-	0.82	0.006	-10	-10	>124
Outlet nozzle	B8810-2	A508 Cl. 2	-	0.79	0.006	-10	-10	>100
Outlet nozzle	B8810-3	A508 Cl. 2	-	0.77	0.006	-10	-10	>102
Outlet nozzle	B8810-4	A508 Cl. 2	-	0.80	0.006	-10	-10	> 75
Nozzle shell	B8804-1	A533B Cl. 1	0.14	0.62	0.011	-10	28	94
Nozzle shell	B8804-2	A533B Cl. 1	0.10	0.58	0.006	-40	15	104
Nozzle shell	B8804-3	A533B Cl. 1	0.14	0.69	0.013	-30	40	92
Intermediate shell[c]	B8805-1	A533B Cl. 1	0.08	0.59	0.004	0	0	90
Intermediate shell[c]	B8805-2	A533B Cl. 1	0.08	0.59	0.004	-10	20	100
Intermediate shell[c]	B8805-3	A533B Cl. 1	0.06	0.60	0.003	-20	30	107
Lower shell[c]	B8606-1	A533B Cl. 1	0.05	0.59	0.005	-50	20	116
Lower shell[c]	B8606-2	A533B Cl. 1	0.05	0.58	0.009	-10	20	113
Lower shell[c]	B8606-3	A533B Cl. 1	0.06	0.64	0.007	-20	10	118
Bottom head torus	B8813-1	A533B Cl. 1	0.13	0.50	0.009	-40	-10	88
Bottom head dome	B8812-1	A533B Cl. 1	0.10	0.53	0.009	-40	-28	122
Intermediate and lower[c] shell vertical weld seams and girth	G1.43	SAW	0.03	0.10	0.007	-80	-80	129

a. This table is based on initial testing of the vessel materials at the time the surveillance capsule program was developed.

b. Normal to major working direction.

c. Denotes materials in reactor vessel closure flange and beltline region.

TABLE 5.3.1-3[a]

VEGP UNIT 2 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

Component	Code No.	Material Spec. No.	Cu (%)	Ni (%)	P (%)	NDT (°F)	RT_{NDT} (°F)	Use NMWD[b] (ft-lb)
Closure head dome	R9-1	A533B Cl. 1	0.07	0.61	0.008	-40	-30	123
Closure head torus[c]	R10-1	A533B Cl. 1	0.07	0.64	0.010	-30	-0	84
Closure head flange[c]	R7-1	A508 Cl. 2	-	0.72	0.011	10	10	130
Vessel flange[c]	R1-1	A508 Cl. 2	-	0.87	0.011	-60	-60	115
Inlet nozzle	B9806-1	A508 Cl. 2	0.07	0.84	0.010	-50	-50	119
Inlet nozzle	B9806-2	A508 Cl. 2	0.06	0.83	0.009	-40	-40	128
Inlet nozzle	R5-1	A508 Cl. 2	0.09	0.87	0.008	-20	-20	147
Inlet nozzle	R5-2	A508 Cl. 2	0.08	0.85	0.009	-20	-20	134
Outlet nozzle	R6-3	A508 Cl. 2	-	0.69	0.011	-10	-10	122
Outlet nozzle	R6-4	A508 Cl. 2	-	0.66	0.010	-10	-10	140
Outlet nozzle	B9807-3	A508 Cl. 2	-	0.66	0.005	-30	-30	116
Outlet nozzle	B9807-4	A508 Cl. 2	-	0.64	0.010	-10	10	132
Nozzle shell	R3-1	A533B Cl. 1	0.20	0.67	0.015	0	-20	79
Nozzle shell	R3-2	A533B Cl. 1	0.20	0.67	0.015	0	-40	79
Nozzle shell	R3-3	A533B Cl. 1	0.15	0.62	0.010	-10	-60	84
Intermediate shell[c]	R4-1	A533B Cl. 1	0.06	0.64	0.009	-20	10	95
Intermediate shell[c]	R4-2	A533B Cl. 1	0.05	0.62	0.009	-10	10	104
Intermediate shell[c]	R4-3	A533B Cl. 1	0.05	0.59	0.009	0	30	84
Lower shell[c]	B8825-1	A533B Cl. 1	0.05	0.59	0.006	-20	40	83
Lower shell[c]	R8-1	A533B Cl. 1	0.06	0.62	0.007	-20	40	87
Lower shell[c]	B8628-1	A533B Cl. 1	0.05	0.59	0.007	-20	50	85
Bottom head torus	R12-1	A533B Cl. 1	0.17	0.64	0.012	-20	-20	89
Bottom head dome	R11-1	A533B Cl. 1	0.10	0.62	0.008	-30	-30	115
Intermediate and lower[c] shell vertical weld seams and girth	G1.60	SAW	0.07	0.13	0.007	-10	-10	147
Intermediate and lower[c] shell vertical weld seams and girth	E3.23	SAW	0.06	0.12	0.007	-50	-30	90

a. This table is based on initial testing of the vessel materials at the time the surveillance capsule program was developed.

b. Normal to major working direction.

c. Denotes materials in reactor vessel closure flange and beltline region.

TABLE 5.3.1-4

VEGP UNIT 1 REACTOR VESSEL CLOSURE HEAD BOLTING MATERIAL PROPERTIES

Closure Head Studs

Heat No.	Material Spec. No.	Bar No.	0.2% Yield Strength (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Energy at 10°F (ft-lb)	Lateral Expansion (mils)
82029	SA540, B24	166	148.2	162.5	17.0	55.5	53, 54, 52	31, 30, 28
82029	SA540, B24	166-1	149.5	163.0	16.5	56.0	50, 51, 50	28, 32, 27
82029	SA540, B24	167	150.2	162.5	17.0	57.3	54, 54, 52	31, 33, 30
82029	SA540, B24	167-1	148.8	162.0	17.0	56.5	50, 51, 50	29, 36, 30
82029	SA540, B24	170	142.0	158.0	17.0	56.0	51, 51, 54	28, 30, 34
82029	SA540, B24	170-1	150.5	164.0	16.5	54.7	49, 49, 50	26, 26, 29
82029	SA540, B24	172	142.0	157.5	17.0	53.8	52, 50, 52	30, 29, 33
82029	SA540, B24	172-1	151.0	164.0	16.5	55.7	49, 50, 51	27, 30, 30
82029	SA540, B24	173	151.5	164.0	16.5	56.5	50, 50, 49	29, 31, 29
82029	SA540, B24	173-1	150.2	165.0	16.0	54.7	49, 50, 52	27, 30, 31
82552	SA540, B24	197	142.5	156.0	18.0	53.8	51, 51, 52	28, 29, 30
82552	SA540, B24	197-1	141.5	155.0	17.0	53.8	50, 51, 50	28, 31, 28
82552	SA540, B24	201	146.0	158.5	15.5	51.7	48, 48, 49	27, 27, 27
82552	SA540, B24	201-1	145.5	159.5	15.5	50.6	47, 49, 47	27, 30, 27
82552	SA540, B24	207	138.0	153.0	17.0	52.5	51, 52, 51	31, 32, 31
82552	SA540, B24	207-1	138.5	153.0	16.5	51.4	51, 50, 49	28, 28, 27
82552	SA540, B24	212	141.0	155.0	17.0	53.0	52, 52, 51	32, 31, 30
82552	SA540, B24	212-1	144.0	157.0	16.0	49.8	51, 50, 49	31, 29, 29

Closure Head Nut and Washers

19632	SA540, B23	75	146.0	159.5	17.0	55.7	49, 52, 49	29, 30, 27
19632	SA540, B23	75-1	142.0	155.0	17.0	54.9	47, 49, 47	30, 29, 29
19632	SA540, B23	78	150.5	162.0	16.0	51.9	48, 48, 48	28, 27, 28
19632	SA540, B23	78-1	143.0	156.0	17.5	54.5	51, 51, 52	32, 31, 34

TABLE 5.3.1-5

VEGP UNIT 2 REACTOR VESSEL CLOSURE HEAD BOLTING MATERIAL PROPERTIES

Closure Head Studs

Heat No.	Material Spec. No.	Bar No.	0.2% Yield Strength (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Energy at 10˚F (ft-lb)	Lateral Expansion (mils)
83090	SA540, B24	265	146.0	159.0	16.5	52.9	50, 50, 51	31, 31, 33
83090	SA540, B24	265-1	146.0	160.0	16.5	51.8	48, 50, 51	30, 31, 33
83090	SA540, B24	271	151.5	164.0	17.5	54.7	47, 48, 47	26, 28, 27
83090	SA540, B24	271-1	150.0	162.0	16.5	51.9	50, 52, 50	31, 33, 30
83090	SA540, B24	273	151.5	163.5	17.0	55.2	47, 46, 48	28, 26, 29
83090	SA540, B24	273-1	152.8	163.0	16.5	53.0	47, 47, 46	27, 29, 26
83090	SA540, B24	278	150.5	162.5	16.0	51.9	47, 46, 48	26, 25, 29
83090	SA540, B24	278-1	152.5	165.0	16.5	51.4	47, 46, 48	27, 29, 29
83090	SA540, B24	283	150.0	163.0	16.5	52.5	48, 48, 49	28, 28, 30
83090	SA540, B24	283-1	145.0	157.0	16.0	50.0	47, 47, 47	29, 29, 27
83090	SA540, B24	286	147.0	158.5	16.5	50.8	48, 47, 48	27, 28, 29
83090	SA540, B24	286-1	142.5	155.0	16.5	52.5	50, 49, 49	32, 30, 30
				Closure Head Nuts and Washers				
83294	SA540, B24	105	148.2	163.0	17.0	52.5	50, 51, 51	29, 31, 28
83294	SA540, B24	105-1	145.2	158.5	18.5	57.3	55, 55, 55	34, 32, 30
83294	SA540, B24	110	148.0	161.5	18.5	57.2	53, 53, 53	30, 31, 32
83294	SA540, B24	110-1	148.0	162.0	18.0	56.2	51, 53, 53	29, 31, 32
83294	SA540, B24	113	146.5	161.0	18.0	54.7	53, 54, 55	34, 33, 33
83294	SA540, B24	113-1	146.2	160.0	18.0	56.5	55, 57, 57	34, 34, 34
83294	SA540, B24	117	143.5	157.0	17.0	57.3	55, 55, 54	36, 34, 32
83294	SA540, B24	117-1	145.0	158.5	17.5	57.8	53, 53, 53	29, 30, 28

TABLE 5.3.1-6

DELETED

TABLE 5.3.1-7

[HISTORICAL]

REACTOR VESSEL MATERIAL SURVEILLANCE PROGRAM

VEGP Unit 1

	Capsules U, V, W, X, Y, and Z		
Material	*Charpy*	*Tensile*	*1/2T-CT*
Plate B8805-3 (long.)	*15*	*3*	*4*
Plate B8805-3 (trans.)	*15*	*3*	*4*
Weld metal (G-1.43)	*15*	*3*	*4*
HAZ	*15*	*-*	*-*

VEGP Unit 2

	Capsules U, V, W, X, Y, and Z		
Material	*Charpy*	*Tensile*	*1/2T-CT*
Plate B8628-1 (long.)	*15*	*3*	*4*
Plate B8628-1 (trans.)	*15*	*3*	*4*
Weld metal (E-3.23)	*15*	*3*	*4*
HAZ	*15*	*-*	*-*

TABLE 5.3.1-8

REACTOR VESSEL MATERIAL SURVEILLANCE PROGRAM WITHDRAWAL SCHEDULE (UNIT 1)

Capsule Number	Vessel Location	Lead Factor[a]	Withdrawal Time EFPY[b]	Approximate Capsule Fluence (n/cm^2 , E > 1.0 MeV) [a]
U	58.5°	4.15	1.14	3.34×10^{18} [c]
Y	241°	3.99	4.85	1.16×10^{19} [c]
V	61°	3.98	8.78	1.97×10^{19} [c] [d]
X	238.5°	4.21	14.33	3.53×10^{19} [c] [e]
W	121.5°	4.17	Standby	(f)
Z	301.5°	4.17	Standby	(f)

a. Updated in Capsule X dosimetry analysis.

b. Effective Full Power Years (EFPY) from plant startup.

c. Plant-specific evaluation.

d. This capsule was withdrawn at approximately the current end-of-license (36 EFPY) peak fluence.

e. This capsule was withdrawn at approximately (60 EFPY) peak fluence.

f. To be withdrawn at a fluence that is not less than once nor greater than twice the peak EOL fluence for an additional 20-year license renewal term to 80 years. Since the lead factor for both capsule W and Z are the same, either one may be withdrawn for 80 years license renewal.

TABLE 5.3.1-9

REACTOR VESSEL MATERIAL SURVEILLANCE PROGRAM WITHDRAWAL SCHEDULE (UNIT 2)

Capsule Number	Vessel Location	Lead Factor[a]	Withdrawal Time EFPY[b]	Approximate Capsule Fluence (n/cm^2 , E > 1.0 MeV) [a]
U	58.5°	4.10	1.20	3.56×10^{18} [c]
Y	241°	3.95	4.98	1.12×10^{19} [c]
X	238.5°	4.25	7.78	1.78×10^{19} [c]
W	121.5°	4.14	13.29	2.98×10^{19} [c] [d]
Z	301.5°	4.15	18.48	4.16×10^{19} [c]
V	61°	3.84	18.48[e]	-------

a. Updated in Capsule W dosimetry analysis.

b. Effective Full Power Years (EFPY) from plant startup.

c. Plant-specific evaluation.

d. This capsule was withdrawn at a fluence not less than once nor greater than twice the peak EOL fluence for a standard license term of 40 years (36 EFPY). In addition, this capsule was withdrawn at a fluence not less than once nor greater than twice the peak EOL fluence for an additional 20-year license renewal term to 60 years (54 EFPY).

e. Capsule V has been removed from the reactor vessel and placed in the spent fuel pool. No testing or analysis has been performed on this capsule. Reinsertion of this capsule may be considered in the future.

TABLE 5.3.2-1

DELETED

TABLE 5.3.2-2 (SHEET 1 OF 2)

VOGTLE UNIT 1 REACTOR VESSEL CORE BELTLINE REGION TOUGHNESS PROPERTIES

Intermediate Shell Course

Plate B8805-1				Plate B8805-2				Plate B8805-3			
Temp. (°F)	Energy (ft lb)	Lat. Exp (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-40	21	17	5	-40	12	9	0	-40	16	12	0
-40	16	14	0	-40	14	11	0	-40	5	6	0
-40	18	15	0	-40	7	5	0	-40	7	7	0
10	31	24	15	10	36	28	20	10	29	21	20
10	28	22	15	10	38	30	20	10	28	22	20
10	36	27	20	10	33	25	15	10	32	24	20
40	50	38	25	40	40	31	20	40	45	32	25
40	40	30	20	40	41	33	20	40	46	34	25
40	44	34	25	40	30	25	15	40	32	24	15
50	44	32	20	70	52	38	25	100	55	43	40
50	50	37	25	70	54	41	30	100	53	46	40
50	52	39	25	70	47	35	20	100	58	49	40
60	58	43	30	80	61	44	35	160	95	70	90
60	57	43	30	80	54	39	30	160	104	74	90
60	72	51	50	80	58	40	30	160	90	67	90
100	77	61	70	100	73	56	50	212	108	73	100
100	70	56	60	100	58	48	40	212	110	77	100
100	75	59	60	100	65	52	50	212	104	75	100
160	83	57	70	160	89	70	90				
160	90	70	95	160	96	73	95				
160	85	65	90	160	92	71	95				
212	80	63	100	212	104	77	100				
212	99	74	100	212	96	72	100				
212	90	68	100	212	99	75	100				

T_{NDT} = 0°F

RT_{NDT} = 0°F

T_{NDT} = -10°F

RT_{NDT} = +20°F

T_{NDT} = -20°F

RT_{NDT} = +30°F

TABLE 5.3.2-2 (SHEET 2 OF 2)

Lower Shell Course

Plate B8606-1				Plate B8606-2				Plate B8606-3			
Temp. (°F)	Energy (ft lb)	Lat. Exp (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-40	9	7	0	-40	8	7	0	-40	6	3	0
-40	8	6	0	-40	7	5	0	-40	7	4	0
-40	8	6	0	-40	7	6	0	-40	8	4	0
10	17	13	5	10	22	22	10	10	24	18	10
10	20	15	5	10	22	20	10	10	19	17	10
10	16	13	5	10	17	17	5	10	24	19	10
40	34	28	15	40	35	27	15	40	40	28	20
40	37	26	20	40	36	29	15	40	35	26	15
40	27	23	15	40	36	26	15	40	43	31	20
70	46	33	25	70	53	40	30	60	39	30	20
70	39	29	20	70	38	29	15	60	54	38	25
70	49	35	25	70	54	42	30	60	49	34	25
80	53	41	30	80	50	36	30	70	51	36	25
80	50	38	25	80	54	40	30	70	58	41	30
80	54	40	30	80	51	37	30	70	54	40	30
100	67	52	40	100	76	56	50	100	74	58	50
100	62	50	40	100	77	59	50	100	61	49	35
100	70	55	40	100	65	52	50	100	72	56	45
160	95	72	90	160	114	78	100	160	105	72	90
160	103	74	95	160	119	76	100	160	113	78	95
160	110	76	95	160	97	74	95	160	113	79	95
212	116	75	100	212	108	76	100	212	118	79	100
212	120	78	100	212	116	79	100	212	120	80	100
212	111	72	100	212	115	80	100	212	117	76	100
T_{NDT} = 50°F				T_{NDT} = 10°F				T_{NDT} = -20°F			
RT_{NDT} = 20°F				RT_{NDT} = 20°F				RT_{NDT} = 10°F			

TABLE 5.3.2-3

VEGP UNIT 1 REACTOR VESSEL CORE BELTLINE
REGION WELD METAL TOUGHNESS PROPERTIES

Intermediate and Lower Shell Long, and Girth Weld Seams
Weld Code No. G1.43

Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-80	6	2	0
-80	5	1	0
-80	6	1	0
-40	65	44	35
-40	10	3	0
-40	8	2	0
10	124	77	80
10	114	66	70
10	96	60	60
60	120	71	70
60	118	70	80
60	119	76	70
100	124	80	80
100	127	78	80
100	123	80	80
160	127	78	100
160	136	82	100
160	140	84	100

T_{NDT} = -80°F

RT_{NDT} = -80°F

TABLE 5.3.2-4 (SHEET 1 OF 2)

VEGP UNIT 2 REACTOR VESSEL CORE BELTLINE REGION TOUGHNESS PROPERTIES

Intermediate Shell Course

Plate R4-1				Plate R4-2				Plate R4-3			
Temp. (°F)	Energy (ft lb)	Lat. Exp (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)	Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-40	11	6	0	-40	6	4	0	-40	9	6	0
-40	10	5	0	-40	7	4	0	-40	7	5	0
-40	12	7	0	-40	8	5	0	-40	7	4	0
10	30	24	15	10	23	17	10	10	19	16	5
10	27	19	10	10	33	27	15	10	18	15	5
10	33	24	15	10	38	31	20	10	19	18	5
40	43	33	25	50	36	28	15	60	37	30	15
40	45	34	25	50	40	34	20	60	40	36	20
40	40	31	20	50	43	35	20	60	55	43	35
60	52	42	35	60	63	49	40	80	42	35	30
60	51	43	35	60	66	50	40	80	44	38	30
60	46	36	30	60	48	34	25	80	48	41	30
70	61	47	40	70	66	53	40	90	59	51	40
70	72	54	50	70	59	47	35	90	52	46	35
70	54	43	40	70	51	39	30	90	50	41	35
100	80	65	80	100	88	67	80	100	70	51	70
100	86	67	90	100	85	64	80	100	67	50	70
100	91	70	90	100	96	65	95	100	74	59	80
160	97	72	100	160	103	69	100	160	86	68	100
160	95	69	100	160	99	70	100	160	89	66	100
160	93	67	100	160	110	74	100	160	78	61	100
T_{NDT} = -20°F				T_{NDT} = -10°F				T_{NDT} = 0°F			
RT_{NDT} = 10°F				RT_{NDT} = 10°F				RT_{NDT} = 30°F			

TABLE 5.3.2-4 (SHEET 2 OF 2)

Lower Shell Course

Plate B8825-1			
Temp. (°F)	Energy (ft lb)	Lat. Exp (mils)	Shear (%)
-40	11	7	0
-40	11	8	0
-40	10	5	0
0	16	10	10
0	14	9	10
0	14	9	10
40	37	25	30
40	28	20	30
40	32	23	30
90	51	38	50
90	49	35	50
90	56	42	50
100	55	41	50
100	60	47	50
100	61	49	50
160	76	60	90
160	79	64	90
160	77	60	90
212	86	66	100
212	82	65	100
212	80	64	100

T_{NDT} = -20°F

RT_{NDT} = 40°F

Plate R8-1			
Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-40	14	7	5
-40	12	5	0
-40	11	4	0
0	22	18	20
0	22	16	20
0	20	16	20
40	33	22	25
40	34	23	25
40	35	25	25
90	49	39	40
90	62	46	60
90	62	46	60
100	67	50	65
100	68	51	65
100	69	52	65
160	85	64	95
160	86	65	95
160	90	66	95
212	83	61	100
212	86	64	100
212	93	68	100

T_{NDT} = -20°F

RT_{NDT} = 40°F

Plate B8628-1			
Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-40	11	4	0
-40	9	3	0
-40	11	4	0
0	15	12	10
0	15	11	10
0	15	11	10
40	26	20	30
40	25	18	30
40	22	17	30
100	55	45	40
100	43	35	30
100	48	36	35
110	61	50	70
110	61	49	70
110	58	45	70
160	76	60	95
160	70	67	95
160	68	58	95
212	76	60	100
212	70	58	100
212	70	57	100
275	76	60	100
275	82	65	100
275	96	68	100

T_{NDT} = -20°F

RT_{NDT} = 50°F

TABLE 5.3.2-5 (SHEET 1 OF 2)

VEGP UNIT 2 REACTOR VESSEL CORE BELTLINE REGION
WELD METAL TOUGHNESS PROPERTIES

Intermediate to Lower Shell Girth Weld Seam

Weld Code No. E3.23			
Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-80	6	4	0
-80	17	13	0
-80	12	8	0
-40	43	30	20
-40	39	28	15
-40	12	6	0
10	53	33	25
10	45	39	20
10	52	30	25
20	28	19	10
20	68	51	50
20	64	50	50
30	58	45	30
30	59	46	35
30	60	46	35
60	68	54	70
60	71	57	70
60	62	44	60
100	92	72	100
100	90	74	80
100	87	68	90
160	88	69	100
160	94	70	100
160	89	68	100

T_{NDT} = 50°F

RT_{NDT} = -30°F

TABLE 5.3.2-5 (SHEET 2 OF 2)

Intermediate and Lower Shell Long Weld Seams

Weld Code No. 1.60

Temp. (°F)	Energy (ft lb)	Lat. Exp. (mils)	Shear (%)
-30	15	10	0
-30	23	15	5
-30	24	17	5
-10	45	31	20
-10	57	41	30
-10	51	37	25
10	120	82	95
10	105	70	70
10	110	76	80
30	118	70	70
30	111	64	60
30	125	76	80
50	142	86	100
50	153	90	100
50	145	87	100
100	156	88	100
100	159	83	100
100	141	79	100

T_{NDT} = 10°F

RT_{NDT} = -10°F

TABLE 5.3.3-1

REACTOR VESSEL DESIGN PARAMETERS

Design/operating pressure (psig)	2485/2235
Design temperature (°F)	650
Overall height of vessel and closure head, bottom head outside diameter to top of control rod mechanism adapter (ft-in.)	43-10
Thickness of reactor pressure vessel head insulation, minimum (in.)	3
Number of reactor closure head studs	54
Diameter of reactor closure head/studs, minimum shank (in.)	6 13/16
Outside diameter of flange (in.)	205
Inside diameter of flange (in.)	167
Outside diameter at shell (in.)	190 1/2
Inside diameter at shell (in.)	173
Inlet nozzle inside diameter (in.)	27 1/2
Outlet nozzle inside diameter (in.)	29
Clad thickness, minimum (in.)	1/8
Lower head thickness, minimum (in.)	5 3/8
Vessel beltline thickness, minimum (in.)	8 5/8
Closure head thickness (in.)	7
Nominal water volume (ft^3)	3700

TABLE 5.3.3-2

UNIT 1 REACTOR VESSEL VALUES FOR ANALYSIS OF POTENTIAL PRESSURIZED THERMAL SHOCK EVENTS[a]

Material	Cu wt-%	Ni wt-%	Initial RT_{NDT} (°F)	10 CFR 50.61 Predicted RT_{PTS} (°F) 36 EFPY	10 CFR 50.61 Predicted RT_{PTS} (°F) 57 EFPY	Initial USE (ft-lb)	Regulatory Guide 1.99 Predicted USE (ft-lb) 36 EFPY	Regulatory Guide 1.99 Predicted USE (ft-lb) 57 EFPY
Intermediate Shell Plate, B8805-1	0.083	0.597	0	98	104	90	72	69
Intermediate Shell Plate, B8805-2	0.083	0.61	20	118	124	100	80	77
Intermediate Shell Plate, B8805-3[b]	0.062	0.598	30	110 121[d]	115 126[d]	107	86[e]	82[e]
Lower Shell Plate, B8606-1	0.053	0.593	20	94	97	116	93	89
Lower Shell Plate, B8606-2	0.057	0.60	20	97	101	113	90	87
Lower Shell Plate, B8606-3	0.067	0.623	10	95	100	118	94	91
Intermediate Shell Longitudinal Weld Seams 101-124 A, B, & C [c]	0.042	0.102	-80	3 -24[d]	11 -21[d]	134	107[e]	103[e]
Lower Shell Longitudinal Weld Seams 101-142 A, B, & C [c]	0.042	0.102	-80	3 -24[d]	11 -21[d]	134	107[e]	103[e]
Intermediate to Lower Shell Girth Weld 101-171 [c]	0.042	0.102	-80	3 -24[d]	11 -21[d]	134	107[e]	103[e]

NOTES:

a. RT_{PTS} values are based on the peak fluences at the vessel inner radius of 2.155 E19 (for 36 EFPY) and 3.485 E19 (for 57 EFPY). USE was predicted using the 1/4T fluence values based on the peak fluence at the vessel inner radius. The vessel wall thickness is 8.625 inches at the beltline region. Copper and nickel values for all materials are the latest Best Estimate values as of April 2005.
b. Limiting vessel material for Pressurized Thermal Shock event.
c. All of the core region welds were fabricated from wire heat 83653, linde 0091 flux, lot # 3536.
d. Determined using 10 CFR 50.61 with credible surveillance capsule data for the welds and noncredible surveillance capsule data for the plate.
e. Conservatively determined using Position 1.2 (without surveillance capsule data) of Regulatory Guide 1.99, Revision 2; however, surveillance data were available.

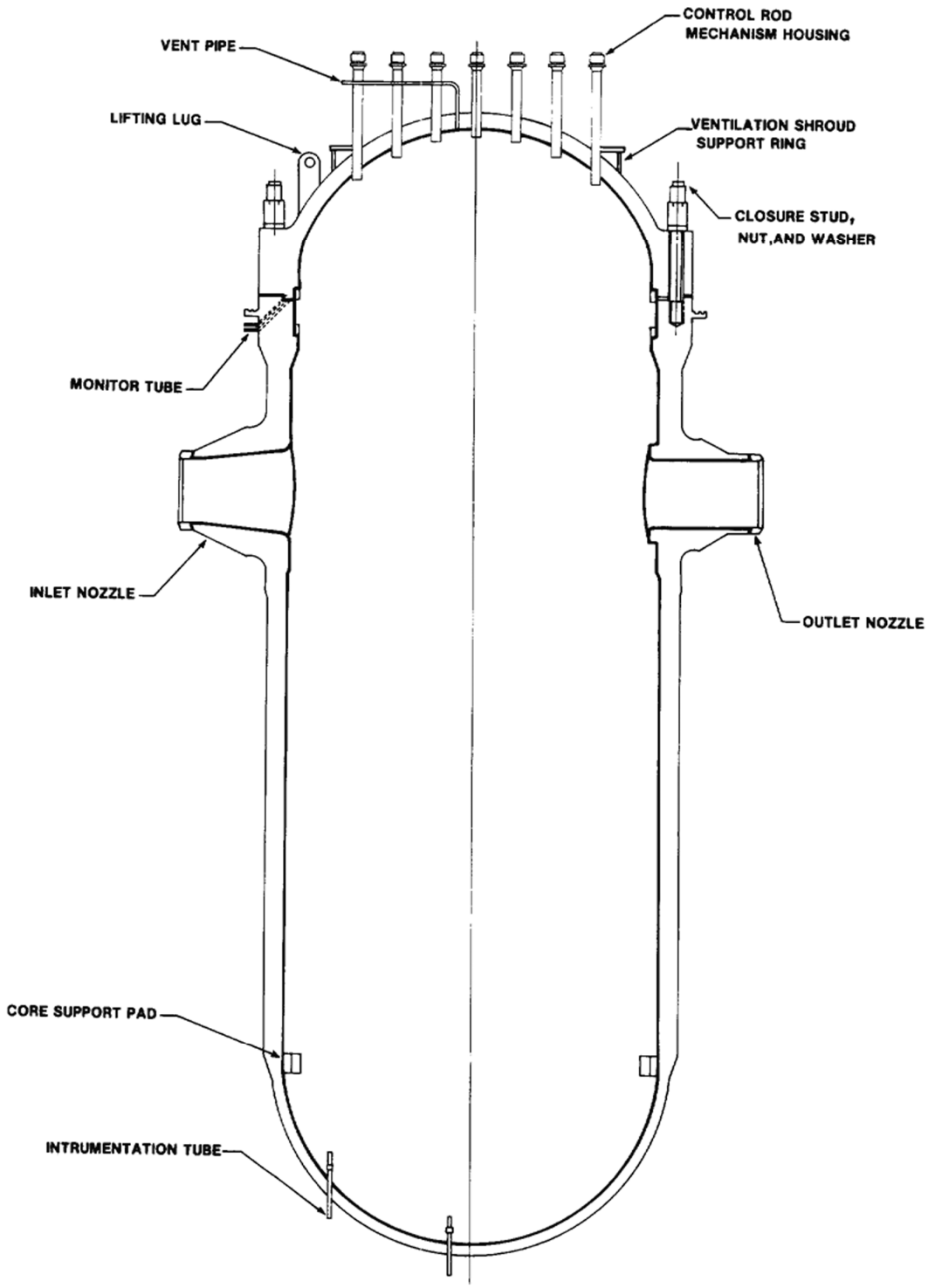
TABLE 5.3.3-3

UNIT 2 REACTOR VESSEL VALUES FOR ANALYSIS OF POTENTIAL PRESSURIZED THERMAL SHOCK EVENTS[a]

Material	Cu wt-%	Ni wt-%	Initial RT_{NDT} (°F)	10 CFR 50.61 Predicted RT_{PTS} (°F) 36 EFPY	57 EFPY	Initial USE (ft-lb)	Regulatory Guide 1.99 Predicted USE (ft-lb) 36 EFPY	57 EFPY
Intermediate Shell Plate, R4-1	0.07	0.63	10	95.9	101.0	95	76	74
Intermediate Shell Plate, R4-2	0.06	0.61	10	87.7	91.9	104	83	81
Intermediate Shell Plate, R4-3	0.05	0.60	30	100.6	104.1	84	67	66
Lower Shell Plate, B8825-1	0.06	0.62	40	117.7	121.9	83	66	65
Lower Shell Plate, R8-1[b]	0.07	0.63	40	125.9	131.0	87	70	68
Lower Shell Plate, B8628-1	0.05	0.59	50	120.6 91.1[d]	124.1 93.4[d]	85	79[e]	79[e]
Intermediate Shell Longitudinal Weld Seams 101-124 A, B, & C [c]	0.05	0.15	-10	92.2 40.8[d]	102.1 45.6[d]	152	140[e]	140[e]
Lower Shell Longitudinal Weld Seams 101-142 A, B, & C [c]	0.05	0.15	-10	92.2 40.8[d]	102.1 45.6[d]	152	140[e]	140[e]
Intermediate to Lower Shell Girth Weld[c]	0.05	0.15	-30	72.2 20.8[d]	82.1 25.6[d]	90	83[e]	83[e]

NOTES:

a. RT_{PTS} values are based on the peak fluence at the vessel inner radius of 1.93 E19 (for 36 EFPY) and 3.06 E19 (for 57 EFPY). USE was predicted using the 1/4T fluence values based on the peak fluence at the vessel inner radius. The vessel wall thickness is 8.625 inches at the beltline region. Copper and nickel values for all materials are the latest Best Estimate values as of April 2005.
b. Limiting vessel material for Pressurized Thermal Shock event.
c. The longitudinal welds were fabricated from wire heat 87005 linde 0091 flux, lot 0145. The girth weld was fabricated from weld wire heat 87005, linde 124 flux, lot 1061.
d. Determined using 10 CFR 50.61 with credible surveillance capsule data.
e. Determined using Position 2.2 (with credible surveillance capsule data) of Regulatory Guide 1.99, Revision 2.



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REACTOR VESSEL

FIGURE 5.3.3–1

5.4 COMPONENT AND SUBSYSTEM DESIGN

5.4.1 REACTOR COOLANT PUMP ASSEMBLY

5.4.1.1 Design Bases

The reactor coolant pump assembly ensures an adequate core cooling flowrate for sufficient heat transfer to maintain a departure from nucleate boiling ratio greater than the design basis limit within the parameters of operation. The required net positive suction head (NPSH) is by conservative pump design always less than that available by system design and operation.

Sufficient pump assembly rotation inertia is provided by a motor flywheel, motor rotor, and pump rotating parts which provide adequate flow during coastdown conditions. This forced flow following an assumed loss of offsite electrical power and the subsequent natural circulation effect provides the core with adequate cooling.

The reactor coolant pump is shown in figure 5.4.1-1. The reactor coolant pump design parameters are given in table 5.4.1-1.

The automatic trip of the reactor coolant pumps discussed in TMI Action Item II.K.3.5 has not been implemented. Generic studies[3][4] performed in response to Nuclear Regulatory Commission Generic Letters 83-10c and d[5] have demonstrated the acceptability of manual tripping of the reactor coolant pumps following accident events postulated to occur at VEGP.

The auxiliary component cooling water (ACCW) pump motors are accessible to the emergency diesel generators as described in section 8.3 and meet the intent of TMI Action Item II.K.3.25.

5.4.1.2 Pump Assembly Description

5.4.1.2.1 Design Description

The reactor coolant pump is a vertical, single-stage, controlled leakage, centrifugal pump designed to pump large volumes of reactor coolant at high temperatures and pressures.

The pump assembly consists of three major sections. They are the hydraulics, the seals, and the motor.

A. The hydraulic section consists of the casing, impeller, turning vane diffuser, and diffuser adapter.

B. The shaft seal section consists of the No. 1 controlled leakage film-riding face seal, No. 2 and No. 3 rubbing face seals, and a shutdown seal assembly (SDS). The seals are contained within the thermal barrier heat exchanger assembly and seal housing. The SDS is housed within the No. 1 seal area and is a passive device actuated by high temperature resulting from a loss of seal injection and ACCW cooling to the thermal barrier heat exchanger.

C. The motor section consists of a drip-proof squirrel cage induction motor with a vertical solid shaft, an oil-lubricated, double-acting Kingsbury type thrust bearing, upper and lower oil-lubricated radial guide bearings, and a flywheel.

Additional components of the pump are the shaft, pump radial bearing, thermal barrier heat exchanger assembly, coupling, spool piece, and motor stand.

In the event of a loss of seal injection and ACCW flow to the thermal barrier heat exchanger, the SDS will actuate only when the No. 1 seal temperature reaches 250ºF to 300ºF. SDS actuation limits leakage from the RCS through the RCP seal package. Leakage is limited when the SDS thermal actuator retracts due to intrusion of hot reactor coolant water into the seal area, which causes the SDS piston and polymer rings to constrict around the shaft.

5.4.1.2.2 Description of Operation

The reactor coolant enters the suction nozzle, is pumped by the impeller through the diffuser, and exits through the discharge nozzle. The diffuser adapter limits the leakage of reactor coolant back to the suction.

Seal injection flow, under slightly higher pressure than the reactor coolant, enters the pump through a connection on the thermal barrier flange and is directed into the plenum between the thermal barrier housing and the shaft. The flow splits, with the major portion flowing down the shaft through the radial bearing and into the RCS. The remaining seal injection flow passes up the shaft through the seals.

The ACCW is provided to the thermal barrier heat exchanger. During normal operation, the thermal barrier limits the heat transfer from hot reactor coolant to the radial bearing and to the seals. In addition, if a loss of seal injection flow should occur, the thermal barrier heat exchanger cools the reactor coolant to an acceptable level before it enters the bearing and seal area.

The reactor coolant pump motor oil-lubricated bearings are of conventional design. The radial bearings are the segmented pad type, and the thrust bearing is a double-acting Kingsbury type.

Auxiliary component cooling water is supplied to the external upper bearing oil cooler and to the integral lower bearing oil cooler.

The oil spillage protection system is attached to the reactor coolant pump motor and is provided to contain and channel oil to a common collection point.

The motor is a drip-proof squirrel cage induction motor with Class F thermalastic epoxy insulation, fitted with external water/air coolers. The rotor and stator are of standard construction. Six resistance temperature detectors are embedded in the stator windings to sense stator temperature. A flywheel and an antireverse rotation device are located at the top of the motor.

The internal parts of the motor are cooled by air. Integral vanes on each end of the rotor draw air in through cooling slots in the motor frame. This air passes through the motor with particular emphasis on the stator end turns. It is then routed to the external water/air coolers, which are supplied with ACCW. Each motor has two such coolers, mounted diametrically opposed to each other. Coolers are sized to maintain optimum motor-operating temperature. The air is finally exhausted to the containment environment.

Each of the reactor coolant pump assemblies is equipped for continuous monitoring of reactor coolant pump shaft and frame vibration levels. Shaft vibration is measured by two relative shaft probes mounted on top of the pump seal housing; the probes are located 90° apart in the same

horizontal plane and mounted near the pump shaft. Frame vibration is measured by two velocity seismoprobes located 90° apart in the same horizontal plane and mounted at the top of the motor support stand. Proximeters and converters linearize the probe output, which is displayed on monitor meters in the control building, with alarm functions available in the control room. The monitor meters automatically indicate the highest output from the relative probes and seismoprobes; manual selection allows monitoring of individual probes. Indicator lights display caution and danger limits of vibration.

The spool piece, which is a removable shaft segment, is located between the motor coupling flange and the pump coupling flange; the spool piece allows removal of the pump seals with the motor in place. The pump internals, motor, and motor stand can be removed from the casing without disturbing the reactor coolant piping. The flywheel is available for inspection by removing the cover.

All parts of the pump in contact with the reactor coolant are austenitic stainless steel except for seals, bearings, and special parts.

5.4.1.2.3 Loss of Seal Injection

Should a loss of seal injection to the reactor coolant pumps occur, the pump radial bearing and seals are lubricated by reactor coolant flowing up through the pump. Under these conditions, the ACCW continues to provide flow to the thermal barrier heat exchanger; this heat exchanger, functioning in its backup capacity, cools the reactor coolant before it enters the pump radial bearing and the shaft seal area. The loss of seal injection flow may result in a temperature increase in the pump bearing area, a temperature increase in the seal area, and a resultant increase in the No. 1 seal leak rate; however, pump operation can be continued provided these parameters remain within the allowable limits. Under certain low seal leak-off conditions it is possible that seal temperatures may increase above allowable limits in 1 to 2 hours requiring shutdown of the pump.

5.4.1.2.4 Loss of Auxiliary Component Cooling Water

Should a loss of ACCW to the reactor coolant pumps occur, the chemical and volume control system continues to provide seal injection flow to the reactor coolant pumps; the seal injection flow is sufficient to prevent damage to the seals with a loss of thermal barrier cooling. However, the loss of ACCW to the motor bearing oil coolers will result in an increase in oil temperature and a corresponding rise in motor bearing metal temperature. It has been demonstrated by testing at the Westinghouse Electromechanical Division that the reactor coolant pumps will incur no damage as a result of an ACCW flow interruption of 10 min.

Safety-related transmitters will be provided to redundantly monitor ACCW flow for the upper and lower reactor coolant pump bearings, as well as to monitor ACCW flow for the reactor coolant pump thermal barriers. These transmitters will provide flow indication and actuate low-flow alarms in the control room.

Operating procedures are provided for a loss of ACCW and seal injection to the reactor coolant pumps and/or motors. Included in these operating procedures is the provision to trip the reactor if ACCW flow, as indicated by the instrumentation discussed above, is lost to the reactor coolant pump motors and cannot be restored within 10 min. The reactor coolant pumps will also be tripped following the reactor trip.

5.4.1.3 Design Evaluation

5.4.1.3.1 Pump Performance

The reactor coolant pumps are sized to deliver flow at rates which equal or exceed the required flowrates. Initial RCS tests confirm the total delivery capability. Thus, assurance of adequate forced circulation coolant flow is provided prior to initial plant operation.

The estimated performance characteristic is shown in figure 5.4.1-2. The knee, at about 25% design flow, introduces no operational restrictions, since the pumps only operate at a speed which corresponds to full flow.

The reactor coolant pump motor is tested, without mechanical damage, at overspeeds up to and including 125% of normal speed. The integrity of the flywheel during a loss-of-coolant accident (LOCA) is demonstrated in reference 1, which is undergoing generic review by the NRC.

The reactor trip system ensures that pump operation is within the assumptions used for loss-of-coolant flow analyses, which also ensures that adequate core cooling is provided to permit an orderly reduction in power if flow from a reactor coolant pump is lost during operation.

An extensive test program has been conducted for several years to develop the controlled leakage shaft seal for pressurized water reactor applications. Long-term tests were conducted on less than full-scale prototype seals as well as on full-size seals. Operating plants continue to demonstrate the satisfactory performance of the controlled leakage shaft seal pump design.

The support of the stationary member of the No. 1 seal (seal ring) is such as to allow deflections, both axial and tilting, while still maintaining its controlled gap relative to the seal runner. Even if all the graphite were removed from the pump bearing, the shaft could not deflect far enough to cause opening of the controlled leakage gap. The spring rate of the hydraulic forces associated with the maintenance of the gap is high enough to ensure that the ring follows the runner under very rapid shaft deflections.

Testing of pumps with the No. 1 seal entirely bypassed (full system pressure on the No. 2 seal) shows that relatively small leakage rates would be maintained for a period of time which is sufficient to secure the pump; even if the No. 1 seal fails entirely during normal operation, the No. 2 seal would maintain these small leakage rates if the proper action is taken by the operator. The plant operator is warned of No. 1 seal damage by an increase in No. 1 seal leakoff rate. Following warning of excessive seal leakage conditions, the plant operator should close the No. 1 seal leakoff line and secure the pump, as specified in the instruction manual.

Gross leakage from the pump does not occur if the proper operator action is taken subsequent to warning of excessive seal leakage conditions.

The effect of loss of offsite power on the pump itself is to cause a temporary stoppage in the supply of injection flow to the pump seals and also of the ACCW for seal and bearing cooling. The emergency diesel generators are started automatically due to loss of offsite electrical power, so that ACCW flow and seal injection flow are automatically restored.

5.4.1.3.2 Coastdown Capability

It is important to reactor protection that the reactor coolant continues to flow for a short time after reactor trip. In order to provide this flow following loss of offsite electrical power, each reactor coolant pump is provided with a flywheel. Thus, the rotating inertia of the pump, motor, and flywheel is employed during the coastdown period to continue the reactor coolant flow. The

coastdown flow transients are provided in the figures in section 15.3. The pump/motor system is designed for the safe shutdown earthquake at the site. Hence, it is concluded that the coastdown capability of the pumps is maintained even under the most adverse case of loss of offsite electrical power coincident with the safe shutdown earthquake. Core flow transients and figures are provided in subsections 15.3.1 and 15.4.4. An inadvertent actuation of the SDS on the rotating assembly will not have any measurable impact on the RCP coastdown.

5.4.1.3.3 Bearing Integrity

The design requirements for the reactor coolant pump bearings are primarily aimed at ensuring a long life with negligible wear, so as to give accurate alignment and smooth operation over long periods of time. The surface bearing stresses are held at very low values and even under the most severe seismic transients do not begin to approach loads that cannot be adequately carried for short periods of time.

Because there are no established criteria for short-time, stress-related failures in such bearings, it is not possible to make a meaningful quantification of such parameters as margins to failure, safety factors, etc. A qualitative analysis of the bearing design, embodying such considerations, gives assurance of the adequacy of the bearing to operate without failure.

Low oil levels in the motor lube oil sumps signal alarms in the control room. Each motor bearing contains embedded temperature detectors, so that initiation of failure, separate from loss of oil, is indicated and alarmed in the control room as a high bearing temperature. This requires pump shutdown. If these indications are ignored and the bearing proceeds to failure, the low melting point of Babbitt metal on the pad surfaces ensures that sudden seizure of the shaft will not occur. In this event, the motor continues to operate, since it has sufficient reserve capacity to drive the pump under such conditions. However, the high torque required to drive the pump will require high current, which leads to the motor being shutdown by the electrical protection systems.

5.4.1.3.4 Locked Rotor

It may be hypothesized that the pump impeller might severely rub on a stationary member and then seize. Analysis has shown that under such conditions, assuming instantaneous seizure of the impeller, the pump shaft fails in torsion just below the coupling to the motor, disengaging the flywheel and motor from the shaft. This constitutes a loss of coolant flow in the loop. Following such a postulated seizure, the motor continues to run without any overspeed, and the flywheel maintains its integrity, since it is still supported on a shaft with two bearings. Flow transients are provided in the figures in subsection 15.3.3 for the assumed locked rotor.

There are no other credible sources of shaft seizure other than impeller rubs. A sudden seizure of the pump bearing is precluded by graphite in the bearing. Any seizure in the seals results in a shearing of the antirotation pin in the seal ring. The motor has adequate power to continue pump operation even after the above occurrences. Indications of pump malfunction in these conditions are initially given by high-temperature signals from the bearing water temperature detector, by excessive No. 1 seal leakoff indications, and by offscale No. 1 seal leakoff indications. Following these signals, pump vibration levels are checked. Excessive vibration indicates mechanical trouble, and the pump is shut down for investigation. Note an inadvertent actuation of the SDS on a rotating assembly will not prevent sufficient cooling flow to the reactor core.

5.4.1.3.5 Critical Speed

The reactor coolant pump shaft is designed so that its operating speed is below its first critical speed. This shaft design, even under the most severe postulated transient, gives low values of actual stress.

5.4.1.3.6 Missile Generation

Precautionary measures taken to preclude missile formation from reactor coolant pump components ensure that the pumps will not produce missiles under any anticipated accident condition.

Appropriate components of the reactor coolant pump have been analyzed for missile generation. Any fragments of the motor rotor would be contained by the heavy stator frame. The same conclusion applies to the pump impeller because the small fragments that might be ejected would be contained by the heavy casing. Further discussion and analysis of missile generation are contained in reference 1.

5.4.1.3.7 Pump Cavitation

The minimum NPSH required by the reactor coolant pump at best estimate flow is approximately a 300-ft head (approximately 133 psi). In order for the controlled leakage seal to operate correctly, it is necessary to require a minimum differential pressure of approximately 200 psi across the No. 1 seal. This corresponds to a primary loop pressure at which the minimum NPSH is exceeded, and no limitation on pump operation occurs from this source.

5.4.1.3.8 Pump Overspeed Considerations

For turbine trips actuated by either the reactor trip system or the turbine protection system, the generator and reactor coolant pumps remain connected to the external network for 30 s to prevent any pump overspeed condition.

An electrical fault requiring immediate trip of the generator (with resulting turbine trip) could result in an overspeed condition. However, the turbine control system and the turbine intercept valves limit the overspeed to less than 120%. As additional backup, the turbine protection system on Unit 1 utilizes redundant overspeed protection with trip manifold assemblies using primary and emergency overspeed trips set at 110% and 110.5%, while Unit 2 has a mechanical overspeed protection trip, usually set at about 110% (of turbine speed). In case a generator trip deenergizes the pump buses, the reactor coolant pump motors are transferred from the unit auxiliary transformers to the reserve auxiliary transformers within 6 to 10 cycles. Further discussion of pump overspeed considerations is contained in reference 1.

5.4.1.3.9 Antireverse Rotation Device

Each of the reactor coolant pumps is provided with an antireverse rotation device in the motor. This antireverse mechanism consists of pawls mounted on the outside diameter of the flywheel, a serrated ratchet plate mounted on the motor frame, a spring return for the ratchet plate, and two shock absorbers.

At an approximate forward speed of 70 rpm, the pawls drop and bounce across the ratchet plate; as the motor continues to slow, the pawls drag across the ratchet plate. After the motor has slowed and come to a stop, the dropped pawls engage the ratchet plate, and as the motor tends to rotate in the opposite direction, the ratchet plate also rotates until it is stopped by the shock absorbers. The rotor remains in this position until the motor is energized again. When the motor is started, the ratchet plate is returned to its original position by the spring return.

As the motor begins to rotate, the pawls drag over the ratchet plate. When the motor reaches sufficient speed, the pawls are bounced into an elevated position and are held in that position by friction resulting from centrifugal forces acting upon the pawls. While the motor is running at speed, there is no contact between the pawls and ratchet plate.

Considerable plant experience with the design of the antireverse rotation device has shown high reliability of operation.

5.4.1.3.10 Shaft Seal Leakage

During normal operation, leakage along the reactor coolant pump shaft is controlled by three shaft seals arranged in series such that reactor coolant leakage to the containment is essentially zero. Seal injection flow is directed to each reactor coolant pump via a seal water injection filter. It enters each pump through a connection on the thermal barrier flange. Here the flow splits; the major portion flows down the shaft to cool the bearing and enters the RCS, and the remainder flows up the shaft through the seals. This seal flow provides a backpressure on the No. 1 seal and a controlled flow through the seal. After passing through the No. 1 seal, most of the flow leaves the pump via the No. 1 seal leakoff line. Minor flow passes through the No. 2 seal to its leakoff line. A backflush injection from a head tank flows into the No. 3 seal between its double dam seal area. At this point the flow divides, with half flushing through one side of the seal and out the No. 2 seal leakoff, while the remaining half flushes through the other side and out the No. 3 seal leakoff. This arrangement ensures essentially zero leakage of reactor coolant or trapped gases to the containment.

In the event of a loss of seal injection and ACCW flow to the thermal barrier heat exchanger, reactor coolant begins to travel along the RCP shaft and displace the cooler seal injection water. Once the temperature of the No. 1 seal reaches 250ºF to 300ºF, the SDS actuates by retraction of a thermal actuator, causing the SDS piston and polymer rings to constrict around the shaft. SDS actuation controls shaft seal leakage and limits the loss of reactor coolant through the RCP seal package.

5.4.1.3.11 Seal Discharge Piping

The No. 1 seal reduces the leakoff pressure to that of the volume control tank. Water from each pump No. 1 seal is piped to a common manifold, through the seal water return filter, and through the seal water heat exchanger where the temperature is reduced to that of the volume control tank. The No. 2 and 3 leakoff lines route No. 2 and 3 seal leakage to the reactor coolant drain tank and the containment sump, respectively.

5.4.1.4 Tests and Inspections

The reactor coolant pumps can be inspected in accordance with the ASME Code, Section XI, for inservice inspection of nuclear RCSs.

The pump casing is cast in one piece. Support feet are cast integrally with the casing to eliminate a weld region.

The design enables disassembly and removal of the pump internals for visual access to the pump casing.

The reactor coolant pump quality assurance program is given in table 5.4.1-2.

Tests and inspections performed under the following license renewal aging management programs are credited as applicable to the reactor coolant pumps and their subcomponents:

- Bolting Integrity Program (see subsection 19.2.2).
- Boric Acid Corrosion Control Program (see subsection 19.2.3).
- Inservice Inspection Program (see subsection 19.2.13).
- Water Chemistry Control Program (see subsection 19.2.28).

5.4.1.5 Pump Flywheel

The integrity of the reactor coolant pump flywheel is ensured on the basis of the following design and quality assurance procedures[a].

5.4.1.5.1 Design Basis

The calculated stresses at operating speed are based on stresses due to centrifugal forces. The stress resulting from the interference fit of the flywheel on the shaft is less than 2000 psi at zero speed, but this stress becomes zero at approximately 600 rpm because of radial expansion of the hub. The reactor coolant pumps run at approximately 1190 rpm and may operate briefly at overspeeds up to 109% (1295 rpm) during loss of offsite electrical power. For conservatism, however, 125% of operating speed was selected as the design speed for the reactor coolant pumps. The flywheels are given a manufacturer's test of 125% of the maximum synchronous speed of the motor.

5.4.1.5.2 Fabrication and Inspection

The flywheel consists of two thick plates bolted together. The flywheel material is produced by a process that minimizes flaws in the material and improves its fracture toughness properties, i.e., an electric furnace with vacuum degassing. Each plate is fabricated from SA-533, Grade B, Class 1 steel. Supplier certification reports are available for all plates and demonstrate the acceptability of the flywheel material on the basis of the requirements of NRC Regulatory Guide 1.14.

Flywheel blanks are flame cut from the SA-533, Grade B, Class 1 plates with at least 1/2 in. of stock left on the outer surface and bore surface for machining to final dimensions. The finished machined bores, keyways, and drilled holes are subjected to magnetic particle or liquid penetrant examinations in accordance with the requirements of Section III of the ASME Code.

[a] Reactor coolant pump flywheel fatigue is evaluated as a time-limited aging analysis (TLAA) for license renewal (see paragraph 19.4.2.3).

The finished flywheels, as well as the flywheel material (rolled plate), are subjected to 100% volumetric ultrasonic inspection using procedures and acceptance standards specified in Section III of the ASME Code.

The reactor coolant pump motors are designed such that, by removing the cover to provide access, the flywheel is available to allow an inservice inspection program which was originally in accordance with the recommendations of Regulatory Guide 1.14, referencing Section XI of the ASME Code for inspection scheduling purposes. Subsequently, the NRC authorized the use of an alternative flywheel inspection as addressed in Westinghouse WCAP-14535A, "Topical Report on Reactor Coolant Pump Flywheel Inspection Elimination." In this alternative inspection program, each flywheel is inspected at least once every 10 years by conducting either: (1) an in-place ultrasonic examination over the volume from the inner bore of the flywheel to the circle of one-half the outer radius, or (2) a surface examination (magnetic particle and/or liquid penetrant) of exposed surfaces of the disassembled flywheel.

5.4.1.5.3 Material Acceptance Criteria

The reactor coolant pump motor flywheel conforms to the following material acceptance criteria:

A. The nil ductility transition temperature (NDTT) of the flywheel material is obtained by two drop weight tests which exhibit no-break performance at 20°F in accordance with ASTM E-208. The above drop weight tests demonstrate that the NDTT of the flywheel material is no higher than 10°F.

B. A minimum of three Charpy V-notch (C_V) impact specimens from each plate shall be tested at ambient (70°F) temperature in accordance with the ASME SA-370 specification. The Charpy V-notch (C_V) energy in both the parallel and normal orientation with respect to the final rolling direction of the flywheel plate material is at least 50 ft-lb and 35-mil lateral expansion at 70°F, and, therefore, the flywheel material has a reference nil ductility temperature (RT_{NDT}) of 10°F. An evaluation of flywheel overspeed has been performed which concludes that flywheel integrity will be maintained.[1]

Thus, it is concluded that flywheel plate materials are suitable for use and can meet Regulatory Guide 1.14 acceptance criteria on the bases of suppliers' certification data. The degree of compliance with Regulatory Guide 1.14 is further discussed in section 1.9.

5.4.1.6 References

1. "Reactor Coolant Pump Integrity in LOCA," WCAP-8163, September 1973.

2. "Safety-Related Research and Development for Westinghouse Pressurized Water Reactor, Program Summaries; Winter 1977 - Summer 1978," WCAP-8768, Revision 2, October 1978.

3. Westinghouse Owners Group Letter OG-110 to Nuclear Regulatory Commission, December 11, 1983.

4. Westinghouse Owners Group Letter OG-117 to Nuclear Regulatory Commission, March 9, 1983.

5. "Resolution of TMI Action Item II.K.3.5, Automatic Trip of Reactor Coolant Pumps," Nuclear Regulatory Commission Generic Letters 83-10c and d, February 8, 1983.

6. "Use of Westinghouse Shield Passive Shutdown Seal for Flex Strategies," TR-FSE-14-1-P, Revision 1, March 2014.

7. "PRA Model for the Generation III Westinghouse Shutdown Seal, "PWROG-14001-P, Revision 0, June 2014.

5.4.2 STEAM GENERATORS

5.4.2.1 Design Bases

Steam generator design data are given in table 5.4.2-1. Code classifications of the steam generator components are given in section 3.2. Although the American Society of Mechanical Engineers (ASME) classification for the secondary side is specified to be Class 2, the current philosophy is to design all pressure-retaining parts of the steam generator, and thus both the primary and secondary pressure boundaries, to satisfy the criteria specified in Section III of the ASME Code for Class 1 components. The design stress limits, transient conditions, and combined loading conditions applicable to the steam generator are discussed in subsection 3.9.1. Estimates of radioactivity levels anticipated in the secondary side of the steam generators during normal operation and the bases for the estimates are given in chapter 11. The accident analysis of a steam generator tube rupture is discussed in chapter 15.

A design objective of the internal moisture separation equipment is that moisture carryover should not exceed 0.25% by weight under the following conditions:

A. Steady-state operation up to 100% of full-load steamflow with water at the normal operating level.

B. Loading or unloading at a rate of 5% of full-power steamflow per minute in the range from 15 to 100% of full-load steamflow.

C. A step load change of 10% of full power in the range from 15- to 100% full-load steamflow.

The water chemistry on the reactor side, selected to provide the necessary boron content for reactivity control, should minimize corrosion of reactor coolant system (RCS) surfaces. The effectiveness of the water chemistry of the steam side in affecting corrosion control is discussed in chapter 10. Compatibility of steam generator tubing with both primary and secondary coolants is discussed further in paragraph 5.4.2.4.3.

The steam generator is designed to minimize unacceptable damage from mechanical or flow-induced vibration. Tube support adequacy is discussed in paragraph 5.4.2.3.3. The tubes and tube sheet are analyzed and confirmed to withstand the maximum accident loading conditions as they are defined in subsection 3.9.1. Further consideration is given in paragraph 5.4.2.3.4 to the effect of tube-wall thinning on accident condition stresses.

5.4.2.2 Design Description

The steam generator is a Model F, vertical-shell, and U-tube evaporator, with integral moisture separating equipment. Figure 5.4.2-1 shows the model, indicating several of its design features which are described in the following paragraphs.

On the primary side, the reactor coolant flows through the inverted U-tubes, entering and leaving through nozzles located in the hemispherical bottom head of the steam generator. The

head is divided into inlet and outlet chambers by a vertical divider plate extending from the apex of the head to the tube sheet.

Steam is generated on the shell side, flows upward, and exits through the outlet nozzle at the top of the vessel. Feedwater enters the steam generator at an elevation above the top of the U-tubes through a feedwater nozzle or the auxiliary feed nozzle. Stratification and striping in the main nozzle region is reduced by utilization of an auxiliary feedwater nozzle during startup, hot standby, and power escalation. During startup, the auxiliary feedwater system supplies the steam generator via the auxiliary nozzle. This mode continues up to about 4% power. At this level the switch is made to the main feedwater pumps which continue to feed into the auxiliary nozzle. Between 12- and 20% power, the main feed pumps feed into the main feed nozzle. The flowrate and temperature of the feedwater above 12% power should be high enough to substantially reduce stratification and striping loads on the main feed nozzle. During normal operation, the bypass line isolation valve is normally open, therefore, some flow will always be directed to the auxiliary nozzle. During hot standby the steam generator is supplied by the auxiliary feed system via the auxiliary nozzle. The water entering through the main feed nozzle is distributed circumferentially around the steam generator by means of a feedwater ring and then flows through an annulus between the tube wrapper and shell. The feedwater enters the ring via a welded thermal sleeve connection and leaves it through inverted J-tubes located at the flow holes which are at the top of the ring. The J-tubes are arranged to distribute the bulk of the colder feedwater to the hot leg side of the tube bundle. The feed ring is designed to minimize conditions which can result in water hammer occurrences in the feedwater piping. Water that enters through the auxiliary feed nozzle is not distributed through a feed ring. At the bottom of the wrapper, the water is directed toward the center of the tube bundle by a flow distribution baffle. This baffle arrangement serves to minimize the tendency in the relatively low-velocity fluid for sludge deposition. Flow-blocking devices discourage the water from flowing up the bypass lane as it enters the tube bundle where it is converted to a steam-water mixture. Subsequently, the steam-water mixture from the tube bundle rises into the steam drum section, where 16 individual centrifugal moisture separators remove most of the entrained water from the steam. The steam continues to the secondary separators for further moisture removal, increasing its quality to a designed minimum of 99.75%. The moisture separators divert the separated water, which is combined with entering feedwater to flow back down the annulus between the wrapper and shell for recirculation through the steam generator. The dry steam exits from the steam generator through the outlet nozzle, which is provided with a steamflow restrictor. (Refer to subsection 5.4.4.) The moisture carryover is expected to be well below the internal moisture separator design objective of 0.25 weight% for the cases with a primary T_{avg} at the high end of the proposed range of 570.7°F to 588.4°F.

With the T_{avg} at the lower end of the range, moisture carryover may exceed 0.25%. However, even for the low T_{avg} condition, moisture carryover will remain below 0.5%, which is the threshold beyond which erosion and corrosion of the piping and valves downstream of the steam generators are of a concern.

5.4.2.3 Design Evaluation

5.4.2.3.1 Forced Convection

The effective heat transfer coefficient is determined by the physical characteristics of the Model F steam generator and the fluid conditions in the primary and secondary systems for the nominal 100% design case. It includes a conservative allowance for fouling and uncertainty. A

designed heat transfer area is provided to permit the achievability of the full-design heat-removal rate.

5.4.2.3.2 Natural Circulation Flow

In the event of loss of offsite power and consequential loss of forced circulation within the RCS, natural circulation functions to remove core decay heat and permit the plant to be stabilized in the hot standby operational mode. Under this condition, pressurizer pressure is maintained by one pressurizer backup heater group, which is powered from one of the emergency electrical buses. To ensure that one backup heater group is available, assuming a single failure, the VEGP is designed with the capability for manual loading of separate backup heater groups (i.e., group A and group B, respectively) on independent emergency, electrical buses (i.e., train A and train B, respectively) within 1 h, following loss of offsite power. One heater group (485 kW Unit 1 group A, 415 kW Unit 2 group A, 485 kW Unit 1 and Unit 2 group B) loaded within 1 h is sufficient to satisfy the minimum heat capacity requirement (150 kW) for natural circulation following loss of offsite power. This minimum heat capacity requirement conservatively covers the pressurizer heat losses at or below normal operating pressure, following loss of offsite power, and permits pressurizer pressure to be stabilized and maintained at any desired value.

The pressurizer heater design is such that following loss of offsite power and assuming a single failure, sufficient heater capacity is available to stabilize pressurizer pressure and preclude boiling in the RCS. If pressurizer heaters are not available to maintain pressurizer pressure, the RCS could be cooled via secondary side steam release. This operation would prevent saturation pressure from being reached in the active portion of the RCS. Depending on the circumstances, saturation conditions could occur in the upper head of the reactor vessel which would lead to the formation of a steam bubble. This would not impede natural circulation flow, however, since any vapor that entered the hot legs and subsequently the steam generators would be condensed by heat transfer to the secondary side of the steam generators. Vapor would be condensed in this manner as long as the steam generator tube bundle remains submerged.

5.4.2.3.3 Mechanical and Flow-Induced Vibration Under Normal Operating Conditions

In the design of the steam generators, the possibility of degradation of tubes due to either mechanical or flow-induced excitation is thoroughly evaluated. This evaluation includes detailed analysis of the tube support systems as well as an extensive research program with tube vibration model tests.

In evaluating degradation due to vibration, consideration is given to sources of excitation such as those generated by primary fluid flowing within the tubes, mechanically-induced vibration, and secondary fluid flow on the outside of the tubes. During normal operation, the effects of primary fluid flow within the tubes and mechanically-induced vibration are considered to be negligible and should cause little concern.

Thus, the primary source of tube vibrations is the hydrodynamic excitation by the secondary fluid on the outside of the tubes. In general, three vibration mechanisms have been identified:

- Vortex shedding.
- Fluidelastic excitation.

- Turbulence.

Vortex shedding does not provide detectable tube bundle vibration. There are several reasons why this happens.

A. Flow turbulence in the downcomer and tube bundle inletregion inhibit the formation of Von Karman's vortex train.

B. The spatial variations of crossflow velocities along the tube preclude vortex shedding at a single frequency.

C. Both axial flow and crossflow velocity components exist on the tubes. The axial flow component disrupts the Von Karman vortices.

Fluid elastic excitation was observed during the testing. The amplitudes of the vibrations were two orders of magnitude smaller than those of the turbulent flow-induced vibrations. Therefore, fluid elastic excitation is excluded from consideration as a factor in steam generator tube bundle vibrations.

Flow-induced vibrations due to flow turbulence cause stresses in the tubes that are two orders of magnitude below the endurance limit (30,000 psi) of the tube material. Therefore, the contribution to fatigue is negligible; and fatigue degradation from flow-induced vibration is not anticipated during normal operation.

Summarizing the results of analyses and tests of the steam generator for vibration, it can be stated that a check of all modes of tube vibration mechanisms has been completed. The conclusions that can be drawn are that the primary source of tube vibration is fluid turbulence and that the magnitude of the vibration is so small that, when combined with its total random nature, its contribution to tube fatigue is negligible. Therefore, fatigue degradation due to flow-induced vibration is not anticipated.

The impact of operation at a power level of 3653 MWt has been evaluated and it is concluded that significant levels of vibration will not occur from the fluid-elastic, vortex shedding, or turbulent mechanisms. The projected level of tube wear as a result of vibration will remain small and will not result in unacceptable tube wear.

5.4.2.3.4 Allowable Tube-Wall Thinning Under All Plant Conditions[a]

An analysis has been performed to define the structural limits for an assumed uniform thinning mode of degradation in both the axial and circumferential directions at a power level of 3653 MWt. The assumption of uniform thinning is generally regarded to result in a conservative structural limit for all flaw types occurring in the field. The allowable tube repair limit, in accordance with Regulatory Guide 1.121, is obtained by incorporating into the structural limit a growth allowance for continued operation until the next scheduled inspection, as well as an allowance for eddy current measurement uncertainty.

Calculations have been performed to establish the tube straight leg (free span) region of the tube for degradation over an unlimited axial length, and for degradation over limited axial extent at the tube support plate (TSP), the flow distribution baffle (FDB), and antivibration bar intersections for the 3653 MWt power level conditions.

The minimum structural limit is calculated to be 60.0% allowable tube wall loss for the high / low T_{avg} straight length location. The straight leg structural limit is also applicable to the tube / AVB

[a] Loss of material from the steam generator tube walls was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in paragraph 19.4.6.3.

tangent points. These tube / AVB tangent points correspond to row 7 for the inner sets of AVBs, row 20 for the middle sets of AVBs, and row 31 for the outer sets of AVBs. The enveloping structural limits at the FDB, TSP, and anti-vibration bars at high / low T_{avg} (other than the tube / AVB tangent points) are 66.5%, 62.0%, and 72.0%, respectively.

Structural integrity performance criteria (SIPC) requirements were evaluated to address circumferential cracks at the TSP and U-bend regions.

Results are provided in% degraded area (reference 3).

5.4.2.4 Steam Generator Materials

5.4.2.4.1 Selection and Fabrication of Materials

All pressure boundary materials used in the steam generator are selected and fabricated in accordance with the requirements of Section III of the ASME Code. A general discussion of materials specifications is given in subsection 5.2.3, with types of materials listed in tables 5.2.3-1 and 5.2.3-2. Fabrication of reactor coolant pressure boundary (RCPB) materials is also discussed in subsection 5.2.3, particularly in paragraphs 5.2.3.3 and 5.2.3.4

Testing has justified the selection of corrosion-resistant Inconel-600, a nickel-chromium-iron alloy (ASME SB-163), for the steam generator tubes. The channel head divider plate is Inconel (ASME SB-168). The interior surfaces of the reactor coolant channel head, nozzles, and manways are clad with austenitic stainless steel. The primary side of the tube sheet is weld clad with Inconel (ASME SFA-5.14). The tubes are then seal welded to the tube-sheet cladding. These fusion welds, performed in compliance with Sections III and IX of the ASME Code, are dye-penetrant inspected and leakproof tested before each tube is hydraulically expanded the full depth of the tube-sheet bore.

Code cases used in material selection are discussed in subsection 5.2.1. The extent of conformance with Regulatory Guides 1.84, Design and Fabrication Code Case Acceptability ASME Section III, Division 1, and 1.85, Materials Code Case Acceptability ASME Section III, Division 1, is discussed in section 1.9.

During manufacture, cleaning is performed on the primary and secondary sides for the steam generator in accordance with written procedures which follow the guidance of Regulatory Guide 1.37, Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants, and American National Standards Institute (ANSI) Standard N45.2.1-1973, Cleaning of Fluid Systems and Associated Components for Nuclear Power Plants. Onsite cleaning and cleanliness control also follow the guidance of Regulatory Guide 1.37 as discussed in section 1.9. Cleaning process specifications are discussed in paragraph 5.2.3.4.

The fracture toughness of the materials is discussed in paragraph 5.2.3.3. Adequate fracture toughness of ferritic materials in the RCPB is provided by compliance with 10 CFR 50, Appendix G, Fracture Toughness Requirements, and Paragraph NB-2300 of Section III of the ASME Code, and by meeting the requirements of General Design Criteria 1, 14, 15, and 31.

5.4.2.4.2 Steam Generator Design Effects on Materials

Several features have been introduced into the Model F steam generator to minimize the deposition of contaminants from the secondary-side flow. Such deposits could otherwise

produce a local environment in which adverse conditions could develop and result in material corrosion. The support plates are made of corrosion resistant stainless steel 405 alloy and incorporate a four-lobe-shaped tube hole design that provides greater flow area adjacent to the tube outer surface and eliminates the need for interstitial flow holes. The resulting increase in flow provides higher sweeping velocities at the tube/tube support plate intersections. These increased sweeping velocities reduce the potential for sludge deposition on the tube support plates. Historically, sludge removal from the tube support plates has not been performed. Figure 5.4.2-2 is an illustration of the quatrefoil broached holes. This modification in the support plate design is a major factor contributing to the increased circulation ratio. The increased circulation results in increased flow in the interior of the bundle, as well as increased horizontal velocity across the tube sheet, which reduces the tendency for sludge deposition. The effect of the increased circulation on the vibrational stability of the tube bundle has been analyzed with consideration given to flow-induced excitation frequencies. The unsupported span length of tubing in the U-bend region and the corresponding optimum number of antivibration bars has been determined. The antivibration bars are fabricated from square Inconel barstock which is then chromium treated to improve frictional characteristics. Because of the increased circulation ratio, the moisture separating equipment has been modified to maintain an adequate margin with respect to the moisture carryover. To provide added strength as well as resistance to vibration, the quatrefoil tube support plate thickness has been increased. In addition, either 8 or 12 peripheral supports provide stability to the plates so that tube fretting or wear due to flow-induced plate vibrations at the tube support contact regions is minimized. For the top tube support plate, there is a continuous ring to distribute the lateral loads.

Assurance against significant flow-induced tube vibration is provided by a combination of analysis and testing.

Combining both vortex shedding and turbulence effects in a conservative manner, the maximum predicted local tube wear depth over a 40-year operating design objective[a] remains less than 0.006 inch with the operation at a power level of 3653 MWt. This value is considerably below the plugging limit for Model F steam generators.

5.4.2.4.3 Compatibility of Steam Generator Tubing With Primary and Secondary Coolants[b]

As mentioned in paragraph 5.4.2.4.1, corrosion tests which subjected the steam generator tubing material, Inconel-600 (ASME SB-163), to simulated steam generator water chemistry have indicated that the loss due to general corrosion over the 40-year operating design objective[a] is insignificant compared to the tube-wall thickness. Testing to investigate the susceptibility of heat exchanger construction materials to stress corrosion in caustic and chloride aqueous solutions has indicated that Inconel-600 has resisted general corrosion in severe operating water conditions. Many reactor years of successful operation have shown the same low general corrosion rates as indicated by the laboratory tests.

Recent operating experience, however, has revealed areas on secondary surfaces where localized corrosion rates were significantly greater than the low general corrosion rates. Both intergranular stress corrosion and tube-wall thinning were experienced in localized areas,

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

[b] The Water Chemistry Control Program is credited as a license renewal aging management program (see subsection 19.2.28).

although not simultaneously at the same location or under the same environmental conditions (water chemistry, sludge composition).

The adoption of the all volatile treatment (AVT) control program minimizes the possibility for recurrence of the tube-wall thinning phenomenon. Successful AVT operation requires maintenance of low concentrations of impurities in the steam generator water, thus reducing the potential for formation of highly concentrated solutions in low-flow zones, which is the precursor of corrosion. By restriction of the total alkalinity in the steam generator and prohibition of extended operation with free alkalinity, the AVT program should minimize the possibility for recurrence of intergranular corrosion in localized areas due to excessive levels of free caustic.

Laboratory testing has shown that the Inconel-600 tubing is compatible with the AVT environment. Isothermal corrosion testing in high-purity water has shown that commercially produced Inconel-600 exhibiting normal microstructures tested at normal engineering stress levels does not suffer intergranular stress corrosion cracking in extended exposure to high-temperature water. These tests also showed that no general type corrosion occurred. A series of autoclave tests in reference secondary water with planned excursions has produced no corrosion attack after 1938 days of testing on any as-produced Inconel-600 tube samples.

Model boiler tests have been used to evaluate the AVT chemistry guidelines adopted in 1974. The guidelines appear to be adequate to preserve tube integrity with one significant alteration: operation with contaminant ingress must be limited.

Additional extensive operating data are presently being accumulated with the conversion to AVT chemistry. A comprehensive program of steam generator inspections, including the recommendations of Regulatory Guide 1.83, Inservice Inspection of Pressurized Water Reactor Steam Generator Tubes, with the exceptions as stated in section 1.9, should provide for detection of any degradation that might occur in the steam generator tubing. Action levels for secondary side water chemistry during power operation are given in the EPRI PWR Secondary Water Chemistry Guidelines.

Increased margin against stress corrosion cracking has been obtained by the use of thermally treated Inconel-600 tubing. Thermal treatment of Inconel tubes has been shown to be particularly effective in resisting caustic corrosion. Tubing used in the Model F is thermally treated in accordance with a laboratory derived treatment process.

The tube support plates used in the Model F are ferritic stainless steel, which has been shown in laboratory tests to be resistant to corrosion in the AVT environment. If corrosion of ferritic stainless steel were to occur due to concentration of contaminants, the volume of the corrosion products is essentially equivalent to the volume of the parent material consumed. This would be expected to preclude denting. The support plates are also designed with quatrefoil tube holes rather than cylindrical holes. The quatrefoil tube-hole design promotes high-velocity flow along the tube and is expected to minimize the accumulation of impurities at the support plate location.

Additional measures are incorporated in the Model F design to prevent areas of dryout in the steam generator and accumulations of sludge in low-velocity areas. Modifications to the wrapper have increased water velocities across the tube sheet. A flow distribution baffle is provided which forces the low-flow area to the center of the bundle. Increased capacity blowdown pipes have been added to enable continuous blowdown of the steam generators at a high volume. The intakes of these blowdown pipes are located below the center cutout section of the flow distribution baffle in the low-velocity region where sludge may be expected to accumulate.

The impact of operating at a power level of 3653 MWt on steam generator water chemistry has been considered. The occurrence of stress corrosion cracking (SCC) and other forms of

degradation that might occur at current and enhanced rates will be found using the nondestructive examination (NDE) techniques specified in the degradation assessment that must be completed for subsequent plant outages.

5.4.2.4.4 Cleanup of Secondary-Side Materials

Several methods are employed to clean operating steam generators of corrosion-causing secondary-side deposits. Sludge lancing, a procedure in which a hydraulic jet inserted through an access opening (handhole) loosens deposits, can be performed when the need is indicated by the results of steam generator tube inspection. Six 6-in. access ports are provided for sludge lancing and inspection. Three of these are located above the tube sheet and three above the flow distribution baffle. Continuous blowdown is performed to monitor water chemistry. The location of the blowdown piping suction, adjacent to the tube sheet and in a region of relatively low-flow velocity, facilitates the removal of particulate impurities to reduce the accumulation on the tube sheet.

5.4.2.5 Steam Generator Inservice Inspection[a]

The steam generator is designed to permit inspection of Class 1 and 2 parts, including individual tubes. The design includes a number of openings to provide access to both the primary and secondary sides of the steam generator. The preservice inspection of the Unit 1 steam generators was completed to the requirements of the 1980 ASME Code through Winter 1980 Addenda. Inservice inspection will comply with the requirements of 10 CFR 50.55a. The openings include four manways, two for access to both chambers of the reactor coolant channel head inlet and outlet sides and two in the steam drum for inspection and maintenance of the moisture separators, and eight 6-in. handholes, three located just above the tube-sheet secondary surface, three located just above the flow distribution baffle, and two located on the tubelane diameter between the upper tube support plate and the row 1 tubes. Additional access to the tube U-bend is provided through each of the three deck plates. For proper functioning of the steam generator, some of the deck-plate openings are covered with welded, but removable, hatch plates. Inspection/access to the primary side is provided by two 16-in. manways located in the channel head.

Regulatory Guide 1.83 provides recommendations concerning the inspection of tubes, which cover inspection equipment, baseline inspections, tube selection, sampling and frequency of inspection, methods of recording, and required actions based on findings. The steam generators are designed to permit access to tubes for inspection and/or repair or plugging, if necessary, per the guidelines described in Regulatory Guide 1.83. Regulatory Guide 1.121, Basis for Plugging Degraded Pressurized Water Reactor (PWR) Steam Generator Tubes, provides recommendations concerning tube plugging. The minimum requirements for inservice inspection of steam generators, including tube plugging criteria, are established as part of the Technical Specifications.

5.4.2.6 Quality Assurance

The steam generator quality assurance program is given in table 5.4.2-2.

[a] The Inservice Inspection Program, Steam Generator Tubing Integrity Program, and Steam Generator Program for Upper Internals are credited as license renewal aging management programs (see subsections 19.2.13, 19.2.26, and 19.2.27).

Radiographic inspection and acceptance standard are in accordance with the requirements of Section III of the ASME Code.

Liquid penetrant inspection is performed on weld-deposited tube-sheet cladding, channel-head cladding, divider-plate-to-tube-sheet and to channel-head weldments, tube-to-tube-sheet weldments, and weld-deposit cladding. Liquid penetrant inspection and acceptance standards are in accordance with the requirements of Section III of the ASME Code.

Some of the tube-to-tube sheet welds of Unit 1 steam generator number 4 were deformed by impacts from a loose part. Based on the completion of a testing and analysis program, it is concluded that tube-to-tubesheet joint integrity remains consistent with the original plant design basis. The minimum requirements for the inservice inspection of steam generator tubes, including tube plugging criteria, are established as part of the Technical Specifications and are not changed as a result of the mechanical deformation of the tube-to-tubesheet welds.

Magnetic particle inspection is performed on the tube-sheet forging, channel-head casting, nozzle forgings, and the following weldments:

A. Nozzle to shell.

B. Support brackets.

C. Instrument connection (secondary).

D. Temporary attachments after removal.

E. All accessible pressure-retaining welds after hydrostatic test.

Magnetic particle inspection and acceptance standards are in accordance with the requirements of Section III of the ASME Code.

Ultrasonic tests are performed on the tube-sheet forgings, tube-sheet cladding, secondary-shell and heat plates, and nozzle forgings.

The heat transfer tubing is subjected to eddy current testing and ultrasonic examination.

Hydrostatic tests are performed in accordance with Section III of the ASME Code.

5.4.2.7 References

1. WCAP-16543-P, Rev. 0, “Regulatory Guide 1.121 Analysis and Structural Integrity Performance Criterion Application for the Vogtle Units 1 & 2 Model F Steam Generators,” August 2006.

5.4.3 REACTOR COOLANT PIPING

5.4.3.1 Design Bases

The reactor coolant system (RCS) piping is designed and fabricated to accommodate the system pressures and temperatures attained under all expected modes of plant operation or anticipated system interactions. Stresses are maintained within the limits of Section III of the American Society of Mechanical Engineers (ASME) Code. Code and material requirements are provided in section 5.2.

Materials of construction are specified to minimize corrosion/ erosion and ensure compatibility with the operating environment.

The piping in the RCS is Safety Class 1 and is designed and fabricated in accordance with ASME Code, Section III, Class 1 requirements.

Stainless steel pipe conforms to American National Standards Institute (ANSI) B36.19 for sizes 1/2 in. through 12 in. and wall thickness schedules 40S through 80S. Stainless steel pipe outside of the scope of ANSI B36.19 conforms to ANSI B36.10.

The minimum wall thickness of the loop piping and fittings are no less than those calculated using the ASME Code, Section III, Class 1 formula of Paragraph NB-3641.1(3) with an allowable stress value of 17,550 psi. The pipe wall thickness for the pressurizer surge line is schedule 160. The minimum pipe bend radius is 5 nominal pipe diameters, and ovality does not exceed 6%.

All butt welds, branch connection nozzle welds for 3 in. nominal pipe sizes and greater, and boss welds are of a full penetration design. Socket-welded connections could be used for branch connection nozzle welds for 2 in. nominal pipe sizes and smaller and for some thermowell connections.

Processing and minimization of sensitization are discussed in subsection 5.2.3.

Flanges conform to ANSI B16.5.

Socket weld fittings and socket joints conform to ANSI B16.11.

Inservice inspection is discussed in subsection 5.2.4.

5.4.3.2 Design Description

The RCS piping includes those sections of piping interconnecting the reactor vessel, steam generator, and reactor cooling pump (RCP). It also includes the following:

A. Charging line and alternate charging line from the designated check valve up to the branch connections on the reactor coolant loop (RCL).

B. Letdown line and excess letdown line from the branch connections on the RCL to the system isolation valve.

C. Pressurizer spray lines from the reactor coolant cold legs to the spray nozzle on the pressurizer vessel.

D. Residual heat removal (RHR) lines to or from the RCLs up to the designated check valve or isolation valve.

E. Safety injection lines from the designated check valve to the RCLs.

F. Accumulator lines from the designated check valve to the RCLs.

G. Deleted.

H. Loop, fill, drain, sample, and instrument[(a)] lines from the designated isolation valve to the RCLs.

I. Pressurizer surge line from one RCL hot leg to the pressurizer vessel surge nozzle.

J. Resistance temperature detector scoop element, pressurizer spray scoop, sample connection[a] with scoop, reactor coolant temperature element installation boss, the temperature element well itself, and the resistance temperature detector (RTD) fast response thermowells.

K. All branch connection nozzles attached to RCLs.

L. Pressure safety and relief lines from nozzles on top of the pressurizer vessel up to and through the pressurizer power-operated relief valves and pressurizer safety valves.

M. Auxiliary spray line from the isolation valve to the main pressurizer spray line.

N. Sample lines[a] from the pressurizer to the isolation valve.

O. Vent line from the reactor vessel head to the system isolation valves.

P. Reactor vessel level instrumentation lines from the isolation valves to the reactor vessel and to two RCL hot legs.

Q. RCP seal water injection lines to or from the RCP.

R. Boron injection lines from the check valve to the RCL.

Principal design data for the reactor coolant piping are given in table 5.4.3-1.

Details of the materials of construction and codes used in the fabrication of reactor coolant piping and fittings are discussed in section 5.2.

The reactor coolant piping and fittings that make up the loops are austenitic stainless steel. Pipe and fittings are cast, seamless without longitudinal or electroslag welds, and comply with the requirements of the ASME Code, Section II (parts A and C), Section III, and Section IX. All smaller piping that is part of the RCS, such as the pressurizer surge line, spray and relief line, loop drains and connecting lines to other systems, are also austenitic stainless steel. The nitrogen supply line for the pressurizer relief tank is carbon steel. All joints and connections are welded, except for the pressurizer code safety valves, where flanged joints are used.

All piping connections from auxiliary systems are above the horizontal centerline of the reactor coolant piping, with the exception of:

A. RHR pump suction lines, which are 45° down from the horizontal centerline. This enables the water level in the RCS to be lowered in the reactor coolant pipe while continuing to operate the RHRS, should this be required for maintenance.

B. Loop drain lines and the connection for temporary level measurement of water in the RCS during refueling and maintenance operation.

C. The differential pressure taps for flow measurement, which are downstream of the steam generators of the first 90° elbow.

D. The pressurizer surge line, which is attached at the horizontal centerline.

E. Two of the three thermowells in each resistance temperature detector hot leg connection.

Penetrations into the coolant flow path are limited to the following:

[a] Lines with a 3/8-in. or less flow restricting orifice qualify as Safety Class 2. In the event of a break in one of these Safety Class 2 lines, the normal makeup system is capable of providing makeup flow while maintaining pressurizer water level.

A. The spray line inlet connections extend into the cold leg piping in the form of a scoop so that the velocity head of the RCL flow adds to the spray driving force.

B. The RCS sample taps protrude into the main stream to obtain a representative sample of the reactor coolant.

C. The hot and cold leg thermowells for the resistance temperature detectors extend into the reactor coolant to measure the RCS temperatures. For the hot legs, the RTDs are located inside of the RTD scoops.

D. The wide-range temperature detectors are located in resistance temperature detector wells that extend into both the hot and cold legs of the reactor coolant piping.

Separate RTDs mounted in thermowells in each RCL hot and cold leg are provided so that individual temperature signals may be developed for use in the reactor control and protection systems. The RTDs are contained in thermowells which extend into the hot leg flow to become exposed to a representative temperature sample of the reactor coolant. Each hot leg is instrumented with three thermowells at locations 120 degrees apart around the reactor coolant piping with one located at the top of the pipe. The temperature measured by the three RTDs is then averaged using electronic weighting to provide the temperature input to the reactor control and protection systems.

An RTD mounted in a thermowell for the cold leg temperature measurement is located downstream of each reactor coolant pump discharge. This connection is located close to the same weld connection at the pump discharge and is in the same relative position in each loop.

Signals from the temperature detectors are used to compute the reactor coolant ΔT (temperature of the hot leg, T_{hot}, minus the temperature of the cold leg, T_{cold}), and an average reactor coolant temperature, T_{avg}. The ΔT and T_{avg} for each loop is indicated on the main control board.

5.4.3.3 Design Evaluation

Piping load and stress evaluation for normal operating loads, seismic loads, blowdown loads, and combined normal, blowdown, and seismic loads is discussed in section 3.9.N.

5.4.3.3.1 Material Corrosion/Erosion Evaluation

The water chemistry is selected to minimize corrosion. A periodic analysis of the coolant chemical composition is performed to verify that the reactor coolant quality meets the specifications. (See subsection 5.2.3.)

Periodic analysis of the coolant chemical composition is performed to monitor the adherence of the system to desired reactor coolant water quality listed in table 5.2.3-3. Maintenance of the water quality to minimize corrosion is accomplished using the CVCS and sampling system which are described in chapter 9.

The design and installation are in compliance with the ASME Code, Section III. Pursuant to this, all pressure-containing welds out to the second valve that delineates the RCS boundary are accessible for inservice examination as required of ASME Code, Section XI, and are fitted with removable insulation.

5.4.3.3.2 Sensitized Stainless Steel

Sensitized stainless steel is discussed in subsection 5.2.3.

5.4.3.3.3 Contaminant Control

Contamination of stainless steel and Inconel by copper, low melting temperature alloys, mercury, and lead is prohibited. Colloidal graphite is the only permissible thread lubricant.

Prior to application of thermal insulation, the austenitic stainless steel surfaces are cleaned and analyzed to a halogen limit of 0.0015 mg chloride/dm^2 and 0.0015 mg fluoride/dm^2.

5.4.3.3.4 Detection Of Thermal Stratification[a]

To assure that unisolable sections of piping connected to the reactor coolant system (RCS) and pressurizer surge line will not be subjected to combined cyclic and thermal stresses that could cause fatigue failure, a program has been implemented that will detect adverse temperature distributions as described below.

5.4.3.3.4.1 Unisolable Section of Piping Connected To The RCS. Unisolable sections of piping for the safety injection, normal and alternate charging, and auxiliary spray lines interconnected with the RCS are instrumented to detect adverse thermal stratification and cycling due to potential isolation valve leakage into the RCS boundary. Fluid leakage is detected by temperature measurements utilizing resistance temperature detectors (RTDs) strapped on the pipe. Temperature data is periodically recorded and evaluated for thermal stratification and cycling to determine its impact on piping structural integrity. Additionally (on Unit 2 only), two 12-in. residual heat removal (RHR) suction lines attached to the reactor coolant loop (RCL) hot leg are instrumented with RTDs.

5.4.3.3.4.2 Pressurizer Surge Line. The Unit 2 pressurizer surge line was instrumented to detect temperature distribution and thermal movement. The instruments consisted of RTDs and displacement transducers. The temperature distribution and thermal movements were monitored and recorded through the first refueling cycle. Analysis utilizing the leak-before-break methodology and the monitoring data taken from the Unit 2 surge line, combined with Unit 1 surge line support design changes and operating procedure changes, are an acceptable program to address the effects of thermal stratification on the pressurizer surge line. Therefore, no further monitoring is required.

5.4.3.4 Tests and Inspections

The RCS piping quality assurance program is given in table 5.4.3-2.

Volumetric examination is performed throughout 100% of the wall volume of each pipe and fitting in accordance with the applicable requirements of Section III of the ASME Code for all

[a] The Fatigue Monitoring Program is credited as a license renewal aging management program (see subsection 19.3.2).

pipe 27 1/2 in. and larger. All unacceptable defects are eliminated in accordance with the requirements of the same section of the code.

A liquid penetrant examination is performed on all accessible surfaces of each finished fitting, in accordance with the criteria of the ASME Code, Section III. Acceptance standards are in accordance with the applicable requirements of the ASME Code, Section III.

The pressurizer surge line conforms to SA-376, Grade 304, 304N, or 316 with supplementary requirements S2 (transverse tension tests) and S6 (ultrasonic test). The S2 requirement applies to each length of pipe. The S6 requirement applies to 100% of the piping wall volume.

The end of pipe sections, branch ends, and fittings are machined back to provide a smooth weld transition adjacent to the weld path.

Tests and inspections performed under the following license renewal aging management programs are credited as applicable to various portions of RCS piping:

- Boric Acid Corrosion Control Program (see subsection 19.2.3).
- CASS RCS Fitting Evaluation Program (see subsection 19.2.5).
- External Surfaces Monitoring Program (see subsection 19.2.8).
- Inservice Inspection Program (see subsection 19.2.13).
- Oil Analysis Program (see subsection 19.2.16).
- One-Time Inspection Program (see subsection 19.2.17).
- One-Time Inspection Program for ASME Class 1 Small Bore Piping (see subsection 19.2.18).
- Water Chemistry Control Program (see subsection 19.2.28).
- Fatigue Monitoring Program (see subsection 19.3.2).

5.4.4 MAIN STEAM LINE FLOW RESTRICTIONS

5.4.4.1 Design Bases

The outlet nozzle of the steam generator is provided with a flow restrictor designed to limit steamflow in the unlikely event of a break in the main steam line. A large increase in steamflow will create a backpressure which limits further increase in flow. The flow restrictor performs the following functions:

- Rapid rise in containment pressure is limited.
- The rate of heat removal from the reactor is such as to keep the cooldown rate within acceptable limits.
- Thrust forces on the main steam line piping are reduced.

- Stresses on internal steam generator components, particularly the tube sheet and tubes, are limited.

The restrictor is configured to minimize the unrecovered pressure loss across the restrictor during normal operation.

5.4.4.2 Design Description

The flow restrictor consists of seven Inconel ASME SB-163 venturi inserts which are installed in holes in an integral steam outlet nozzle forging. The inserts are arranged with one venturi at the centerline of the outlet nozzle and the other six equally spaced around it. After insertion into the nozzle forging holes, the Inconel venturi inserts are welded to the Inconel cladding on the inner surface of the forging.

5.4.4.3 Design Evaluation

The flow restrictor design has been analyzed to determine its structural adequacy. The equivalent throat diameter of the steam generator outlet is 16 in. and the resultant pressure drop through the restrictor at 100% steamflow is approximately 2.78 psig. This is based on a design flowrate of 3.78×10^6 lb/h. Materials of construction and manufacturing of the flow restrictor are in accordance with Code Class 1 Section III of the ASME Code. The method for seismic analysis is dynamic.

5.4.4.4 Inspections

Since the restrictor is not part of the steam system boundary, no inspections beyond those performed during fabrication are anticipated.

5.4.5 MATERIALS AND INSPECTIONS

This subsection is not applicable to VEGP.

5.4.6 REACTOR VESSEL DESIGN DATA

This subsection is not applicable to VEGP.

5.4.7 RESIDUAL HEAT REMOVAL SYSTEM

The residual heat removal system (RHRS) transfers heat from the reactor coolant system (RCS) to the nuclear service cooling water system via the component cooling water (CCW) system to reduce the temperature of the reactor coolant to the cold shutdown temperature at a controlled rate during the second part of normal plant cooldown and maintains this temperature until the plant is started up again.

Refer to subsection 9.2.2 for a description of the CCW system. Parts of the RHRS also serve as parts of the emergency core cooling system (ECCS) for accident mitigation (section 6.3).

In addition, the RHRS is used to transfer refueling water between the refueling cavity and the refueling water storage tank at the beginning and end of the refueling operations.

Nuclear plants employing the same RHRS design as the VEGP are given in section 1.3.

5.4.7.1 Design Bases

The RHRS design parameters are listed in table 5.4.7-1.

The following establishes a design bases supporting a 50°F/h cooling rate. The RHR system and support systems have been reviewed and evaluated to document the acceptability of an operational cooling rate of up to 100°F/h not to exceed 100°F in any 1 h period.

The RHRS is placed in operation approximately 2 to 4 hours after reactor shutdown, when the temperature and pressure of the RCS are approximately 350°F and 365 psig, respectively. Assuming that two heat exchangers and two pumps are in service and that each heat exchanger is supplied with 5000 gal/min CCW initially at 105°F, the RHRS is designed to reduce the temperature of the reactor coolant to 140°F within 20 h following reactor shutdown. Under these conditions, the time required to reduce the reactor coolant temperature from 350°F to 200°F is approximately 3 h. The heat load handled by the RHRS during the cooldown transient includes residual and decay heat from the core and reactor coolant pump heat. The design heat load is based on the decay heat fraction that exists at 20 h following reactor shutdown from an extended run at full power.

Assuming that only one heat exchanger and pump are in service and that the heat exchanger is supplied with CCW at 5000 gal/min and initially at 105°F, the RHRS is capable of reducing the temperature of the reactor coolant from 350°F to 200°F within approximately 30 h. The time required under these conditions to reduce reactor coolant temperature from 350°F to 212°F is approximately 20 h. The RHRS is designed to be isolated from the RCS whenever the RCS pressure exceeds the RHRS design pressure. The RHRS is isolated from the RCS on the suction side by two motor-operated valves in series on each suction line. Each motor-operated valve is interlocked to prevent its opening if RCS pressure is greater than approximately 365 psig. The valves have a control room alarm which alerts the operators if one or both of the valves is not fully closed and the RCS pressure exceeds 420 psig. The RHRS is isolated from the RCS on the discharge side by two check valves in each return line. Also provided on the discharge side is a normally open motor-operated valve downstream of each RHRS heat exchanger. (These check valves and motor-operated valves are not considered part of the RHRS; they are shown as part of the ECCS. See drawing 1X4DB121.)

Each inlet line to the RHRS is equipped with a pressure relief valve designed to prevent RHRS overpressurization assuming the most severe overpressure transients. These relief valves protect the system from inadvertent overpressurization during plant startup, shutdown, and cold shutdown decay heat-removal operations.

Each discharge line from the RHRS to the RCS is equipped with a pressure relief valve designed to relieve the maximum possible backleakage through the valves isolating the RHRS from the RCS. These valves are considered part of the ECCS, as depicted in drawing 1X4DB120. Relief capacity of these valves is given in table 6.3.2-2.

The RHRS is designed for a single nuclear power unit and is not shared between units.

The RHRS is designed to be fully operable from the control room for normal operation except as described in paragraph 5.4.7.2.7. By nature of its redundant design, the RHRS is designed to accept all major component single failures, with the only effect being an extension in the

required cooldown time. There are no motor-operated valves in the RHRS that are subject to flooding following a secondary side break or a LOCA. Although considered to be of low probability, spurious operation of a single motor-operated valve can be accepted without loss of function as a result of the redundant two-train design.

Missile protection, protection against dynamic effects associated with the postulated rupture of piping, and seismic design are discussed in section 3.5 and subsections 3.6.2 and 3.7.N.2, respectively.

5.4.7.2 System Design

5.4.7.2.1 Schematic Piping and Instrumentation Diagrams

The RHRS, as shown in drawings 1X4DB122 and 2X4DB122 and figure 5.4.7-1, consists of two residual heat exchangers, two RHR pumps, and the associated piping, valves, and instrumentation necessary for operational control. Notes to figure 5.4.7-1 identify modes of operation, valve alignments, and process conditions at various points in the system. The inlet lines to the RHRS are connected to the hot legs of two reactor coolant loops (RCLs), while the return lines are connected to the cold legs of each of the RCLs. These return lines are also the ECCS low-head injection lines (drawings 1X4DB119, 2X4DB119, 1X4DB120, and 1X4DB121). The RHRS suction lines are isolated from the RCS by two motor-operated valves in series located inside the containment. Each discharge line is isolated from the RCS by two check valves located inside the containment and by a normally open motor-operated valve, with power lockout capability, located outside the containment. (The check valves and the motor-operated valve on each discharge line are not part of the RHRS; these valves are shown as part of the ECCS (drawings 1X4DB119, 2X4DB119, 1X4DB120, and 1X4DB121).

During RHRS operation, reactor coolant flows from the RCS to the RHR pumps, through the tube side of the residual heat exchangers, and back to the RCS. The heat is transferred to the CCW circulating through the shell side of the residual heat exchangers.

Coincident with operation of the RHRS, a portion of the reactor coolant flow may be diverted from downstream of the residual heat exchangers to the chemical and volume control system (CVCS) low-pressure isolation letdown line for cleanup and/or pressure control. By regulating the diverted flowrate and the charging flow, the RCS pressure may be controlled. Pressure regulation is necessary to maintain the pressure range dictated by the fracture prevention criteria requirements of the reactor vessel and by the No. 1 seal differential pressure and net positive suction head requirements of the reactor coolant pumps.

The RCS cooldown rate is manually controlled by regulating the reactor coolant flow through the tube side of the residual heat exchangers. The flow control valve in the bypass line around each residual heat exchanger automatically maintains a constant return flow to the RCS. Instrumentation is provided to monitor system pressure, temperature, and total flow.

The RHRS is also used for filling the refueling cavity before refueling. After refueling operations, the RHRS is used to pump the water back to the refueling water storage tank until the water level is brought down to the flange of the reactor vessel. The remainder of the water is removed via a drain connection at the bottom of the refueling canal.

When the RHRS is in operation, the water chemistry is the same as that of the reactor coolant. Provision is made for the process sampling system to extract samples from the flow of reactor coolant downstream of the residual heat exchangers. A local sampling point is also provided on each RHR train between the pump and heat exchanger.

The RHRS also functions, in conjunction with the high-head portion of the ECCS, to provide injection of borated water from the refueling water storage tank into the RCS cold legs during the injection phase following a loss-of-coolant accident (LOCA).

In its capacity as the low-head portion of the ECCS, the RHRS provides long-term recirculation capability for core cooling following the injection phase of the LOCA. This function is accomplished by aligning the RHRS to take fluid from the containment sump, cool it by circulation through the residual heat exchangers, and supply it to the core directly as well as via the centrifugal charging pumps and safety injection pumps.

The use of the RHRS as part of the ECCS is more completely described in section 6.3.

The RHR pumps, in order to perform their ECCS function, are interlocked to start automatically on receipt of a safety injection signal (section 6.3).

The RHRS suction isolation valves in each inlet line from the RCS are separately interlocked to prevent both of them from being opened when RCS pressure is greater than approximately 365 psig, and the valves have a control room alarm, which alerts the operators if one or both of the valves is not fully closed and the RCS pressure exceeds 420 psig. These interlocks are described in more detail in paragraph 5.4.7.2.4 and subsection 7.6.2.

The RHRS suction isolation valves are also interlocked to prevent their being opened unless the isolation valves in the following lines are closed:

A. Recirculation lines from the residual heat exchanger outlets to the suctions of the safety injection pumps and centrifugal charging pumps.

B. RHR pump suction line from the refueling water storage tank.

C. RHR pump suction line from the containment sump.

The motor-operated valves in the RHRS miniflow bypass lines are interlocked to open when the RHRS pump discharge flow is less than the open setpoint (824 gpm at 350 °F, 780 gpm at 100 °F) and to close when the flow exceeds the closed setpoint (1944 gpm at 350 °F, 1841 gpm at 100 °F).

5.4.7.2.2 Equipment and Component Descriptions

The materials used to fabricate RHRS components are in accordance with the applicable code requirements. All parts of components in contact with borated water are fabricated or clad with austenitic stainless steel or equivalent corrosion resistant material. Component parameters are given in table 5.4.7-2.

5.4.7.2.2.1 Residual Heat Removal Pumps. Two pumps are installed in the RHRS. The pumps are sized to deliver reactor coolant flow through the RHR heat exchangers to meet the plant cooldown requirements. The use of two separate RHR trains ensures that cooling capacity is only partially lost should one pump become inoperative.

The RHR pumps are protected from overheating and loss of suction flow by miniflow bypass lines that ensure flow to the pump suction. A valve located in each miniflow line is regulated by a signal from the flow transmitters located in each pump discharge header. The control valves open when the residual pump discharge flow is less than the open setpoint (824 gpm at 350 °F, 780 gpm at 100 °F) and close when the flow exceeds the closed setpoint (1944 gpm at 350 °F, 1841 gpm at 100 °F).

Although both valves could be closed by operator error, the design of the miniflow system would preclude any pump damage. The miniflow bypass valve is a fast operating motor-operated gate valve which will open or close in 10 s or less. The residual heat removal pump can operate safely without damage during this period with no flow.

A pressure sensor in each pump discharge header provides a signal for an indicator in the control room. A high-pressure alarm is also actuated by the pressure sensor.

The two pumps are vertical, centrifugal units with mechanical seals on the shafts. All pump surfaces in contact with reactor coolant are austenitic stainless steel or equivalent corrosion resistant material.

The RHR pumps also function as the low-head safety injection pumps in the ECCS (section 6.3).

A pump performance curve is provided in figure 5.4.7-2.

5.4.7.2.2.2 Residual Heat Exchangers. Two residual heat exchangers are installed in the system. The heat exchanger design is based on heat load and temperature differences between reactor coolant and CCW existing 20 h after reactor shutdown when the temperature difference between the two systems is small.

The installation of two heat exchangers in separate and independent RHR trains ensures that the heat-removal capacity of the system is only partially lost if one train becomes inoperative.

The residual heat exchangers are of the shell and U-tube type. Reactor coolant circulates through the tubes, while CCW circulates through the shell. The tubes are welded to the tube sheet to prevent leakage of reactor coolant.

The residual heat exchangers also function as part of the ECCS (section 6.3).

5.4.7.2.2.3 RHRS Valves. Valves that perform a modulating function are equipped with two sets of packing and an intermediate leakoff connection that discharges to a local equipment drain.

Manual and motor-operated valves have backseats to facilitate repacking and to limit stem leakage when the valves are open. Leakage connections are provided where required by valve size and fluid conditions. The RHR discharge cross-connect valves are provided with bonnet vents to the RHR pump side of the valves. RHR loop suction valves 1/2HV8701A, 1/2HV8702A, 1/2HV8702B, and 1HV8701B are provided with bonnet vents to the upstream side of the valves. The RHR loop suction valve 2HV8701B bonnet vent to the upstream side of the valve has been removed.

5.4.7.2.3 System Operation

5.4.7.2.3.1 Plant Startup. Generally, while in the cold shutdown condition, decay heat from the reactor core is being removed by the RHRS. The number of pumps and heat exchangers in service depends upon the heat load at the time.

At the beginning of plant startup, at least one RHR pump is operating, and a portion of the discharge flow may be directed to the CVCS.

This arrangement augments RCS pressure control during startup. When the reactor coolant pumps are started, the RHR pump is stopped. The thermal input of the reactor coolant pumps heats the reactor coolant inventory. Once the pressurizer steam bubble formation is complete, the RHRS is isolated from the RCS and aligned for operation as part of the ECCS.

5.4.7.2.3.2 Power Generation and Hot Standby Operation. During power generation and hot standby operation, the RHRS is not in service but is aligned for operation as part of the ECCS.

5.4.7.2.3.3 Plant Shutdown. Plant shutdown is defined as the operation that brings the plant from no-load temperature and pressure to cold conditions.

5.4.7.2.3.4 Normal Cold Shutdown. The initial phase of plant shutdown is accomplished by transferring heat from the RCS to the steam and power conversion system through the use of the steam generators.

When the reactor coolant temperature and pressure are reduced to approximately 350°F and 365 psig, approximately 2 to 4 hours after reactor shutdown, the second phase of cooldown starts with the RHRS being placed in operation.

Startup of the RHRS includes a warmup period, during which time reactor coolant flow through the heat exchangers is limited to minimize thermal shock. The rate of heat removal from the reactor coolant is manually controlled by regulating the coolant flow through the residual heat exchangers. By adjusting these control valves downstream of the residual heat exchangers, the mixed mean temperature of the return flows is controlled. Coincident with the manual adjustment of flow through the heat exchangers, each heat exchanger bypass valve is automatically regulated to give the required total flow.

The reactor cooldown rate is limited by RCS equipment cooling rates based on allowable stress limits and the operating temperature limits of the CCW system. As the reactor coolant temperature decreases, the reactor coolant flow through the residual heat exchangers is increased by adjusting the control valve in each heat exchanger's tube side outlet line.

Should both RHR heat exchanger outlet and bypass flow control valves fail simultaneously (i.e., loss of instrument air), then the maximum cooldown rate may be increased. The maximum cooldown rate depends on many factors, including the time of failure, the RHR flowrate, the CCW flowrates and temperatures, and other heat loads on the CCW system. One of the key factors is the RCS water temperature, since the cooldown rate depends upon the temperature difference between the RHR (RCS) flow and the CCW flow in the RHR heat exchanger. Even with the maximum flow through the RHR heat exchangers, it is typically impossible to maintain a cooldown rate as high as the technical specification design rate of 100°F/h when the RCS temperature is less than 250°F. The operator can significantly limit the cooldown rate by merely stopping one of the RHR pumps.

During plant shutdown, pressurizer steam bubble operation is maximized to control RCS pressure. When the RHRS is in operation, RCS control is augmented by regulating the charging flowrate and the rate of letdown from the RHRS to the CVCS.

After the reactor coolant pressure is reduced and the temperature is 140°F or lower, the RCS may be opened for refueling or maintenance.

A failure modes and effects analysis for normal cooldown operations is provided in table 5.4.7-3.

5.4.7.2.3.5 Safety-Grade Cold Shutdown. It is expected that the systems normally used for cold shutdown will be available anytime the operator chooses to perform a reactor cooldown. However, to ensure that the plant can be taken to cold shutdown at anytime, the safety-grade cold shutdown design enables the RCS to be taken from no-load temperature and pressure to cold conditions using only safety-grade systems, with only onsite or offsite power available, and assuming the most limiting single failure.

Should portions of normal shutdown systems be unavailable, the operator will maintain the plant in a hot standby condition while making those normal systems functional. Local manual actions are performed as described in table 5.4.7-4. Appropriate procedures are provided for the use of safety-grade backups contingent upon the inability to make normal systems available. The operator should use any of the normal systems that are available in combination with the safety-grade backups for the systems that cannot be made operable. The safety-grade provisions are to be used only upon the inability to make available the equipment normally used for the given function.

The safety-grade cold shutdown design enables the operator to maintain the plant in hot standby for approximately 4 h. Since it is assumed that the reactor coolant pumps are not available, circulation of the reactor coolant is provided by natural circulation with the reactor core as the heat source and the steam generators as the heat sink. Heat removal is accomplished via the steam generator power-operated relief valves and auxiliary feedwater system.

The boration of the RCS is initiated prior to cooling the RCS. The charging pumps are used to provide borated water to the RCS at a rate of approximately 0 to 50 gal/min. The borated water is delivered to the RCS via the safety-grade charging line or the high-head safety injection lines. Both flow paths have provisions for flow control. Reactor coolant pump seal injection is also maintained. To accommodate this addition to RCS inventory, continuous letdown is discharged from the reactor vessel head letdown line to the pressurizer relief tank.

During boration to cold shutdown concentration, the safety-grade cooldown is accomplished by increasing the steam dump from the steam generator power-operated relief valves to attain a rate of primary side cooling of approximately 35°F/h. In conjunction with this portion of the cooldown, the charging pumps are used to deliver borated water to makeup for primary contraction due to cooling. Makeup is also provided for the RCS inventory discharged when the reactor vessel head letdown path is periodically cycled to provide head cooling. Upon approaching the end of this phase of cooldown, the RCS is depressurized by venting or letdown through the reactor vessel head vent valves to the pressurizer relief tank.

To ensure that the accumulators do not repressurize the RCS, the accumulator discharge valves are closed prior to the RCS pressure dropping below the accumulator discharge pressure. Additionally, each accumulator is provided with two Class 1E solenoid-operated valves in parallel to ensure that the accumulator may be vented should it fail to be isolated from the RCS.

When the reactor coolant temperature and pressure are reduced to approximately 350°F and 365 psig, respectively, and the RCS borated to cold shutdown concentration, the second phase of cooldown starts with the RHRS being placed in operation.

As safety-grade cooldown continues, the reactor vessel head letdown line is periodically opened to increase head cooling and to accommodate any additional input to the RCS, such as reactor

coolant pump seal injection. Since loss of nonsafety-grade equipment results in a loss of the air supply to the flow control valves that are normally used to limit the initial RHRS cooldown rate, the operator may choose to use only one of the RHR subsystems. Should a single failure, such as that of an RHRS component or of an emergency power train (when only onsite power is available), limit operation to one of the RHR subsystems, the operator would open the series isolation valves in the suction of only the operable RHR subsystem. In this case, the operator would also close the cross-connect isolation valves between the subsystems. RHR would continue under these conditions until the redundant subsystem could be made available.

A failure modes and effects analysis for safety-grade cold shutdown operations is provided in table 5.4.7-4.

5.4.7.2.3.6 Refueling. The RHRS is utilized during refueling to transfer borated water from the refueling water storage tank to the refueling cavity. During this operation, one RHR train is selected for fill, the isolation valves in the suction lines from the RCS are closed, the isolation valves from the refueling water storage tank are opened and the RHR pump may be started or the refueling water storage tank (RWST) water is allowed to gravity drain.

The refueling cavity is prepared for flooding and the vessel head is removed to its storage pedestal using the containment polar crane. The refueling water is then transferred into the reactor vessel through the RHRS hot-leg or cold-leg return lines and into the refueling cavity through the open reactor vessel. After the water level reaches the normal refueling level, the RHR pump is stopped, the refueling water storage tank supply valves are closed, and the suction isolation valves from the RCS are opened.

During refueling, the RHRS is maintained in service with the number of pumps and heat exchangers in operation as required by the heat load. Also during refueling, an RHR pump not being applied for shutdown cooling can be used to satisfy the Technical Requirements Manual requirement that a pump be available for boration when the refueling cavity water level is ≥ 23 feet above the flange. The flowpath associated with this application of the RHR pump is the pump taking suction from the RWST and injecting into the RCS cold legs.

Following refueling, the RHRS is used to drain the refueling cavity to the top of the reactor vessel flange by pumping water from the RCS to the refueling water storage tank. The vessel head is then replaced and the normal RHRS flow path reestablished. The remainder of the water is removed from the refueling canal via a drain connection in the bottom of the canal.

An alternative methodology may be used for raising the vessel head. In this method, the reactor vessel head is lifted slightly. Then the cavity is filled as discussed above, while reactor vessel head is gradually raised as the water level in the refueling cavity increases. The filling of the cavity is also terminated as discussed above. Following refueling, placement of the head on the vessel is accomplished by maintaining the vessel head just above the water in the cavity, as the level lowers while the refueling water is being pumped from the cavity to the RWST by the RHR pumps.

5.4.7.2.3.7 Mid-loop and Drain Down Operations. The RHR system is used to provide core cooling when the RCS must be partially drained to allow maintenance or inspection of the reactor head, steam generators, or reactor coolant pump seals.

The level in the primary system is lowered to near the mid-line of the hot and cold legs. At this water level the air/water interface is at close proximity to the RHR suction nozzles located on the hot legs of loops 1 and 4, and care must be taken to avoid air entrainment into the RHR

pump suction. Air ingestion by an RHR pump can cause loss of pump function, creating the potential for loss of residual heat removal.

Instrumentation has been provided to assist the operator in safely maintaining adequate level in the RCS hot legs during mid-loop and drain down operations. This instrumentation is shown in drawings 1X4DB111, 2X4DB111, 1X4DB112, 2X4DB112, and 1X4DB113. Instrumentation has also been provided to assist the operator in quickly identifying air ingestion in the RHR pumps.

Two differential pressure transmitters are connected to the RCS to provide independent level indications in the main control room. One transmitter is connected to the RCS loop 1 hot leg and provides narrow range indication. This narrow range indicator spans from the bottom of the hot leg upward 8 feet. This instrument loop also provides annunciation of the low hot leg level. The other transmitter is connected to the RCS loop 4 hot leg and provides wide range indication. This wide range indicator spans from the bottom of the hot leg upward 20 feet. The instrument loops are powered from separate breakers to maximize the availability of the indication.

Local RCS level indication is available via a sight glass located in the containment building. The sight glass spans from the top of the pressurizer heaters down to just below the bottom of the RCS hot legs. The piping for this sight glass is connected to the RCS as required during modes 5 and 6. The tubing used for the sight glass is vacuum resistant.

Current transducers monitor the 4,160 V power feeders to each RHR pump. The output of these transducers is routed to the plant computer. Historical traces of the pump motor current can be obtained at any plant computer terminal. The logic associated with modes 5 and 6 core cooling critical safety function status trees provides a visual alarm at the plant computer safety parameter display system (SPDS) terminal in the main control room, if the motor current becomes unstable. The critical safety function status trees alarm the control room annunciator when any adverse condition occurs.

5.4.7.2.4 Control

Each inlet line to the RHRS is equipped with a relief valve to prevent RHRS overpressurization during plant startup, shutdown, and cold shutdown decay heat-removal operation. Each valve has a relief capacity of 900 gal/min at a set pressure of 450 psig. An analysis has been conducted to confirm the capability of the RHRS relief valve to prevent overpressurization in the RHRS. All credible events were examined for their potential to overpressurize the RHRS. These events included normal operating conditions, infrequent transients, and abnormal occurrences. The analysis confirmed that one relief valve has the capability to maintain the RHRS maximum pressure within code limits. The above capacities of the RHRS suction line relief valves are adequate to provide relief protection necessary for the RHRS and the RCS as part of the cold overpressure mitigating system. For a discussion of the cold overpressure mitigating system and the overpressure events examined, refer to WCAP-10529.

Each discharge line from the RHRS to the RCS is equipped with a pressure relief valve to relieve the maximum possible back-leakage through the valves separating the RHRS from the RCS. Each valve has a relief flow capacity of 20 gal/min at a set pressure of 600 psig. These relief valves are located in the ECCS (drawing 1X4DB121).

The fluid discharged by the suction-side relief valves is collected in the pressurizer relief tank. The fluid discharged by the discharge side relief valves is collected in the recycle holdup tank.

The design of the RHRS includes two motor-operated gate isolation valves in series on each inlet line between the high-pressure RCS and the lower pressure RHRS. They are closed during normal operation and are opened only for RHR during a plant shutdown after the RCS

pressure is reduced to approximately 365 psig or lower and RCS temperature is reduced to approximately 350°F. During a plant startup, the inlet isolation valves are shut after drawing a bubble in the pressurizer and prior to increasing RCS pressure above 425 psig. These isolation valves are provided with independent and diverse "prevent-open" interlocks and main annunciator alarms which are designed to prevent possible exposure of the RHRS to normal RCS operating pressure. The two inlet isolation valves in each RHR subsystem are independently interlocked with diverse pressure signals to prevent their being opened whenever the RCS pressure is greater than approximately 365 psig. Additionally, during plant startup, the valves are interlocked such that they alarm in the main control room when RCS pressure increases to 420 psig to alert the operators that one or both of the valves is not fully closed.

The use of two independently powered motor-operated valves in each of the two inlet lines, with an independently powered pressure transmitter for each valve, along with two independent and diverse interlock signals for each function, ensures a design which meets applicable single-failure criteria. These protective interlock designs, in combination with plant operating procedures, provide the means of accomplishing the protective function. For further information on the instrumentation and control features, see subsection 7.6.2.

The RHR inlet isolation valves are provided with control switches with integral red-green position indicator lights on the main control board and an alarm window on a main control board annunciator panel.

Isolation of the low-pressure RHRS from the high-pressure RCS is provided on the discharge side by two check valves in series. These check valves are located in the ECCS, and their testing is described in paragraph 6.3.4.2.

5.4.7.2.5 Applicable Codes and Classifications

The entire RHRS is designed as Nuclear Safety Class 2 with the exception of the suction isolation valves, which are Safety Class 1. Component codes and classifications are given in section 3.2.

5.4.7.2.6 System Reliability Considerations

General Design Criterion 34 requires that a system to remove residual heat be provided. The safety function of this system is to transfer fission product decay heat and other residual heat from the core at a rate sufficient to prevent fuel or pressure boundary design limits from being exceeded. Safety-grade systems are provided in the plant design, both nuclear steam supply system scope and balance of plant scope, to perform this function. The safety-grade systems that perform this function for all plant conditions except a LOCA are the RCS and steam generators, which operate in conjunction with the auxiliary feedwater system and the steam generator safety and power-operated relief valves, and the RHRS, which operates in conjunction with the CCW system and the nuclear service cooling water (NSCW) system. For LOCA conditions, the safety-grade system which performs the function of removing residual heat from the reactor core is the ECCS, which operates in conjunction with the CCW system and the NSCW system.

The auxiliary feedwater system, along with the steam generator safety and power-operated relief valves, provides a completely separate, independent, and diverse means of performing the safety function of removing residual heat, which is normally performed by the RHRS when RCS temperature is less than 350°F. The auxiliary feedwater system is capable of performing this function for an extended period of time following plant shutdown.

The RHRS is provided with two RHR pumps and two residual heat exchangers arranged in two separate, independent flow paths. To ensure reliability, each RHR pump is connected to a different vital bus. Each RHR train is isolated from the RCS on the suction side by two motor-operated valves in series, with each valve receiving power via a separate motor control center and from a different vital bus. The suction isolation valves prevent exposure of the RHRS to the normal operating pressure of the RCS (paragraph 5.4.7.2.4).

The RHRS operation for normal conditions, even with a major failure, is accomplished completely from the control room except as described in paragraph 5.4.7.2.7. The redundancy in the RHRS design provides the system with the capability to maintain its cooling function, even with a major single failure such as failure of a pump, valve, or heat exchanger, since the redundant train can be used for continued heat removal.

Status-indicating lights are provided at the control board for the RHR pump, the RHRS suction isolation valves, and the miniflow isolation valves.

The major portion of the RHRS is located in the auxiliary building. Leakages resulting from a passive failure of the RHRS piping are collected by the floor drain system. See subsection 9.3.3 for a discussion of this system and the alarms and instrumentation provided to detect any radioactive leaks, should they occur.

5.4.7.2.7 Manual Actions

The RHRS is designed to be fully operable from the control room for normal operation except for restoring power to the suction isolation valves, and for restoring air to the heat exchanger flow control valves prior to RHR initiation. Manual operations required of the control room operator are restoring power to and opening the suction isolation valves, restoring air to and positioning the flow control valves downstream of the residual heat exchangers, and starting the RHR pumps. The power lockout of the suction isolation valves is discussed in subsection 7.6.2. The air is isolated from the RHR heat exchanger flow control valves by closing the air supply isolation valve. During normal cooldown, there is adequate time and accessibility to perform these actions. Refer to paragraph 6.3.2.8 for the manual actions required during ECCS operations.

5.4.7.3 Performance Evaluation

The performance of the RHRS in reducing reactor coolant temperature is evaluated through the use of heat balance calculations on the RCS, RHRS, and CCW system at stepped intervals following the initiation of removal operation. Heat removal through the RHR and component cooling heat exchangers is calculated at each interval by use of standard water-to-water heat exchanger performance correlations; the resultant fluid temperatures for the RCS, RHRS, and CCW system are calculated and used as input to the next interval's heat balance calculation.

Assumptions utilized in the series of heat balance calculations describing plant RHRS cooldown are as follows:

A. RHR operation is initiated 4 h after reactor shutdown.[(a)]

B. RHR operation begins at a reactor coolant temperature of 350°F.

[(a)] This value is used in design bases criteria and analyses to support the plant maximum cold shutdown condition. The RHR system and support systems have been reviewed and evaluated to document the acceptability of an operational cooling rate up to 100°F/h.

C. Thermal equilibrium is maintained throughout the RCS during the cooldown (i.e., one reactor coolant pump in operation whenever the reactor coolant temperature is above 160°F).

D. CCW temperature during cooldown is limited to a maximum of 120°F.

E. RCS cooldown rates of 50°F/h are not exceeded.[(b)]

Cooldown curves calculated using this method are provided in figure 5.4.7-3.

5.4.7.4 Preoperational Testing

Preoperational testing of the RHRS is addressed in chapter 14.

5.4.7.5 Reliability Tests and Inspections

As the RHRS functions as part of the ECCS, periodic tests and inspections are conducted in conjunction with those conducted on ECCS components. (See paragraph 6.3.4.2 for a discussion of the tests and inspections. See the Technical Specifications for the selection of test frequency, acceptability of testing, and measured parameters. A description of the inservice inspection program is included in section 6.6.)

5.4.8 REACTOR WATER CLEANUP SYSTEM

This subsection is not applicable to VEGP.

5.4.9 MAIN STEAM LINE AND FEEDWATER PIPING

The main steam and feedwater systems of VEGP are not part of the reactor coolant system. The main steam system, including the main steam piping, is described in section 10.3. The feedwater system, including piping, is described in subsection 10.4.7.

5.4.10 PRESSURIZER

5.4.10.1 Design Bases

The pressurizer provides a point in the reactor coolant system (RCS) where liquid and vapor are maintained in equilibrium under saturated conditions for control of pressure of the RCS during steady-state operations and transients.

The volume of the pressurizer is equal to, or greater than, the minimum volume of steam, water, or total of the two which satisfies all of the following requirements:

A. The combined saturated water volume and steam expansion volume is sufficient to provide the desired pressure response to system volume changes.

B. The water volume is sufficient to prevent the heaters from being uncovered during a step-load increase of 10% at full power.

C. The steam volume is large enough to accommodate the surge resulting from a 50% reduction of full load with automatic reactor control and a 40% steam dump without the water level reaching the high level reactor trip point.

D. The steam volume is large enough to prevent water relief through the safety valves following a loss of load with the high water level initiating a reactor trip, without reactor control or steam dump.

E. The pressurizer will not empty following reactor trip and turbine trip.

F. A low pressurizer pressure safety injection signal will not be activated because of a reactor trip and turbine trip.

The surge line is sized to maintain the pressure drop between the RCS and the safety valves within allowable limits during a design discharge flow from the safety valves.

The surge line is designed to withstand the thermal stresses resulting from volume surges occurring during operation. In accordance with TMI Action Item II.K.3.9, the derivative action setting of the proportional integral derivative controller has been set to zero to prevent spurious power-operated relief valves (PORV) operation.

5.4.10.2 Design Description

5.4.10.2.1 Pressurizer and Surge Line

The pressurizer is a vertical, cylindrical vessel having hemispherical top and bottom heads constructed of carbon steel, with austenitic stainless steel cladding on all internal surfaces exposed to the reactor coolant. The surge line is constructed of stainless steel.

The general configuration of the pressurizer is shown in figure 5.4.10-1. The design data for the pressurizer are given in table 5.4.10-1. A comparison of the design basis operating conditions determined that the conditions originally qualified for the pressurizer will envelop the operation at a power level of 3653 MWt. Codes and material requirements are provided in section 5.2.

The pressurizer surge line connects the pressurizer to one reactor coolant hot leg, thus enabling continuous coolant volume and pressure adjustments between the RCS and the pressurizer.

The surge line nozzle and electric heaters are located in the bottom head of the pressurizer. The heaters are designed to be removable for maintenance or replacement.

The pressurizer surge line nozzle diameter is given in table 5.4.10-1, and the pressurizer surge line size is shown in drawing 1X4DB112.

A retaining screen is located above the nozzle to prevent passage of any foreign matter from the pressurizer to the RCS. Baffles in the lower section of the pressurizer prevent an insurge of cold water from flowing directly to the steam/water interface and also assist in mixing.

The spray line nozzles and the relief and safety valve connections are located in the top head of the pressurizer vessel. Spray flow is modulated by automatically controlled air-operated valves. The spray valves also can be operated manually from the control room.

A small continuous spray flow is provided through a manual bypass valve around each power-operated spray valve to minimize the boron concentration difference between the pressurizer liquid and the reactor coolant and to prevent excessive cooling of the spray piping.

During an outsurge of water from the pressurizer, flashing of water to steam and generation of steam by automatic actuation of the heaters keep the pressure above the low-pressure reactor trip setpoint. During an insurge from the RCS, the spray system, which is fed from two cold legs, condenses steam in the pressurizer to prevent the pressurizer pressure from reaching the setpoint of the PORVs. The heaters are energized on high water level during insurge to heat the subcooled surge water that enters the pressurizer from the reactor coolant loop. The skirt-type support is attached to the lower head and extends for a full 360° around the vessel. The lower part of the skirt terminates in a bolting flange with bolt holes for securing the vessel to its foundation. The support is provided with ventilation holes to ensure free convection of ambient cooling air past the heater and connector ends.

Material specifications are provided in table 5.2.3-1 for the pressurizer and the surge line. Design transients for the components of the RCS are discussed in subsection 3.9.N.1.

5.4.10.2.2 Pressurizer Spray and Relief Line Instrumentation

Refer to chapter 7 for details of the instrumentation associated with pressurizer pressure, level, and temperature.

Temperatures in the spray lines from the cold legs of two loops are measured and indicated. Alarms from these signals are actuated to warn the operator of low spray water temperature or to indicate insufficient flow in the spray lines.

Temperatures in the pressurizer safety and relief valves discharge lines are measured and indicated. An increase in a discharge line temperature is an indication of leakage or relief through the associated valve. High temperature alarms are initiated if the leakage is abnormal.

5.4.10.3 Design Evaluation

5.4.10.3.1 System Pressure Control

The RCS pressure is controlled by the pressurizer whenever a steam volume is present in the pressurizer.

A design basis safety limit has been set such that the RCS pressure does not exceed the maximum transient value allowed under the American Society of Mechanical Engineers (ASME) Code, Section III. Evaluation of plant conditions of operation considered for design indicates that this safety limit is not reached.

During startup and shutdown, the rate of temperature change in the RCS is controlled by the operator. Heatup rate is controlled by energy input from the reactor coolant pumps and by the pressurizer electrical heating capacity.

The pressurizer heaters are powered from four electrical panels: one panel for each heater group. Two groups of heaters can be administratively loaded onto the non-Class 1E emergency buses (drawings 1X3D-AA-A01A and 2X3D-AA-A01A). The Class 1E 4160-V breakers supplying the non-Class 1E buses are automatically opened upon a safety injection signal. The non-Class 1E buses can be manually reenergized under administrative procedure. Actions associated with these two groups of heaters can be controlled from either the control room or the shutdown panels. Engineered safety features loads need not be shed to manually load a pressurizer heater group.

Pressurizer heater controls are described in paragraph 7.7.1.D.

Analysis performed for a four-loop plant with an 1800-ft^3 pressurizer similar to VEGP indicates that a heater capacity of 150 kW is adequate to maintain subcooled conditions in the RCS during natural circulation. Each VEGP heater panel has a heater capacity of 483 kW. Furthermore, the analysis demonstrates that the RCS sensible heat capacity is such that adequate subcooling can be maintained in the RCS for 4 h without heat input from the pressurizer heaters. Thus the recommendations of Action Item II.E.3.1 of NUREG-0737 are satisfied.

When the pressurizer is filled with water, i.e., during initial system heatup or near the end of the second phase of plant cooldown, RCS pressure is controlled by the letdown flowrate via the residual heat removal system.

5.4.10.3.2 Pressurizer Level Control

The normal operating water volume at full-load conditions is approximately 60% of the free internal vessel volume. Under part-load conditions the water volume in the pressurizer is reduced proportionally with reductions in plant load to approximately 25% of the free internal vessel volume at the zero-power condition.

5.4.10.3.3 Pressure Setpoints

The RCS design and operating pressure, together with the safety valve setpoints, PORV setpoints, pressurizer spray valve setpoints, and the protection system pressure setpoints, are listed in table 5.4.10-2. The design pressure allows for operating transient pressure changes. The selected design margin considers core thermal lag, coolant transport times and pressure drops, instrumentation and control response characteristics, and system relief valve characteristics. The low pressurizer pressure reactor trip does not require a coincident low water level signal. This is in accordance with the recommendations of Action Item II.K.1.17 of NUREG-0660.

Temperature changes which may affect the relief valve setpoints have been considered. Normal ambient air temperature variations have no significant effects. However, cold valves relieving hot fluid may show reduced setpoints; therefore, this has been considered in the design of such valves.

5.4.10.3.4 Pressurizer Spray

Two separate automatically controlled spray valves with remote manual overrides are used to initiate pressurizer spray. In parallel with each spray valve is a manual throttle valve which permits a small continuous flow through both spray lines to reduce thermal stresses and thermal shock when the spray valves open and to help maintain uniform water chemistry and temperature in the pressurizer. Temperature sensors with low alarms are provided in each spray line to alert the operator to insufficient bypass flow. The layout of the common spray line piping routed to the pressurizer forms a water seal which prevents the steam buildup back to the control valves. The design spray rate is selected to prevent the pressurizer pressure from reaching the operating setpoint of the PORVs during a step reduction in power level of 10% of full load.

The pressurizer spray lines and valves are large enough to provide the required spray flowrate under the driving force of the differential pressure between the surge line connection in the hot leg and the spray line connection in the cold leg. The spray line inlet connections extend into the cold leg piping in the form of a scoop in order to utilize the velocity head of the reactor coolant loop flow to add to the spray driving force. The spray valves and spray line connections are arranged so that the spray operates, although at a reduced capacity when one reactor coolant pump is not operating. The spray line also assists in equalizing the boron concentration between the reactor coolant loops and the pressurizer.

A flowpath from the chemical and volume control system to the pressurizer spray line is also provided. This path provides auxiliary spray to the vapor space of the pressurizer during cooldown when the reactor coolant pumps are not operating. The pressurizer spray connection and the spray piping are designed to withstand the thermal stresses resulting from the introduction of cold spray water.

5.4.10.4 Tests and Inspections

The pressurizer is designed and constructed in accordance with the ASME Code, Section III.

To implement the requirements of the ASME Code, Section XI, the following welds are designed and constructed to present a smooth transition surface between the parent metal and the weld metal. The weld surface is ground smooth for ultrasonic inspection.

- Support skirt to the pressurizer lower head.
- Surge nozzle to the lower head.
- Safety, relief, and spray nozzles to the upper head.
- Nozzle to safe end attachment welds.
- All girth and longitudinal full-penetration welds.
- Manway attachment welds.

The liner within the safe end nozzle region extends beyond the weld region to maintain a uniform geometry for ultrasonic inspection.

Peripheral support rings are furnished for the removable insulation modules.

The pressurizer quality assurance program is given in table 5.4.10-3.

Tests and inspections performed under the following license renewal aging management programs are credited as applicable to the pressurizer and its subcomponents:

- Bolting Integrity Program (see subsection 19.2.2).
- Boric Acid Corrosion Control Program (see subsection 19.2.3).
- Inservice Inspection Program (see subsection 19.2.13).
- Nickel Alloy Management Program for Nonreactor Vessel Closure Head Penetration Locations (see subsection 19.2.14).

- Water Chemistry Control Program (see subsection 19.2.28).
- Fatigue Monitoring Program (see subsection 19.3.2).

5.4.11 PRESSURIZER RELIEF DISCHARGE SYSTEM

5.4.11.1 Design Bases

The pressurizer relief discharge system collects, cools, and directs the steam and water discharged from various safety and relief valves in the containment for processing. The system consists of the pressurizer relief tank (PRT), the pressurizer safety and relief valve discharge piping, the relief tank internal spray header and associated piping, the tank nitrogen supply, and the drain to the liquid waste processing system.

The system design, including the PRT design volume, is based on the requirement to condense and cool a discharge of steam equivalent to 110% of the full-power pressurizer steam volume, without exceeding a pressure/temperature condition of 50 psig/200°F in the PRT. These values are well below the PRT design conditions of 100 psig and 340°F. Additional design data for the tanks are given in table 5.4.11-1.

The minimum volume of water in the PRT is determined by the energy content of the steam to be condensed and cooled, by the assumed initial temperature of the water, and by the desired final temperature of the water volume. The initial water temperature is assumed to be 120°F, which corresponds to the design maximum expected containment temperature for normal conditions. Provision is made to permit cooling of the water in the tank should the water temperature rise above 120°F during plant operation. The design final temperature, following a design discharge to the tank, is 200°F, which allows the contents of the tank to be drained directly to the liquid waste processing system without cooling.

The PRT saddle supports and anchor bolt arrangement are designed to withstand the loadings resulting from the vessel seismic, static, and nozzle loadings.

The pressurizer safety and relief valve piping and support arrangement is designed such that the effect of thrust forces on the piping system from valve operations is minimized. The piping analysis is discussed in section 3.9.N.

The design and location of the PRT rupture disks are such that they do not pose a missile threat to any safety-related equipment.

5.4.11.2 System Description

The piping and instrumentation diagram for the pressurizer relief discharge system is given in drawing 1X4DB112.

The steam and reactor grade water discharged from the various safety and relief valves inside the containment is routed to the PRT. Table 5.4.11-2 provides an itemized list of the discharges to the tank, together with references to the corresponding piping and instrumentation diagrams.

The pressurizer safety and relief valve piping and support arrangement in figure 5.4.11-2 shows the valve discharge piping, as well as the piping upstream of the safety and relief valves. The

piping upstream of the valves, which is not considered part of the pressurizer relief discharge system, includes the following:

A. Three lines with loop seal arrangements connecting the pressurizer nozzles to the three safety valves.

B. A line from the pressurizer relief nozzle branching to the two power-operated relief valves (PORVs), which have individual water seals and motor-operated isolation valves.

The pressurizer safety and relief valve discharge piping consists of:

A. A common piping manifold (supported over the top of the pressurizer) into which the safety and relief valves discharge.

B. Safety valve discharge lines to the manifold.

C. Relief valve discharge lines to the manifold.

D. A manifold downcomer discharge pipe.

E. Piping to the PRT.

The main support structure for the safety and relief valve piping consists of four column members equally spaced around the common manifold coupled to the valve support brackets on the pressurizer. No welding to the pressurizer is required. To increase the natural frequency of the system, auxiliary crossmembers are provided from the common manifold to the main support columns. The safety valves are provided with a bottom saddle type support coupled to the auxiliary crossmembers. The relief valves are positioned above the manifold, and the relief valve lines are supported at various points along the manifold.

The pressurizer safety and relief valve piping is constructed of austenitic stainless steel. Design data for the pressurizer safety and relief valve piping are given in table 5.4.3-1.

The piping upstream of the safety and relief valves is part of the reactor coolant system (RCS) and is designed and fabricated in accordance with American Society of Mechanical Engineers (ASME) Code, Section III, Class 1 requirements. The piping between these valves and the downcomer tee connection is nonnuclear safety related but is designed and fabricated to ASME Code, Section III, Class 2, to the extent practical. The support structure for the piping from the pressurizer to the downcomer tee connection is designed and fabricated to ASME Code, Section III, Subsection NF.

The piping from the pressurizer to the PRT is designed to Seismic Category 1 requirements. The principal design codes are indicated in table 3.2.2-1.

The general configuration of the PRT is shown in figure 5.4.11-1. The tank is a horizontal, cylindrical vessel with elliptical dished heads. The vessel is constructed of austenitic stainless steel and is overpressure protected by means of two safety heads with stainless steel rupture discs. Also shown in figure 5.4.11-1 are the flanged connection for the pressurizer safety and relief valve discharge line, the spray water inlet, the bottom drain connection, the gas vent connection, and the vessel supports. Although the tank is classified as nonnuclear safety related, it is designed and fabricated to Section III, Division 1, Class 3 of the ASME Code.

The tank normally contains water and a predominantly nitrogen atmosphere. In order to obtain effective condensing and cooling of the discharged steam, the tank is installed horizontally so that the steam can be discharged through a sparger pipe located near the bottom, under the water level. The sparger holes are designed to ensure good mixing of the discharged steam with the water initially in the tank.

A nitrogen gas blanket is used to control the atmosphere in the tank and to allow room for the expansion of the original water, plus the condensed steam discharge. The tank gas volume is sized such that the pressure following a design basis steam discharge does not exceed 50 psig, assuming an initial pressure of 3 psig. This pressure is low enough to prevent opening of the rupture discs. Provisions are made to permit the gas in the tank to be periodically analyzed to determine the concentration of hydrogen and/or oxygen.

The internal spray and bottom drain on the PRT function to cool the water when the temperature exceeds 120°F, as in the case following a steam discharge. The contents are cooled by a feed-and-bleed process, with cold reactor makeup water entering the tank through the spray water inlet and the warm mixture draining to the reactor coolant drain tank (RCDT). The contents may also be cooled by recirculation through the RCDT heat exchanger of the liquid waste processing system.

5.4.11.3 Safety Evaluation

The pressurizer relief discharge system does not constitute part of the reactor coolant pressure boundary in accordance with 10 CFR 50.2, since all of its components are downstream of the RCS safety and relief valves; thus, General Design Criteria 14 and 15 are not applicable. Furthermore, complete failure of the auxiliary systems serving the PRT will not impair the capability for safe plant shutdown.

The design of the system piping layout and piping restraints is consistent with the hazards protection requirements discussed in section 3.6 and appendix 3F. The safety and relief valve discharge piping is restrained so that the integrity and operability of the valves are maintained in the event of a rupture. Regulatory Guide 1.67 is not applicable since the system is not an open discharge system.

The pressurizer relief discharge system is capable of handling the design discharge of steam without exceeding the design pressure and temperature. The volume of nitrogen in the PRT is that required to limit the maximum pressure accompanying the design basis discharge to 50 psig, half the design pressure of the tank. The volume of water in the PRT is capable of absorbing the heat from the assumed discharge while maintaining the water temperature below 200°F.

If a discharge results in a pressure that exceeds the design, the rupture discs on the tank would pass the discharge through the tank to the containment. The rupture discs on the relief tank have a relief capacity equal to or greater than the combined capacity of the pressurizer safety valves. The tank and rupture discs holders are also designed for full vacuum to prevent tank collapse, if the contents cool following a discharge without nitrogen being added.

The discharge piping from the pressurizer safety and relief valves to the PRT is sufficiently large to prevent backpressure at the safety valves from exceeding 20% of the setpoint pressure at full flow.

The recommendations of NUREG-0737, Action Items II.G.1 and II.K.3.1, are met as discussed below. The pressurizer is equipped with two Class 1E PORVs (solenoid operated) and two Class 1E PORV block valves (motor operated). The PORV and associated block valve on one line are supplied with control and motive power from train A, while the other PORV and associated block valve on the other line are powered from train B (drawings 1X4DB111, 2X4DB111, 1X4DB112, 2X4DB112, and 1X4DB113).

The PORV block valves 1HV-8000A and 1HV-8000B are powered from Class 1E 480-V buses. These buses are normally supplied from offsite power. In the event of a loss of offsite power,

these buses are automatically loaded onto the diesels (drawings 1X3D-AA-A01A and 2X3D-AA-A01A). PORVs 455A and 456A are Class 1E dc solenoid valves and are powered from redundant Class 1E 125-V dc trains A and B, respectively. The train assignment for power to the PORVs and block valves is based on:

A. The ability to open one of the parallel pressurizer vent paths in conjunction with a single failure.

B. The ability to close both parallel paths in conjunction with a single failure. (Capability to isolate both parallel paths in conjunction with a single failure is based upon the fact that the solenoid-operated PORVs are qualified, dc powered, and designed to fail closed.)

Pressurizer pressure is interlocked with the PORV block valves, which provides automatic closure of the block valves upon low pressurizer pressure.

5.4.11.4 Instrumentation Requirements

The following instrumentation is provided on the main control board:

A. The PRT pressure transmitter provides a signal to an indicator. An alarm is provided to indicate high tank pressure.

B. The PRT level transmitter supplies a signal to an indicator. High- and low-level alarms are also provided.

C. The temperature of the water in the PRT is displayed by an indicator. An alarm actuated by high temperature informs the operator that cooling of the tank contents is required.

D. The temperature of the safety and relief valve discharge lines is displayed by indicators. Alarms actuated by high temperature notify the operator of steam discharge due to either leakage or valve actuation.

5.4.11.5 Inspection and Testing Requirements

The nondestructive examinations performed during fabrication of the forged piping from the pressurizer to the downcomer tee connection are identified in table 5.4.11-3.

The PRT is subject to nondestructive and hydrostatic testing during construction and after installation in accordance with Section III, Division 1, Class 3 of the ASME Code.

The downcomer piping to the PRT is subject to nondestructive and hydrostatic testing during construction.

Periodic visual inspections and preventive maintenance are conducted on the system components according to normal industrial practice.

5.4.12 VALVES

5.4.12.1 Design Bases

As noted in section 5.2, all valves out to and including the second valve that is normally closed or capable of automatic or remote closure, larger than 3/4 in., are American Society of Mechanical Engineers (ASME) Code, Section III, Class 1 valves. Valves 3/4-in. or smaller in lines connected to the reactor coolant system (RCS) are Class 1, unless the Class 1 piping interface is provided with suitable flow limiting orificing, then the valves are Class 2.

For a check valve to qualify as part of the RCS, it must be located inside the containment system. When the second of two normally closed check valves is considered part of the RCS (as defined in section 5.1), means are provided to periodically assess backflow leakage of the first valve when closed.

To ensure that the valves will meet the design objectives, the materials of construction minimize corrosion/erosion and are compatible with the environment. Leakage is minimized to the extent practicable by design.

5.4.12.2 Design Description

All manual- and motor-operated valves of the RCS which are 3 in. and larger are provided with double-packed stuffing boxes and intermediate lantern ring leakoff connections or a reduced packing configuration with the valve stem leakoff line removed. Throttling control valves over 2 in. are provided with double-packed stuffing boxes and stem leakoff connections. In general, RCS leakoff connections are piped to a closed collection system. Leakage to the atmosphere is essentially zero for these valves.

Gate valves at the engineered safety features interface are wedge design and are essentially straight through. The wedges are flex-wedge or solid. All gate valves have backseats.

Globe valves are "T" and "Y" styles of outside screw and yoke construction.

Check valves are swing type for sizes 2 1/2 in. and larger. All check valves which contain radioactive fluid are stainless steel and do not have body penetrations other than the inlet, outlet, and bonnet. The check hinge is serviced through the bonnet. All operating parts are contained within the valve body. The disc has limited rotation to provide a change of seating surface and alignment after each valve opening.

5.4.12.3 Design Evaluation

The design requirements for Class 1 valves, as discussed in section 5.2, limit stresses to levels which ensure the structural integrity of the valves. In addition, the testing programs described in section 3.10.N demonstrate the ability of the valves to operate, as required, during anticipated and postulated plant conditions.

Reactor coolant chemistry parameters are specified in the design specifications to ensure the compatibility of valve construction materials with the reactor coolant. To ensure that the reactor coolant continues to meet these parameters, the chemical composition of the coolant will be analyzed periodically, as discussed in the Technical Requirements Manual.

The above requirements and procedures, coupled with the previously described design features for minimizing leakage, ensure that the valves will perform their intended functions.

5.4.12.4 Tests and Inspections

Technical Specification 3.4.14, RCS Pressure Isolation Valve (PIV) Leakage, requires leakage from reactor coolant pressure isolation valves to be within limits. The valves to which Technical specification 3.4.14 applies and which are required to satisfy leakage limits are stated in FSAR table 5.4.12-3.

FSAR table 5.4.12-3 shows the following:

- The valves to which Technical Specification 3.4.14 applies and the associated function.
- The size of each RCS pressure isolation valve, and
- The maximum allowable leakage for each RCS pressure isolation valve.

Hydrostatic shell test and seat leakage and functional tests are performed on all RCS valves. The tests and inspections discussed in section 3.10.N are performed to ensure the operability of the active valves.

There are no full-penetration welds within the valve body walls. Valves are accessible for disassembly and internal visual inspection to the extent practical. Plant layout configurations determine the degree of inspectability. The valve nondestructive examination program is given in table 5.4.12-2. Inservice inspection is discussed in subsection 5.2.4.

Tests and inspections performed under the following license renewal aging management programs are credited as applicable to various RCS valves:

- Bolting Integrity Program (see subsection 19.2.2).
- Boric Acid Corrosion Control Program (see subsection 19.2.3).
- External Surfaces Monitoring Program (see subsection 19.2.8).
- Inservice Inspection Program (see subsection 19.2.13).
- Oil Analysis Program (subsection 19.2.16).
- One-Time Inspection Program (see subsection 19.2.17).
- Water Chemistry Control Program (see subsection 19.2.28).

5.4.13 SAFETY AND RELIEF VALVES

5.4.13.1 Design Bases

The combined capacity of the pressurizer safety valves can accommodate the maximum pressurizer surge resulting from complete loss of load. Sizing of the pressurizer safety valves is discussed in subsection 5.2.2.

The pressurizer power-operated relief valves (PORVs) are designed to limit pressurizer pressure to a value below the high pressure reactor trip setpoint. They are designed to fail in the closed position on loss of actuating power.

5.4.13.2 Design Description

The pressurizer safety valves are of the pop type. The valves are spring loaded and self-actuated by direct fluid pressure and have backpressure compensation features.

The pipe connecting each pressurizer nozzle to its safety valve is shaped in the form of a loop seal. Condensate resulting from normal heat losses accumulates in the loop. This loop seal minimizes any leakage of hydrogen gas or steam through the safety valve seats. If the pressurizer pressure exceeds the set pressure of the safety valves, they start lifting, and the water from the seal discharges during the actuation period.

The pressurizer PORVs are solenoid-operated valves which respond to a signal from a pressure-sensing system or to manual control. Remotely operated block valves are provided to isolate the inlets to the PORVs if excessive leakage develops.

The PORVs and their associated block valves are interlocked by a pressurizer low-pressure interlock. Actuation of the interlock prevents the relief valves from opening and closes the block valves. Manual control may override this interlock.

In the event that a pressurizer PORV open signal actuation is sent due to a failure in a pressure channel associated with normal PORV operation, the interlock is provided to close the PORV as pressure decreases below the interlock pressure setpoint. The pressure signal associated with the interlock originates in the narrow range pressurizer pressure instrumentation. This signal and interlock operate separately from the cold overpressure pressure control signal, which originates in the wide range pressure instrumentation in the reactor coolant system loops. The overpressure protection system would not become disabled in the event of a single failure. The logic diagrams for the PORV interlocks are shown in drawing 1X6AA02-235.

In accordance with the requirements of NUREG-0737, TMI Action Item II.D.3, positive position indication is provided for the primary safety and relief valves.

Position indication on the PORV is accomplished through electrical reed switches. A magnetic rod, actuated by the valve plug, is located inside a projection above the top face of the bonnet and operates the reed switches contained in a switch assembly mounted externally on the bonnet. Safety valve indication is also accomplished through reed switches.

Temperatures in the pressurizer safety and relief valve discharge lines are measured, and an indication and a high alarm are provided on the main control board. An increase in a discharge line temperature is an indication of leakage or relief through the associated valve.

The PORVs provide the safety-related means for reactor coolant system depressurization to achieve cold shutdown. For a discussion of the use of these valves to achieve safety-grade cold shutdown, see subsection 5.4.7.

Design parameters for the pressurizer safety valves and power relief valves are given in table 5.4.13-1.

Relief and safety valve failures to close will promptly be reported to the Nuclear Regulatory Commission. This will meet the requirements of NUREG-0737, Action Item II.K.3.3.

5.4.13.3 Design Evaluation

The pressurizer safety valves prevent reactor coolant system pressure from exceeding 110% of system design pressure, in compliance with the ASME Code, Section III.

The slight time delay associated with the discharge of water from the loop seal piping configuration is accounted for in the limiting case analyses discussed in chapter 15.

The limiting pressure transient is the turbine trip event as described in subsection 15.2.3 and in the ASME Code Overpressure Protection Report. In the event, the pressurizer safety valves are assumed to open following a 2% tolerance in the set pressure, plus a 1% shift resulting from the presence of the loop seal, plus the time delay required to purge the loop seal. After these effects have been accounted for, the safety valves are assumed to relieve at their full capacity. As described in subsection 15.2.3, there is margin to the 110% of design pressure limit of 2748.5 psia. The results from the limiting case from subsection 15.2.3 are also presented in figure 5.4.13-1.

The pressurizer PORVs prevent actuation of the reactor high-pressure trip for all design transients up to and including the design step-load decreases with steam dump. The relief valves also limit undesirable opening of the spring-loaded safety valves.

5.4.13.4 Tests and Inspections

All safety and relief valves are subjected to hydrostatic tests, seat leakage tests, operational tests, and inspections, as required. For safety valves that are required to function during a faulted condition, additional tests are performed. These tests are described in section 3.10.N. There are no full-penetration welds within the valve body walls. Valves are accessible for disassembly and internal visual inspection. Refer to subsection 5.4.12 for the list of license renewal aging management programs credited to manage aging of various RCS valves.

Safety and relief valves similar to those at VEGP have been tested within the Electric Power Research Institute safety and relief test program and have been found adequate for steamflow and waterflow. The completion of this program addresses the requirements of Action Item II.D.1 of NUREG-0737 as related to valve testing.

5.4.14 COMPONENT SUPPORTS

5.4.14.1 Design Bases

Component supports allow unrestrained lateral thermal movement of the loop during plant operation and provide restraint to the loops and components during accident and seismic conditions. The loading combinations and design stress limits are discussed in paragraph 3.9.B.3.[(a)] Support design is in accordance with the American Society of Mechanical Engineers

[(a)] Reference 1 provides the original criteria for postulating breaks in the reactor coolant loop. The basis for eliminating eight of these postulated large pipe breaks in the reactor coolant loop is provided in reference 2. The RCL component support design configuration of Unit 1 is unchanged from that provided in reference 1. However, the elimination of the large pipe breaks led to partial installation of the primary loop whip restraints on Unit 1 and the removal of the primary loop whip restraints on Unit 2 and the reduction from five large bore hydraulic snubbers to two in the steam generator upper support assemblies

(ASME) Code, Section III, Subsection NF. The design maintains the integrity of the RCS boundary for normal, seismic, and accident conditions and satisfies the requirements of the piping code. The results of piping and supports stress evaluation are presented in section 3.9.

Conformance with Regulatory Guides 1.124 and 1.130 is discussed in section 1.9.

5.4.14.2 Description

The support structures are welded structural steel sections. Linear-type structures (tension and compression struts, columns, and beams) are used in all cases except for the reactor vessel supports, which are plate-type structures. Attachments to the supported equipment are nonintegral type that are bolted to or bear against the components. The supports-to-concrete attachments are either anchor bolts or embedded fabricated assemblies.

The supports permit virtually unrestrained thermal growth of the supported systems but restrain vertical, lateral, and rotational movement resulting from seismic and pipe break loadings. This is accomplished using spherical bushings in the columns for vertical support and girders, bumper pedestals, hydraulic snubbers, and tie rods for lateral support.

Because of manufacturing and construction tolerances, ample adjustment in the support structures is provided to ensure proper erection alignment and fit-up. This is accomplished by shimming or grouting at the supports-to-concrete interface and by shimming at the supports-to-equipment interface.

The supports for the various components are described in the following paragraphs.

5.4.14.2.1 Reactor Pressure Vessel

Supports for the reactor vessel (figure 5.4.14-1) are individual, air-cooled, rectangular box structures beneath the vessel nozzles bolted to the primary shield wall concrete. Each box structure consists of a horizontal top plate that receives loads from the reactor vessel shoe, a horizontal bottom plate that transfers the loads to the primary shield wall concrete, and connecting vertical plates. The supports are air-cooled to maintain the supporting concrete temperature within acceptable levels.

5.4.14.2.2 Steam Generator

As shown in figure 5.4.14-2, the steam generator supports consist of the following elements:

A. Vertical Support

Four individual columns provide vertical support for each steam generator. These are bolted at the top to the steam generator and at the bottom to the concrete structure. Spherical ball bushings at the top and bottom of each column allow unrestrained lateral movement of the steam generator during heatup and cooldown. The column base design permits both horizontal and vertical adjustment of the steam generator for erection and adjustment of the system.

B. Lower Lateral Support

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for Unit 2. The structural analysis of the RCL component supports for Units 1 and 2 considers the elimination of loads due to RCL breaks.

Lateral support is provided at the generator tube sheet by fabricated steel girders and struts. These are bolted to the compartment walls and include bumpers that bear against the steam generator but permit unrestrained movement of the steam generator during changes in system temperature. Stresses in the beams caused by wall displacements during compartment pressurization are considered in the design.

C. Upper Lateral Support

Upper lateral support of the steam generator is provided by a builtup ring plate girder at the operating deck. The 2-way acting snubbers restrain sudden seismic or blowdown-induced motion but permit the normal thermal movement of the steam generator.

Movement perpendicular to the thermal growth direction of the steam generator is prevented by struts.

5.4.14.2.3 Reactor Coolant Pump

Three individual columns, similar to those used for the steam generator, provide the vertical support for each pump. Lateral support for seismic and blowdown loading is provided by three lateral tension tie bars. The pump supports are shown in figure 5.4.14-3.

5.4.14.2.4 Pressurizer

The supports for the pressurizer, as shown in figure 5.4.14-4, consist of:

A. A steel ring plate between the pressurizer skirt and the supporting structure. The ring serves as leveling and adjusting member for the pressurizer.

B. The upper lateral support consists of struts cantilevered off the compartment walls that bear against the lugs provided on the pressurizer.

5.4.14.2.5 Control Rod Drive Mechanism (CRDM) Supports

The support system for the CRDM provides lateral restraint to limit CRDM deflections due to seismic or pipe break loadings. The CRDM support system consists of the following:

A. A support platform extends across the top of the control rod drive housing columns. The CRDM rod travel housing extensions protrude through holes in this platform, thus limiting lateral deflection of the CRDM housing.

B. Vertical support of the support platform is provided by three reactor vessel head lifting legs. The lifting legs are vertical columns which are pinned to the reactor vessel head.

C. Horizontal support of the support platform is provided by lateral tension tie rods which are pinned to the refueling cavity wall.

5.4.14.3 Evaluation

Detailed evaluation ensures the design adequacy and structural integrity of the reactor coolant loop and the primary equipment supports system. This detailed evaluation is made by comparing the analytical results with established criteria for acceptability. Structural analyses are performed to demonstrate design adequacy for safety and reliability of the plant in case of a large or small seismic disturbance and/or loss-of-coolant accident conditions. Loads which the system is expected to encounter often during its lifetime (thermal, weight, and pressure) are applied, and stresses are compared to allowable values. The modeling and analysis methods are discussed in paragraph 3.9.N.1.4

5.4.14.4 Tests and Inspections

Nondestructive examinations are performed in accordance with the procedures of the ASME Code, Section V, except as modified by the ASME Code, Section III, Subsection NF. The Inservice Inspection Program is credited as a license renewal aging management program for RCS primary equipment supports (see subsection 19.2.13). The Structural Monitoring Program is also credited as a license renewal aging management program for ASME piping and component supports (see subsection 19.2.32).

5.4.14.5 References

1. "Pipe Breaks for the LOCA Analysis of the Westinghouse Primary Coolant Loop," WCAP-8082-P-A (proprietary) and WCAP-8172-A (nonproprietary), January 1975.
2. Federal Register, Vo. 50, No. 27, February 8, 1985.

5.4.14.6 Bibliography

DeRosa, P., et al., "Evaluation of Steam Generator Tube, Tube Sheet and Divider Plant Under Combined LOCA Plus SSE Conditions," WCAP-7832, December 1973.

Eggleston, F. T., "Safety-Related Research and Development for Westinghouse Pressurized Water Reactor, Program Summaries-Winter 1976," WCAP-8768, Revision 1, June 1977.

"Reactor Coolant Pump Integrity in LOCA," WCAP-8163, September l973.

5.4.15 REACTOR VESSEL HEAD VENT SYSTEM

The reactor vessel head vent system (RVHVS) (figure 5.1.2-1) removes noncondensable gases or steam from the reactor vessel head. This system is designed to mitigate a possible condition of inadequate core cooling or impaired natural circulation resulting from the accumulation of noncondensable gases in the reactor coolant system (RCS). The design of the RVHVS is in accordance with the requirements of TMI action plan item II.B.1 of NUREG 0737 as discussed below.

5.4.15.1 Design Bases

The RVHVS is designed to remove noncondensable gases or steam from the RCS via remote manual operations from the control room. The system discharges to the pressurizer relief tank. Additionally, a letdown flow path is provided from the reactor vessel head vent to the excess letdown heat exchanger in the chemical and volume control system. The RVHVS is designed to vent a volume of hydrogen at system design pressure and temperature approximately equivalent to one-half of the RCS volume in 1 h.

The system provides for venting the reactor vessel head by using only safety grade equipment. The RVHVS satisfies applicable requirements and industry standards, including ASME Code classification, safety classification, single-failure criteria, and environmental qualification.

All piping and equipment from the vessel head vent up to and including the second isolation valve in each flow path are designed and fabricated in accordance with ASME Section III, Class 1 requirements. The piping and equipment in the flow paths from the isolation valve to the modulating valves and from the isolation valves to the excess letdown heat exchanger are designed and fabricated in accordance with ASME Section III, Class 2 requirements. The remainder of the piping and equipment is Seismic Category 1, nonnuclear safety.

All supports and support structures conform with the requirements of the ASME Code.

The analysis of the reactor vessel head vent piping is based on the following plant operating conditions defined in the ASME Code, Section III:

A. Normal Condition

   Pressure, deadweight, and thermal expansion analysis of the vent piping during:

   1. Normal reactor operation with the vent isolation valves closed.
   2. Post-refueling venting.

B. Upset Condition

   Loads generated by the operating basis earthquake (OBE).

C. Faulted Condition

   Loads generated by the safe shutdown earthquake (SSE). Loads generated by valve thrust during venting. In accordance with ASME III, faulted conditions are not included in fatigue evaluations.

The Class 1 piping used for the reactor vessel head vent is 1 in. schedule 160 and, therefore, in accordance with ASME III, is analyzed following the procedures of NC-3600 for Class 2 piping.

For all plant operating conditions listed above, the piping stresses are shown to meet the requirements of equations 8, 9, and 10 or 11 of ASME III, NC-3600, with a design temperature of 650°F and a design pressure of 2,485 psig.

5.4.15.2 System Description

The RVHVS consists of a single active failure proof flow path with redundant isolation valves. The equipment design parameters are listed in table 5.4.15-1.

The active portion of the system consists of four 1-in. open/close solenoid-operated isolation valves connected to the existing 1-in. vent pipe, which is located near the center of the reactor vessel head. The system design with two valves in series in each flow path minimizes the possibility of reactor coolant pressure boundary leakage. The isolation valves in one flow path

are powered by one vital power supply and the valves in the second flow path are powered by a second vital power supply. The isolation valves are fail closed normally closed valves. The valves are included in the valve operability program and will be qualified to IEEE-323-1975, -344-A75 and 382-1972. The control valves are also normally closed, fail closed.

The vent system piping is supported to ensure that the resulting loads and stresses on the piping and on the vent connection to vessel head are acceptable.

5.4.15.3 Safety Evaluation

If one single active failure prevents a venting operation through one flow path, the redundant path is available for venting. The two isolation valves in each flow path provide a similar method of isolating the venting system. With two valves in series, the failure of any one valve or power supply will not inadvertently open a vent path or prevent opening and closing a flow path. Thus, the combination of safety grade train assignments and valve failure modes will not prevent vessel head venting nor venting isolation with any single active failure.

The RVHVS has two normally deenergized valves in series in each flow path. This arrangement eliminates the possibility of an opened flow path due to the spurious movement of one valve. As such, power lockout to any valve is not considered necessary.

A break of the RVHVS line would result in a small loss-of-coolant accident (LOCA) of not greater than 1-in. diameter. Such a break is similar to those analyzed in WCAP-9600 (1979). Since a break in the head vent line would behave similarly to the hot leg break case presented in WCAP-9600, the results presented therein are applicable to a RVHVS line break. Therefore, this postulated vent line break results in no calculated core uncovery.

5.4.15.4 Inspection and Testing Requirements

Inservice inspection is conducted in accordance with section 6.6.

5.4.15.5 Instrumentation Requirements

The system is operated from the control room or the shutdown panels. The isolation valves have stem position switches. The position indication from each valve is monitored at the control room by status lights.

TABLE 5.4.1-1 (SHEET 1 OF 2)

REACTOR COOLANT PUMP DESIGN PARAMETERS

Unit design pressure (psig)	2485
Unit design temperature (°F)	650[a]
Unit overall height (ft)	27.4
Seal water injection (gal/min)	8
Seal water return (gal/min)	3
Component cooling waterflow (gal/min)	596
Maximum continuous component cooling water inlet temperature (°F)	105
Total weight, dry (lb)	201,300
Pump	
Design flow (gal/min)	100,600
Developed head (ft)	288
NPSH required (ft)	Figure 5.4.1-2
Suction temperature, thermal design (°F)	558.2
Pump discharge nozzle, inside diameter (in.)	27-1/2
Pump suction nozzle, inside diameter (in.)	31
Speed (rpm)	1187
Water volume (ft^3)	80[b]
Motor	
Type	Drip-proof squirrel cage induction, with water/air coolers
Power (hp)	7000
Voltage (V)	13,200
Phase	3
Frequency (Hz)	60
Insulation class	Class F thermalastic epoxy insulation
Current (amp)	
Starting	1750 at 13,200 V
Nominal input, hot reactor coolant	253
Nominal input, cold reactor coolant	336

TABLE 5.4.1-1 (SHEET 2 OF 2)

Pump moment of inertia, maximum (lb/ft^2)	
Flywheel	70,000
Motor	22,500
Shaft	520
Impeller	1980

a. Design temperature of pressure-retaining parts of the pump assembly exposed to the reactor coolant and injection water on the high-pressure side of the controlled leakage seal is that temperature determined for the parts for a reactor coolant loop temperature of 650°F.

b. Composed of reactor coolant in the casing and of seal injection and cooling water in the thermal barrier.

TABLE 5.4.1-2

REACTOR COOLANT PUMP QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]
Castings	Yes		Yes	
Forgings				
Main shaft		Yes	Yes	
Main studs		Yes		Yes
Plate				
Flywheel		Yes	Yes[b]	Yes[b]
Weldments				
Circumferential	Yes		Yes	
Instrument connections			Yes	

a. RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.

b. Of machined bores keyways and drilled holes (either PT or MT).

TABLE 5.4.2-1

STEAM GENERATOR DESIGN DATA

Design pressure, reactor coolant side (psig)	2485
Design pressure, steam side (psig)	1185
Design pressure, primary to secondary (psi)	1600
Design temperature, reactor coolant side (°F)	650
Design temperature, steam side (°F)	600
Design temperature, primary to secondary (°F)	650
Total heat transfer surface area (ft^2)	55,000
Maximum moisture carryover (weight percent)	0.25
Overall height (ft-in.)	67-8
Number of U-tubes	5626
U-tube nominal diameter (in.)	0.688
Tube-wall nominal thickness (in.)	0.040
Number of manways	4
Inside diameter of manways (in.)	16
Number of handholes	8
Design fouling factor (ft^2-h-°F/Btu)	0.00006
Steamflow (lb/h)	4.055×10^6 to 4.080×10^6

TABLE 5.4.2-2 (SHEET 1 OF 2)

STEAM GENERATOR QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]	ET[a]
Tube Sheet					
Forging		Yes		Yes	
Cladding		Yes[b]	Yes		
Channel Head (if fabricated)					
Fabrication	Yes[c]	Yes[d]		Yes	
Cladding			Yes		
Secondary Shell and Head					
Plates		Yes			
Tubes		Yes			Yes
Nozzles (Forgings)		Yes		Yes	
Weldments					
Shell, longitudinal	Yes			Yes	
Shell, circumferential	Yes			Yes	
Cladding (channel head-tube sheet joint cladding restoration)			Yes		
Primary nozzles to fab head	Yes			Yes	
Manways to fab head	Yes			Yes	
Steam and feedwater nozzles to shell	Yes			Yes	
Support brackets			Yes		
Tube to tube sheet			Yes		
Instrument connections (primary and secondary)				Yes	
Temporary attachments after removal				Yes	

TABLE 5.4.2-2 (SHEET 2 OF 2)

	RT[a]	UT[a]	PT[a]	MT[a]	ET[a]
After hydrostatic test (all major pressure boundary welds and complete cast channel head - where accessible)				Yes	
					Yes
Nozzle safe ends (if weld deposit)	Yes		Yes		

a. RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.
ET - Eddy current.

b. Flat surfaces only.

c. Weld deposit.

d. Base material only.

TABLE 5.4.3-1

REACTOR COOLANT PIPING DESIGN PARAMETERS

Reactor Inlet Piping Inside diameter (ID) (in.)	27 1/2
Reactor Inlet Piping Nominal wall thickness (in.)	2.32
Reactor Outlet Piping ID (in.)	29
Reactor Outlet Piping Nominal wall thickness (in.)	2.45
Coolant Pump Suction Piping ID (in.)	31
Coolant Pump Suction Piping Nominal wall thickness (in.)	2.60
Pressurizer Surge Line Piping Nominal pipe size (in.)	16 reduced to 14
Pressurizer Surge Line Piping	
Nominal wall thickness, for 16 in. nominal pipe size, schedule 160 (in.)	1.594
Nominal wall thickness, for 14 in. nominal pipe size, schedule 160 (in.)	1.406
Nominal Water Volume, all four loops including surge line (ft^3)	1200-1300
RCL Piping	
Design/operating pressure (psig)	2485/2235
Design temperature (°F)	650
Pressurizer Surge Line	
Design pressure (psig)	2485
Design temperature (°F)	680
Pressurizer Safety Valve Inlet Line	
Design pressure (psig)	2485
Design temperature (°F)	680
Pressurizer Power-Operated Relief Valve Inlet Line	
Design pressure (psig)	2485
Design temperature (°F)	680

TABLE 5.4.3-2

REACTOR COOLANT PIPING QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]
Fittings and Pipe (Castings)	Yes		Yes
Fittings and Pipe (Forgings)		Yes	Yes
Weldments			
Circumferential	Yes		Yes
Nozzle to runpipe (except no RT for nozzles less than 6 in.)	Yes		Yes
Instrument connections			Yes
Castings	Yes		Yes (after finishing)
Forgings			Yes (after finishing)

a. RT - Radiographic; UT - ultrasonic; PT - dye penetrant.

TABLE 5.4.7-1

DESIGN BASES FOR RHRS OPERATION

RHRS initiation, hours after reactor shutdown	~4
RCS initial pressure (psig)	~365
RCS initial temperature (°F)	~350
CCW design temperature (°F)	105[(a)]
Cooldown time, hours after initiation of RHRS operation (two trains in service)	~16[(b)]
RCS temperature at end of cooldown (°F)	140[(b)]
Decay heat generation at 20 h after reactor shutdown (Btu/h)	80.4×10^6
Residual heat removal heat exchanger UA	2.5×10^6 Btu/h°F
Fouling factor	0.0003 ft^2h°F/Btu

a. The maximum CCW temperature during cooldown with offsite power available is 120°F.

b. The design bases for cold shutdown are 32 h (one train) to 200°F.

TABLE 5.4.7-2

RHRS COMPONENT DATA

RHR Pumps

Number	2
Design pressure (psig)	600
Design temperature (°F)	400
Design flow (gal/min)	3000
Design head (ft)	375
Material	Austenitic stainless steel

Residual Heat Exchangers

Number		2
Design heat removal capacity (Btu/h)		32.8×10^6
Estimated UAF_{LMTD} (Btu/h)		2.5×10^6
	Tube Side	Shell Side
Design pressure (psig)	600	150
Design temperature (°F)	400	200
Design flow (lb/h)	1.48×10^6	2.48×10^6
Inlet temperature (°F)	140	105
Outlet temperature (°F)	117.8	118.2
Material	Austenitic stainless steel	Carbon steel
Fluid	Reactor coolant	CCW

TABLE 5.4.7-3 (SHEET 1 OF 5)

FAILURE MODES AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM
ACTIVE COMPONENTS - NORMAL COOLDOWN OPERATION

Component	Failure Mode	Effect on System Operation[a]	Failure Detection Method[b]	Remarks
1. Motor-operated gate valve HV-8701A (HV-8701B analogous).	a. Fails to open on demand (open manual mode CB switch selection).	Failure blocks reactor coolant flow from hot leg of RC loop 1 through train A of RHRS. Fault reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements will be met by reactor coolant flow from hot leg of RC loop 4 flowing through train B of RHRS. However, time required to reduce RCS temperature will be extended.	Valve position indication (closed to open position change) at CB; RC loop 1 or 4 hot leg pressure indication at CB; RHR train A discharge flow indication; and RHR pump discharge pressure indication at CB.	Valve is electrically interlocked with the containment sump suction valve HV-8811A, with RWST isolation valve HV-8812A, with RHR to charging pump suction isolation valve HV-8804A and with a "prevent-open" pressure interlock PT-438 (PT-408). The valve cannot be opened remotely from the CB if one of the indicated isolation valves is open or if RC loop pressure exceeds 365 psig. If both trains of RHRS are unavailable for plant cooldown due to multiple component failures, the auxiliary feedwater system and SG power-operated relief valves can be used to perform the safety function of removing residual heat.
2. Motor-operated gate valve HV-8702A (HV-8702B analogous).	Same failure modes as those stated for item 1.	Same effect on system operation as that stated for item 1.	Same methods of detection as those stated for item 1.	Same remarks as those stated for item 1, except for pressure interlock PT-418 (PT-428) control.

TABLE 5.4.7-3 (SHEET 2 OF 5)

Component	Failure Mode	Effect on System Operation	Failure Detection Method	Remarks
3. RHR pump 1, (RHR pump 2 analogous).	Fails to deliver working fluid.	Failure results in loss of reactor coolant flow from hot leg of RC loop 1 through train A of RHRS. Fault reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg or RC loop 4 flowing through train B of RHRS. However, time required to reduce RCS temperature will be extended.	Open pump switchgear circuit breaker indication at CB; circuit breaker close position monitor light for group monitoring of components at CB; common breaker trip alarm at CB; RC loop 1 hot leg pressure indication at CB; RHR train A discharge flow indication and low flow alarm at CB; and pump discharge pressure indication at CB.	The RHRS shares components with the ECCS. Pumps are tested as part of the ECCS testing program. (See subsection 6.3.4.) Pump failure may also be detected during ECCS testing.
4. Motor-operated globe valve FV-610 (FV-611 analogous).	a. Fails to open on demand (open manual mode CB switch selection).	Failure blocks miniflow line to suction of RHR pump 1 during cooldown operation. No effect on safety for system operation. Operator may establish miniflow for RHR pump 1 operation by opening of CVCS letdown control valve HCV-128 and manual valve 1205-U4-021 to allow flow to CVCS.	Valve position indication (closed to open position change) at CB.	Valve is automatically controlled to open when pump discharge is less than the open setpoint (824 gpm at 350°F, 780 gpm at 100°F) and close when the discharge exceeds the closed setpoint (1944 gpm at 350°F, 1841 gpm at 100°F). The valve protects the pump from deadheading during ECCS operation CB switch set to "Auto" position for automatic control of valve positioning.
	b. Fails to close on demand (auto mode CB switch selection).	Failure allows for a portion of RHR heat exchanger discharge flow to be bypassed to suction of RHR pump 1. RHRS train A is degraded for the regulation of coolant temperature by RHR heat exchanger 1. No effect on safety for system operation. Cooldown of RCS within established specification cooldown rate may be accomplished through operator action of throttling flow control valve HCV-606 and controlling cooldown with redundant RHRS train B.	Valve position indication (open to closed position change) and RHRS train A discharge flow indication at CB.	

TABLE 5.4.7-3 (SHEET 3 OF 5)

Component	Failure Mode	Effect on System Operation	Failure Detection Method	Remarks
5. Air diaphragm-operated butterfly valve FCV-618 (FCV-619 analogous).	a. Fails to open on demand (Auto mode CB switch selection).	Failure prevents coolant discharged from RHR pump 1 from bypassing RHR heat exchanger 1resulting in mixed mean temperature of coolant flow to RCS being low. RHRS train A is degraded for the regulation of controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve HCV-606 and controlling cooldown with redundant RHRS train B.	RHR pump 1 discharge flow temperature and RHRS train A discharge to RCS cold leg flow temperature recorded on the plant computer; and RHRS train A discharge to RCS cold leg flow indication at CB.	Valve is designed to fail closed and is electrically wired so that electrical solenoid of the air diaphragm operator is energized to open the valve. Valve is normally closed to align RHRS for ECCS operation during plant power operation and load follow.
	b. Fails to close on demand (Auto mode CB switch selection).	Failure allows coolant discharge from RHR pump 1 to bypass RHR heat exchanger 1, resulting in mixed mean temperature of coolant flow to RCS being high. RHRS train A is degraded for the regulation of controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve HCV-606 and controlling cooldown with redundant RHRS train B. However, cooldown time will be extended.	Same methods of detection as those stated for item 5.a.	

TABLE 5.4.7-3 (SHEET 4 OF 5)

Component	Failure Mode	Effect on System Operation	Failure Detection Method	Remarks
6. Air diaphragm-operated butterfly valve HCV-606 (HCV-607 analogous).	a. Fails to close on demand for flow reduction.	Failure prevents control of coolant discharge flow from RHR heat exchanger 1, resulting in loss of mixed mean temperature coolant flow adjustment to RCS. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished by operator action of controlling cooldown with redundant RHRS train B.	Same methods of detection as those stated for item 5. In addition, monitor light and alarm (valve closed) for group monitoring of components at CB.	Valve is designed to fail open. Valve is normally open to align RHRS for ECCS operation during plant power operation and load follow.
	b. Fails to open on demand for increased flow.	Same effect on system operation as that stated for item 6.a.	Same methods of detection as those stated for item 6.a.	
7. Motor-operated gate valve HV-8812A (HV-8812B analogous).	Fails to close on demand.	Failure reduces the redundancy of isolation valves provided to isolate RHRS train A from RWST. No effect on safety for system operation. Check valve in series with motor-operated valve provides the primary isolation against the bypass of RCS coolant flow from the suction of RHR pump A to RWST.	Valve position indication (open to closed position change) at CB and valve (closed) monitor light and alarm at CB.	Valve is a component of the ECCS that performs an RHR function during plant cooldown. Valve is normally open to align RHRS for ECCS operation during plant power operation and load follow.

TABLE 5.4.7-3 (SHEET 5 OF 5)

Component	Failure Mode	Effect on System Operation	Failure Detection Method	Remarks
8. Motor-operated gate valve HV-8716A (HV-8716B analogous).	Fails to close on demand.	Failure reduces the redundancy of isolating, train A from train B for single train operation of RHRS. No effect on safety for system operation. Isolation will be provided by closing valve HV-8716B (HV-8716A).	Same as item 7.	

a. List of acronyms and abbreviations.

Auto - Automatic.
CB - Main control board.
CVCS - Chemical and volume control system.
ECCS - Emergency core cooling system.
RC - Reactor coolant.
RCS - Reactor coolant system.
RHR - Residual heat removal.
RHRS - Residual heat removal system.
RWST - Refueling water storage tank.
SG - Steam generator.

b. As part of plant operation; periodic tests, surveillance inspections, and instrument calibrations are conducted to monitor equipment and performance. Failures may be detected during such monitoring of equipment, in addition to detection methods noted.

TABLE 5.4.7-4 (SHEET 1 OF 5)

RESIDUAL HEAT REMOVAL SYSTEM - SAFETY GRADE COLD SHUTDOWN OPERATIONS - FAILURE MODES AND EFFECTS ANALYSIS

Component[a]	Failure Mode	Function[b]	Effect on System Operation	Failure Detection Methods[c]	Remarks
1. Motor-operated gate valve HV-8701A (HV-8701B analogous).	a. Fails to open on demand.	Provides isolation of fluid flow from the RCS to the suction of RHR pump 1.	a. Failure blocks reactor coolant flow from hot leg of RC loop 1 through train A of RHRS. Failure reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg of RC loop 4 flowing through train B of RHRS, however time required to reduce RCS temperature is extended.	a. Valve open/close position indication at CB; RC loop 1 or 4 hot leg pressure indication at CB; RHR train A discharge flow indication, and RHR pump 1 discharge pressure indication at CB.	Valve is electrically interlocked with a containment suction valve HV-8811A RWST to RHR suction line isolation valve HV-8812A, with RHR to charging pump suction line isolation valve HV-8804A and with a "prevent-open" pressure interlock PT-438 (PT-408). The valve can not be opened remotely from the CB if one of the indicated isolation valves is open or if RC loop pressure exceeds 365 psig. The valve can be manually opened.
	b. Once the valves are open the main control board annunciator alarm fails and RCS pressure exceeds 420 psig.		b. Failure reduces redundancy of main control board annunciator alarm at 420 psig. No effect on safety system operation. Plant operating procedures require that operators close both valves prior to an RCS pressure of 420 psig. Alternate alarm is provided by valve HV-8701B (HV-8701A).	b. Valve is interlocked with pressure interlock PT-438 (PT-408) to alarm on the main control board annunciator panel if one or both of the valves is not fully closed and RCS pressure exceeds 420 psig.	

TABLE 5.4.7-4 (SHEET 2 OF 5)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Methods	Remarks
2. Motor-operated gate valve HV-8702A (HV-8702B analogous).	Same as item 1.	Same as item 1.	Same as item 1.	Same as item 1.	Same as item 1, except for pressure interlock PT418 (PT428) control.
3. RHR pump 1 (RHR pump 2 analogous).	Fails to deliver working fluid.	Provides fluid flow of reactor coolant through RHR heat exchanger 1 to reduce RCS temperature during cooldown operation.	Failure results in loss of reactor coolant flow from hot leg of RC loop 1 through train A of RHRS. Failure reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg of RC loop 4 flowing through train B of RHRS, however, time required to reduce RCS temperature is extended.	Open pump switchgear circuit breaker indication at CB; circuit breaker close position monitor light for group monitoring of components at CB; common breaker trip alarm at CB; RC loop 1 hot leg pressure indication at CB; RHR train A discharge flow indication and low flow alarm at CB; and pump discharge pressure indication at CB.	The RHRS shares components with the ECCS. Pumps are tested as part of the ECCS testing program. (See subsection 6.3.4.)
4. Motor-operated globe valve FV-610 (FV-611 analogous).	a. Fails closed.	Provides regulation of fluid flow through mini-flow bypass line to suction of RHR pump 1 to protect against overheating of the pump and loss of discharge flow from the pump.	a. Failure blocks mini-flow line to suction of RHR pump 1 during cooldown operation. No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg of RC loop 4 flowing through train B of RHRS. However, time required to reduce RCS temperature is extended.	a. Valve open/close position indication at CB; and RHRS train A discharge flow indication at CB.	Valve is automatically controlled to open when pump discharge is less than the open setpoint (824 gpm at 350°F, 780 gpm at 100°F) and close when the discharge exceeds the closed setpoint (1944 gpm at 350°F, 1841 gpm at 100°F).

TABLE 5.4.7-4 (SHEET 3 OF 5)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Methods	Remarks
	b. Fails open.		b. Failure allows for a portion of RHR heat exchanger 1 discharge flow to be bypassed to suction of RHR pump 1. RHRS train A is degraded for the regulation of coolant temperature by RHR heat exchanger 1. No effect on safety for system operation. Cool-down of RCS remains within established specification cooldown rate.	Same as item 4.a.	
5. Air - diaphragm-operated - butterfly valve FCV-618 (FCV-619 analogous).	a. Fails to open on demand for flow increase ("Auto" mode CB switch selection).	Controls rate of fluid flow bypassed around RHR heat exchanger 1 during cooldown operation.	a. Failure prevents cool ant discharged from RHR pump 1 from bypassing RHR heat exchanger 1 resulting in mixed mean temperature of coolant flow to RCS being low. RHRS train A is degraded for the regulation of controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve HCV-606 and controlling cooldown with redundant RHRS train B.	a. RHR pump 1 discharge flow temperature and RHRS train A discharge to RCS cold leg flow temperature recorded on the plant computer; and RHRS train A discharge to RCS cold leg flow indication at CB.	Valve is designed to fail closed and is electrically wired so that electrical solenoid of the air diaphragm operator is energized to open the valve. Valve is normally closed to align RHRS for ECCS operation during plant power operation and load follow. Valve is designed for normal plant cooldown operation. It is not required for safety grade cold shutdown operations.

TABLE 5.4.7-4 (SHEET 4 OF 5)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Methods	Remarks
	b. Fails to close on demand for flow reduction (Auto mode CB switch selection).		b. Failure allows coolant discharged from RHR pump 1 to bypass RHR heat exchanger 1 resulting in mixed mean temperature of coolant flow to RCS being high. RHRS train A is degraded for the regulation controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve HCV-606 and controlling cooldown with redundant RHRS train B; however, cooldown time is extended.	b. Same as item 5.a.	
6. Air diaphragm-operated butterfly valve HCV-606 (HCV-607 analogous).	a. Fails to close on demand for flow reduction.	Controls rate of fluid flow through RHR heat exchanger 1 during cool-down operation.	a. Failure prevents control of coolant discharge flow from RHR heat exchanger 1 resulting in loss of mixed mean temperature coolant flow adjustment to RCS. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished by operator action of controlling cooldown with redundant RHS train B.	a. Same methods of detections as those stated for item 5.a. addition, monitor light and alarm (valve closed) for group monitoring of components at CB.	Valve is designed to fail open. Valve is normally open to align RHRS for ECCS operation during plant power operation and load follow.
	b. Fails to open on demand for flow increase.		b. Same as item 6.a.	b. Same as item 6.a.	

TABLE 5.4.7-4 (SHEET 5 OF 5)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Methods	Remarks
7. Motor-operated gate valve HV-8812A (HV-8812B analogous).	Fails to close on demand.	Provides isolation of fluid from the RWST to suction of RHR pump 1 during cooldown operation.	No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg loop 4 flowing through train B of RHRS; however, time required to reduce RCS temperature is extended.	Valve open/closed position indication at CB and valve (closed) monitor light and alarm at CB.	Valve is normally open to align RHRS for ECCS operation during plant power operation and load follow. Valve must be closed during plant cooldown to satisfy electrical interlock to permit valves HV-8701A and B (HV-8702A, B) to be opened.
8. Motor-operated gate valve HV-8716A (HV-8716B analogous).	Fails to close on demand.	Provides separation between the two RHR trains during cooldown operation.	Failure reduces the redundancy for isolating RHR trains during cooldown. Negligible effect on system operation. Isolation valve HV-8716B (HV-8716A) provides backup isolation between the two RHR trains.	Same as item 7.	

a. Component 7 is a component of the ECCS that performs a safety-grade cold shutdown function.

b. List of acronyms and abbreviations.

Auto	-	Automatic.	HELB	-	High-energy line break.	RWST	-	Refueling water storage tank.
BAT	-	Boric acid tank.	MELB	-	Moderate-energy line break.	RV	-	Reactor vessel.
BIT	-	Boron injection tank (Unit 1 only).	PRT	-	Pressurizer relief tank.	SI	-	Safety injection.
CB	-	Main control board.	RC	-	Reactor coolant.	VCT	-	Volume control tank.
CVCS	-	Chemical and volume control system.	RCS	-	Reactor coolant system.			
ECCS	-	Emergency core cooling system.	RHR	-	Residual heat removal.			
			RHRS	-	Residual heat removal system.			

c. As part of plant operation, periodic tests, surveillance inspections, and instrument calibrations are made to monitor equipment and performance. Failures may be detected during such monitoring of equipment in addition to detection methods noted.

TABLE 5.4.10-1

PRESSURIZER DESIGN DATA

Design pressure (psig)	2485
Design temperature (°F)	680
Surge line nozzle diameter (in.)	14
Heatup rate of pressurizer using heaters only (°F/h)	55
Internal volume (ft^3)	1800
Nominal conditions at 100-percent rated load	
Steam volume (ft^3)	720
Water volume (ft^3)	1080

TABLE 5.4.10-2

REACTOR COOLANT SYSTEM DESIGN PRESSURE SETTINGS

	Psig
Hydrostatic test pressure	3106
Design pressure	2485
Safety valves (begin to open)	2460
High pressure reactor trip	2385
High pressure alarm	2310
Power-operated relief valves	
PV-0455A	2345[(a)]
PV-0456A	2335[(b)]
Pressurizer spray valves (full open)	2310
Pressurizer spray valves (begin to open)	2260
Proportional heaters (begin to operate)	2250
Operating pressure	2235
Proportional heater (full operation)	2220
Backup heaters on	2210
Low pressure alarm	2210
Pressurizer power-operated relief valve interlock	2185
Low pressure reactor trip	1870

a. At 2345 psi, a pressure signal initiates actuation (opening) of this valve. Remote manual control is also provided.

b. At 2335 psi, a pressure signal initiates actuation (opening) of this valve. Remote manual control is also provided.

TABLE 5.4.10-3

PRESSURIZER QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]
Heads				
Plates		Yes		
Cladding			Yes	
Shell				
Plates		Yes		
Cladding			Yes	
Heaters				
Tubing		Yes[b]	Yes	
Centering of element	Yes			
Nozzle (Forgings)		Yes	Yes[c]	Yes[c]
Weldments				
Shell,longitudinal	Yes			Yes
Shell, circumferential	Yes			Yes
Cladding			Yes	
Nozzle safe end	Yes		Yes	
Instrument connection			Yes	
Support skirt, longitudinal seam	Yes			Yes
Support skirt to lower head		Yes		Yes
Temporary attachments (after removal)				Yes
All external pressure boundary welds after shop hydrostatic tests				Yes

a. RT - Radiographic.
UT - Ultrasonic.
PT - Dye Penetrant.
MT - Magnetic Particle.

b. Eddy current testing can be used in lieu of UT.

c. MT or PT.

TABLE 5.4.11-1

PRESSURIZER RELIEF TANK DESIGN DATA

Design pressure (psig)	100
Normal operating pressure (psig)	3
Final operating pressure (psig) [(a)]	50
Rupture disc release pressure (psig)	
Nominal	91
Range	86 to 100
Normal water volume (ft^3)	1350
Normal gas volume (ft^3)	450
Design temperature (°F)	340
Initial operating water temperature (°F) [(a)]	120
Final operating water temperature (°F) [(a)]	200
Total rupture disc relief capacity at 100 psig (lb/h)	1.6×10^6
Cooling time required following maximum discharge, approximately (h)	
Spray feed and bleed	1
Utilizing RCDT heat exchanger	8

a. For the design basis pressurizer steam discharge.

TABLE 5.4.11-2

DISCHARGES TO THE PRESSURIZER RELIEF TANK

Reactor coolant system (drawings 1X4DB111, 2X4DB111, 1X4DB112, 2X4DB112, and 1X4DB113)

Two pressurizer PORVs
Three pressurizer safety valves
One reactor vessel head vent

Residual heat removal system (drawing 1X4DB122)

Two suction line relief valves from the RCS hot legs

Chemical and volume control system (drawing 1X4DB114)

One seal water return line relief valve
One RCS letdown line relief valve

TABLE 5.4.11-3

PRESSURIZER RELIEF DISCHARGE SYSTEM NONDESTRUCTIVE TESTING PROGRAM

Components	Radiographic	Ultrasonic	Dye Penetrant
Fittings and pipe (castings)	Yes		Yes
Fittings and pipe (forgings)		Yes	Yes
Weldments			
Circumferential	Yes		Yes
Nozzle to runpipe (except no radiographic for nozzles less than 4 in.)	Yes		Yes
Instrument connections			Yes

TABLE 5.4.12-1

REACTOR COOLANT SYSTEM VALVE DESIGN PARAMETERS

Design pressure (psig)	2485
Preoperational plant hydrotest (psig)	3106
Design temperature (°F)	650

TABLE 5.4.12-2

REACTOR COOLANT SYSTEM VALVES NONDESTRUCTIVE EXAMINATION PROGRAM

	RT[a]	UT[a]	PT[a]
Castings			
Larger than 4 in.	Yes		Yes
2 to 4 in.	Yes[b]		Yes
Forgings			
Larger than 4 in.	[c]	[c]	Yes
2 to 4 in.			Yes

a. RT - Radiographic
UT - Ultrasonic
PT - Dye Penetrant

b. Weld ends only.

c. Either RT or UT.

TABLE 5.4.12-3 (SHEET 1 OF 2)

REACTOR COOLANT SYSTEM PRESSURE ISOLATION VALVES

	VALVE NUMBER	VALVE SIZE (in.)	FUNCTION	MAXIMUM ALLOWABLE LEAKAGE(gpm)
1.	HV-8701A	12	RHR Suction (gate valve)	5.0
2.	HV-8701B	12	RHR Suction (gate valve)	5.0
3.	HV-8702A	12	RHR Suction (gate valve)	5.0
4.	HV-8702B	12	RHR Suction (gate valve)	5.0
5.	1204-U4-120	2	SI-Hot Leg 2nd Isolation Valve	1.0
6.	1204-U4-121	2	SI-Hot Leg 2nd Isolation Valve	1.0
7.	1204-U4-122	2	SI-Hot Leg 2nd Isolation Valve	1.0
8.	1204-U4-123	2	SI-Hot Leg 2nd Isolation Valve	1.0
9.	1204-U6-079	10	Accumulator 2nd Isolation Valve	5.0
10.	1204-U6-080	10	Accumulator 2nd Isolation Valve	5.0
11.	1204-U6-081	10	Accumulator 2nd Isolation Valve	5.0
12.	1204-U6-082	10	Accumulator 2nd Isolation Valve	5.0
13.	1204-U6-083	10	Injection Line 1st Isolation Valve	5.0
14.	1204-U6-084	10	Injection Line 1st Isolation Valve	5.0
15.	1204-U6-085	10	Injection Line 1st Isolation Valve	5.0

TABLE 5.4.12-3 (SHEET 2 OF 2)

REACTOR COOLANT SYSTEM PRESSURE ISOLATION VALVES

	VALVE NUMBER	VALVE SIZE (in.)	FUNCTION	MAXIMUM ALLOWABLE LEAKAGE(gpm)
16.	1204-U6-086	10	Injection Line 1st Isolation Valve	5.0
17.	1204-U6-124	6	SI-Hot Leg 1st Isolation Valve	3.0
18.	1204-U6-125	6	SI-Hot Leg 1st Isolation Valve	3.0
19.	1204-U6-126	6	SI-Hot Leg 1st Isolation Valve	3.0
20.	1204-U6-127	6	SI-Hot Leg 1st Isolation Valve	3.0
21.	1204-U6-128	8	RHR-Hot Leg 2nd Isolation Valve	4.0
22.	1204-U6-129	8	RHR-Hot Leg 2nd Isolation Valve	4.0
23.	1204-U4-143	2	SI-Cold Leg 2nd Isolation Valve	1.0
24.	1204-U4-144	2	SI-Cold Leg 2nd Isolation Valve	1.0
25.	1204-U4-145	2	SI-Cold Leg 2nd Isolation Valve	1.0
26.	1204-U4-146	2	SI-Cold Leg 2nd Isolation Valve	1.0
27.	1204-U6-147	6	RHR Cold Leg 2nd Isolation Valve	3.0
28.	1204-U6-148	6	RHR Cold Leg 2nd Isolation Valve	3.0
29.	1204-U6-149	6	RHR Cold Leg 2nd Isolation Valve	3.0
30.	1204-U6-150	6	RHR Cold Leg 2nd Isolation Valve	3.0

TABLE 5.4.13-1

PRESSURIZER SAFETY AND RELIEF VALVES DESIGN PARAMETERS

Pressurizer safety valves	
Number	3
Minimum relieving capacity at 2560 psig per valve, ASME flowrate (lb/h)	420,000
Set pressure (psig)	2460
Design temperature (°F)	650
Fluid	Saturated steam
Backpressure	
Normal (psig)	3 to 5
Expected maximum during discharge (psig)	500
Environmental conditions	
Ambient temperature (°F)	50 to 120
Relative humidity (percent)	0 to 100
Pressurizer power-operated relief valves	
Number	2
Design pressure (psig)	2485
Design temperature (°F)	650
Saturated steam-relieving capacity at 2385 psig, per valve (lb/h)	210,000
Saturated water-relieving capacity at 2485 psig, per valve (gal/min)	230

TABLE 5.4.15-1

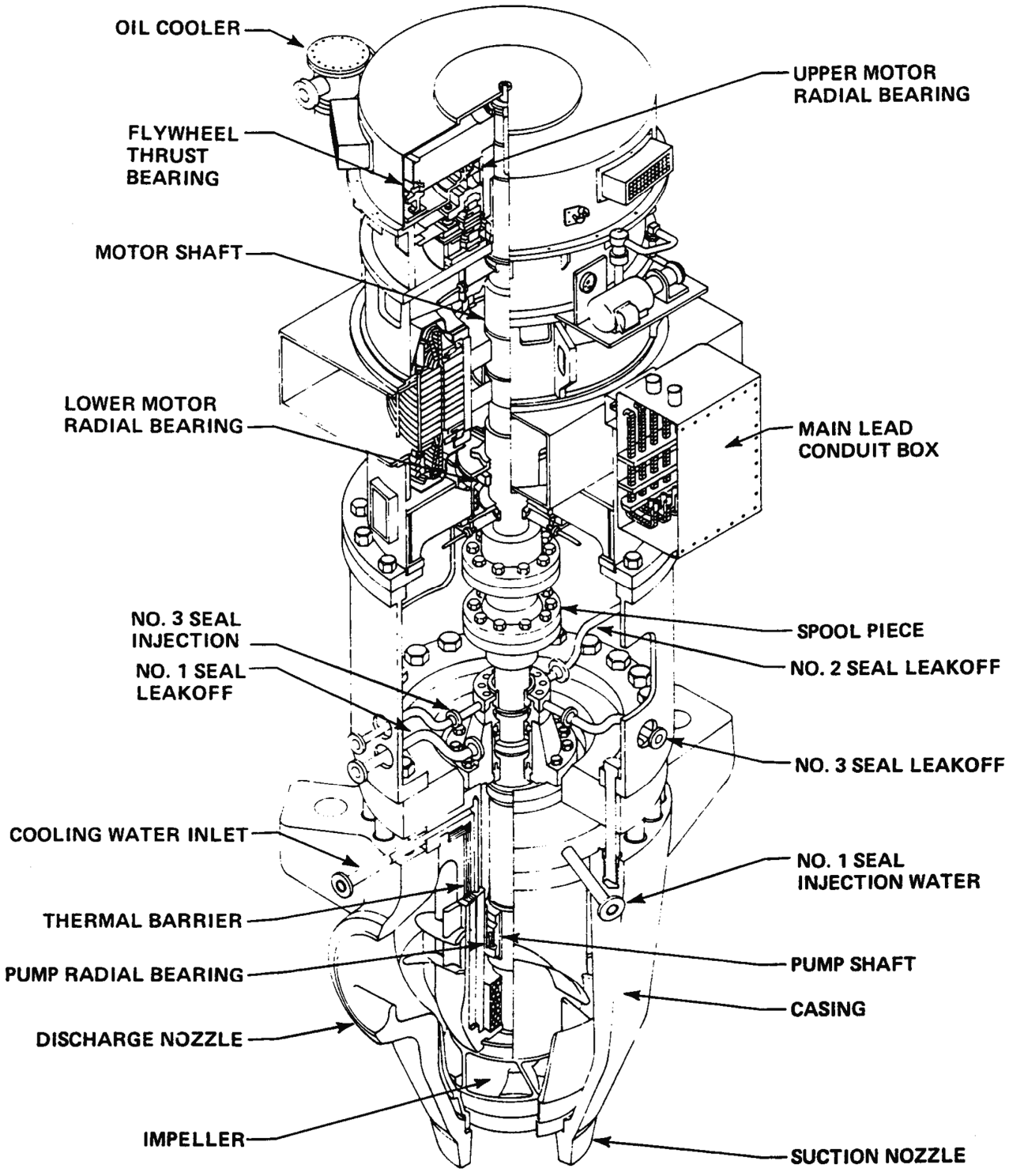
REACTOR VESSEL HEAD VENT SYSTEM EQUIPMENT
DESIGN PARAMETERS

Valves

Number (includes one manual valve)	7
Design pressure, psig	2485
Design temperatures, °F	650

Piping

Vent line, nominal diameter, in.	1
Design pressure, psig	2485
Design temperature, °F	650
Maximum operating temperature	620



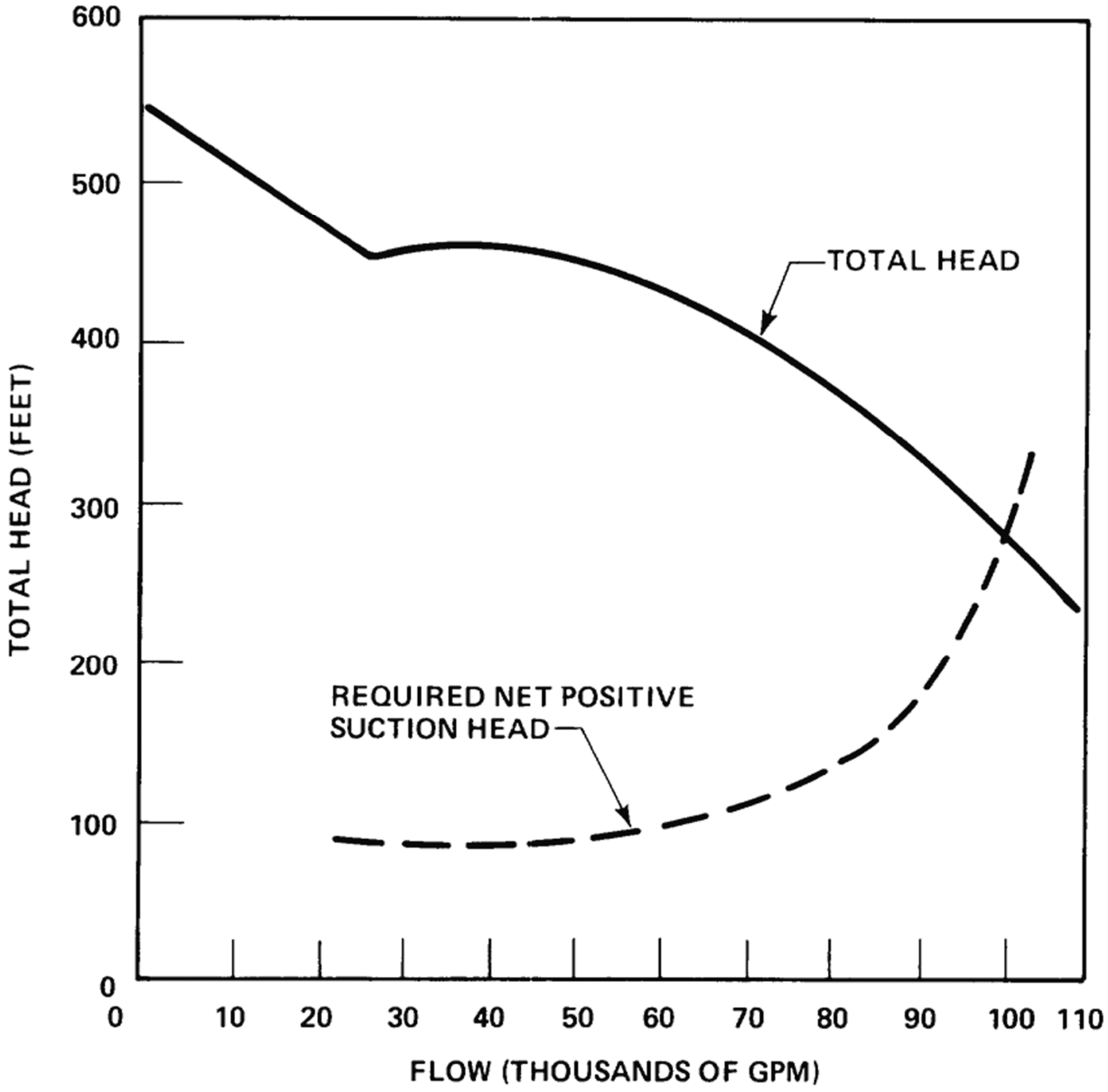
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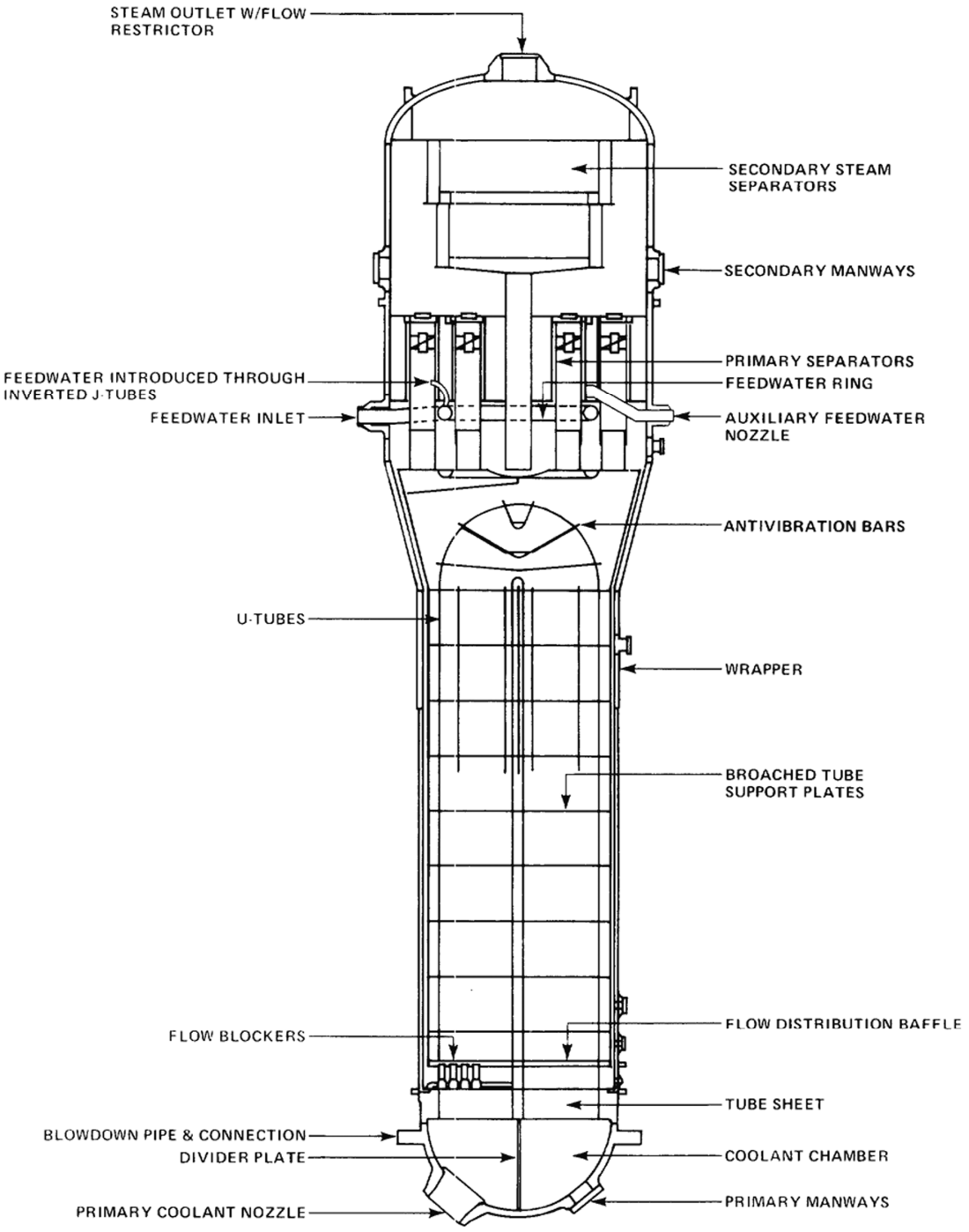
REACTOR COOLANT PUMP

FIGURE 5.4.1–1



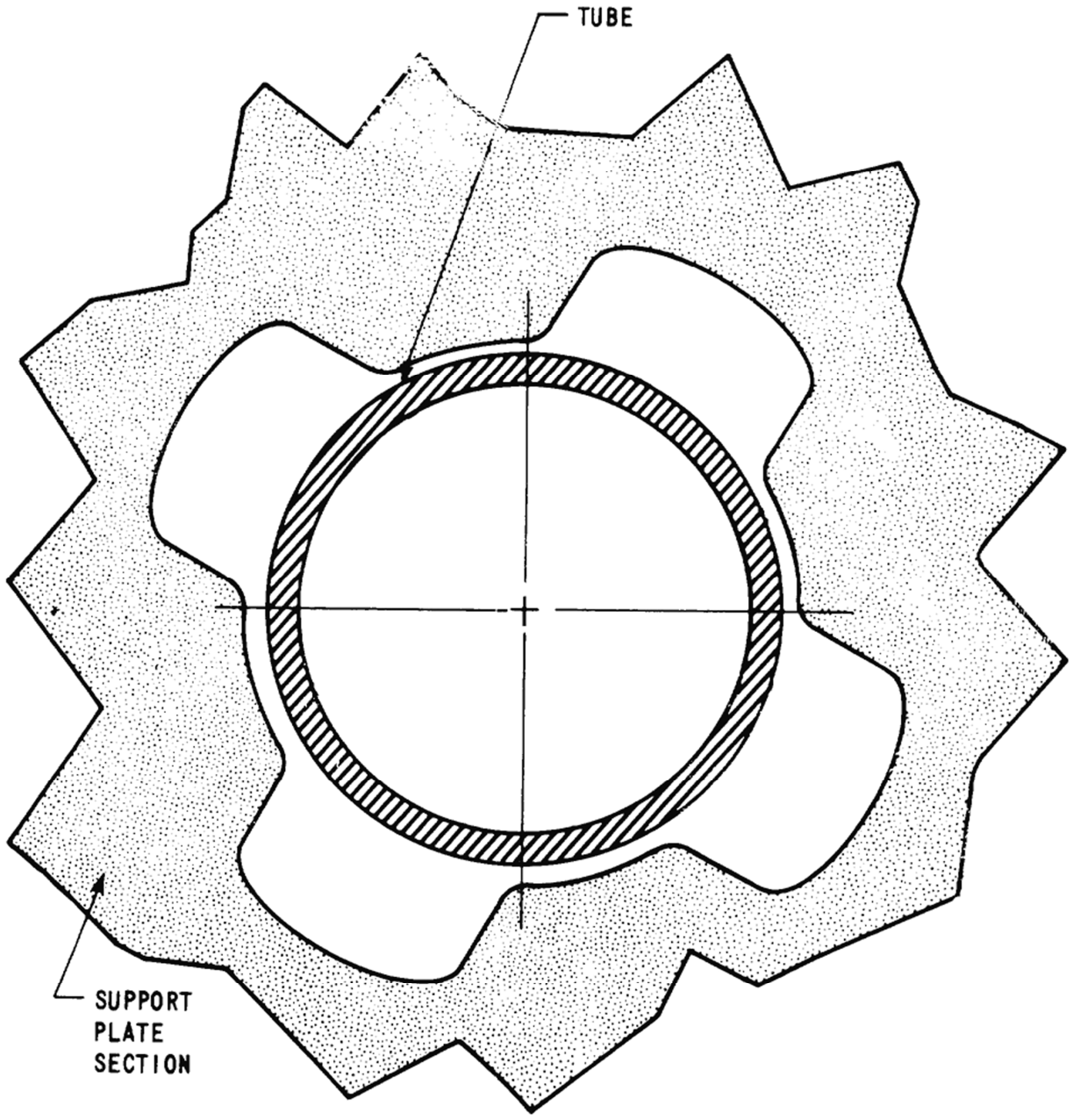
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SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR COOLANT PUMP ESTIMATED PERFORMANCE CHARACTERISTIC FIGURE 5.4.1–2



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	MODEL F STEAM GENERATOR
	FIGURE 5.4.2–1



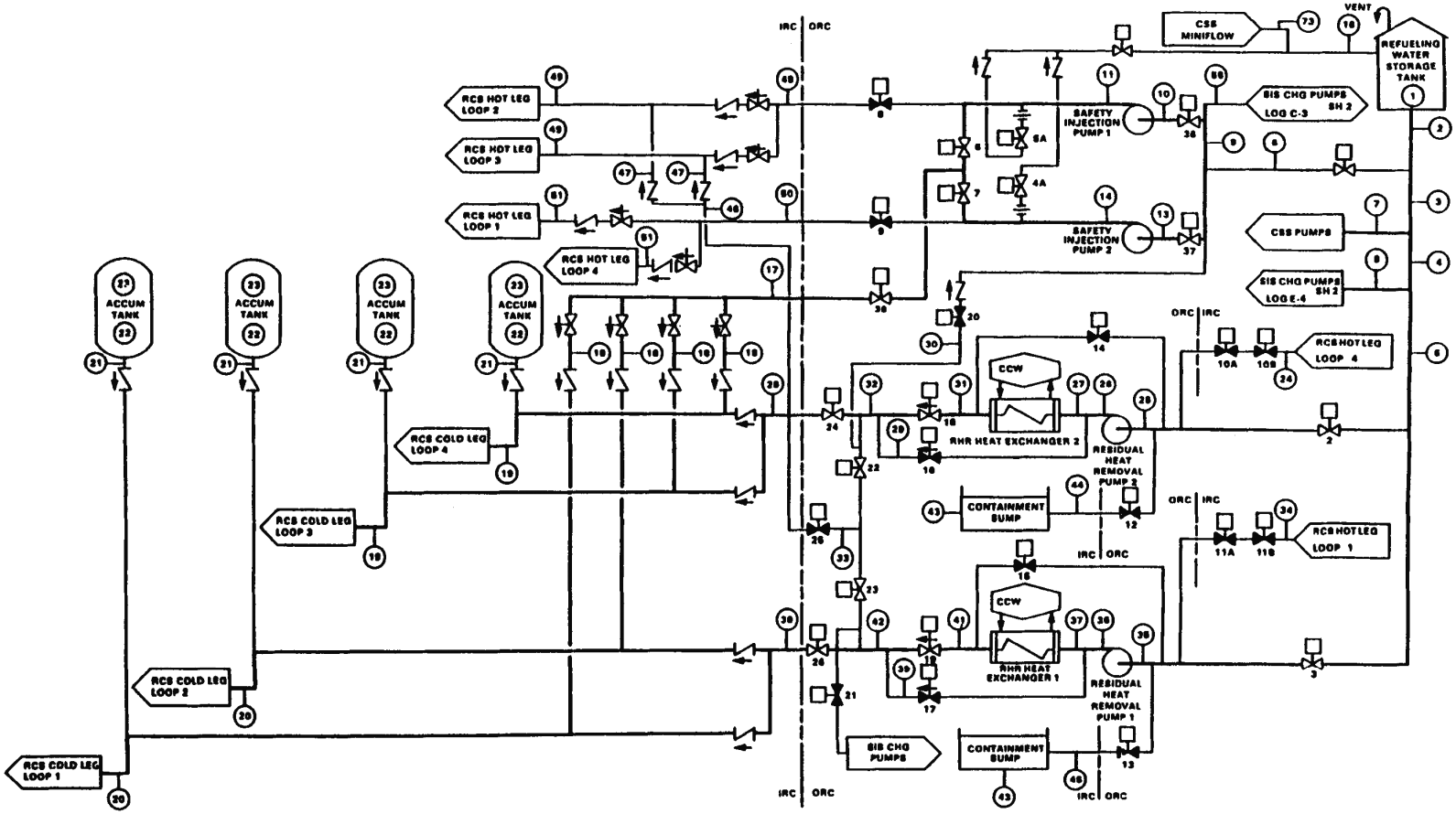
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QUATREFOIL TUBE SUPPORT PLATES

FIGURE 5.4.2–2



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	RHRS PROCESS FLOW DIAGRAM
	FIGURE 5.4.7–1 (SHEET 1 OF 5)

MODES OF OPERATION

MODE A INITIATION OF RHR OPERATION

When the reactor coolant temperature and pressure are reduced to 350°F and 365 psig, approximately 2 to 4 hours after reactor shutdown, the second phase of plant cooldown starts with the RHRS being placed in operation. Before starting the pumps, the inlet isolation valves are opened, the heat exchanger flow control valves are set at minimum flow, and the outlet valves are verified open. The automatic miniflow valves are open and remain so until the pump flow exceeds the closed setpoint (1944 gpm at 350°F, 1841 gpm at 100°F), at which time they trip closed. Should the pump flow drop below the open setpoint (824 gpm at 350°F, 780 gpm at 100°F), the miniflow valves open automatically.

Startup of the RHRS includes a warmup period, during which reactor coolant flow through the heat exchangers is limited to minimize thermal shock on the RCS. The rate of heat removal from the reactor coolant is controlled manually by regulating the reactor coolant flow through the residual heat exchangers. The total flow is regulated automatically by control valves in the heat exchanger bypass line to maintain a constant total flow. The cooldown rate is limited to 100°F/h based on equipment stress limits and a 120°F maximum component cooling water temperature.

MODE B END CONDITIONS OF A NORMAL COOLDOWN

This situation characterizes most of the RHRS operation. As the reactor coolant temperature decreases, the flow through the residual heat exchanger is increased until all of the flow is directed through the heat exchanger to obtain maximum cooling.

NOTE

For the safeguards functions performed by the RHRS, refer to section 6.3.

VALVE ALIGNMENT CHART

Valve No.	Flow Diagram Valve No.	Operational Mode[a] A	B
HV-8812B	2	C	C
HV-8812A	3	C	C
HV-8702A/B	10	O	O
HV-8701A/B	11	O	O
HV-8811B	12	C	C
HV-8811A	13	C	C
FV-0611	14	C	C
FV-0610	15	C	C
FV-0619	16	P	C
FV-0618	17	P	C
HV-0607	18	P	P
HV-0606	19	P	P
HV-8804B	20	C	C
HV-8804A	21	C	C
HV-8716B	22	C	C
HV-8716A	23	C	C
HV-8809B	24	O	O
HV-8809A	26	O	O

a. O = Open.
C = Closed.
P = Partially open.

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MODE A INITIATION OF RHR OPERATION

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)[a]	Flow (lb/h × 10^6)
24	RC	400	350	3000	1.340
25	RC	417	350	3000	1.340
26	RC	565	350	3000	1.340
27	RC	541	350	1657	0.740
31	RC	536	140	1657	0.740
29	RC	492	350	1343	0.600
32	RC	492	234	3000	1.340
28	RC	432	234	3000	1.340
19 Loop 4	RC	400	234	1552	0.680
19 Loop 3	RC	400	234	1478	0.660
34	RC	400	350	3000	1.340
35	RC	418	350	3000	1.340
36	RC	567	350	3000	1.340
37	RC	543	350	1657	0.740
41	RC	536	140	1657	0.740
39	RC	492	350	1343	0.600
42	RC	492	234	3000	1.340
38	RC	435	234	3000	1.340
20 Loop 1	RC	400	234	1487	0.664
20 Loop 2	RC	400	234	1513	0.676

a. At reference conditions 350°F and 400 psig.

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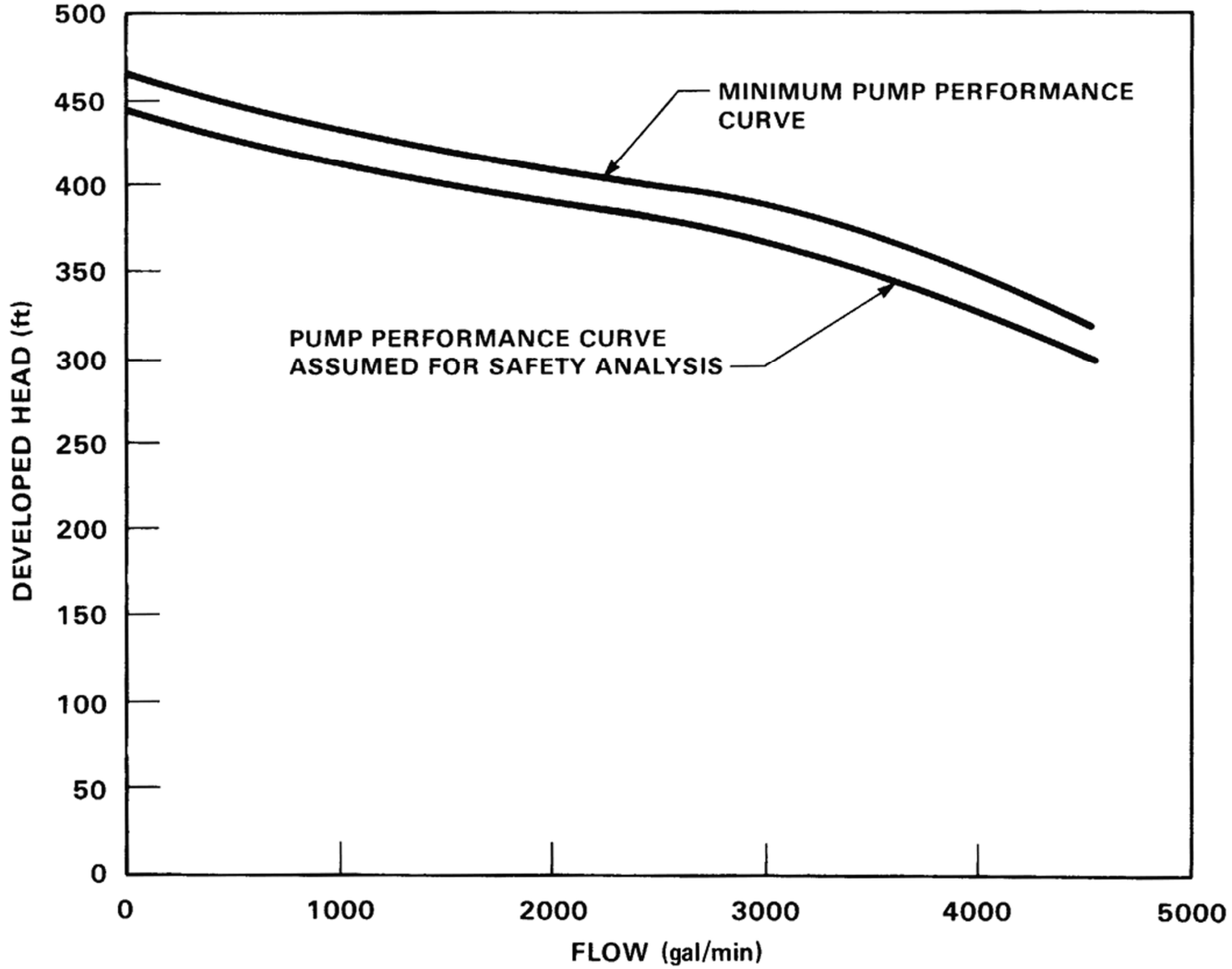
RHRS PROCESS FLOW DIAGRAM

FIGURE 5.4.7–1 (SHEET 4 OF 5)

MODE B END CONDITIONS OF A NORMAL COOLDOWN

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)[a]	Flow (lb/h x 10^6)
24	RC	0	140	3000	1.480
25	RC	19	140	3000	1.480
26	RC	181	140	3000	1.480
27	RC	152	140	3000	1.480
31	RC	134	115	3000	1.480
29	RC	105	115	0	0
32	RC	105	115	3000	1.480
28	RC	38	115	3000	1.480
19 Loop 4	RC	0	115	1570	0.750
19 Loop 3	RC	0	115	1480	0.730
34	RC	0	140	3000	1.480
35	RC	19	140	3000	1.480
36	RC	184	140	3000	1.480
37	RC	154	140	3000	1.480
41	RC	136	115	3000	1.480
39	RC	105	115	0	0
42	RC	105	115	3000	1.480
38	RC	42	115	3000	1.480
20 Loop 1	RC	0	115	1486	0.733
20 Loop 2	RC	0	115	1514	0.747

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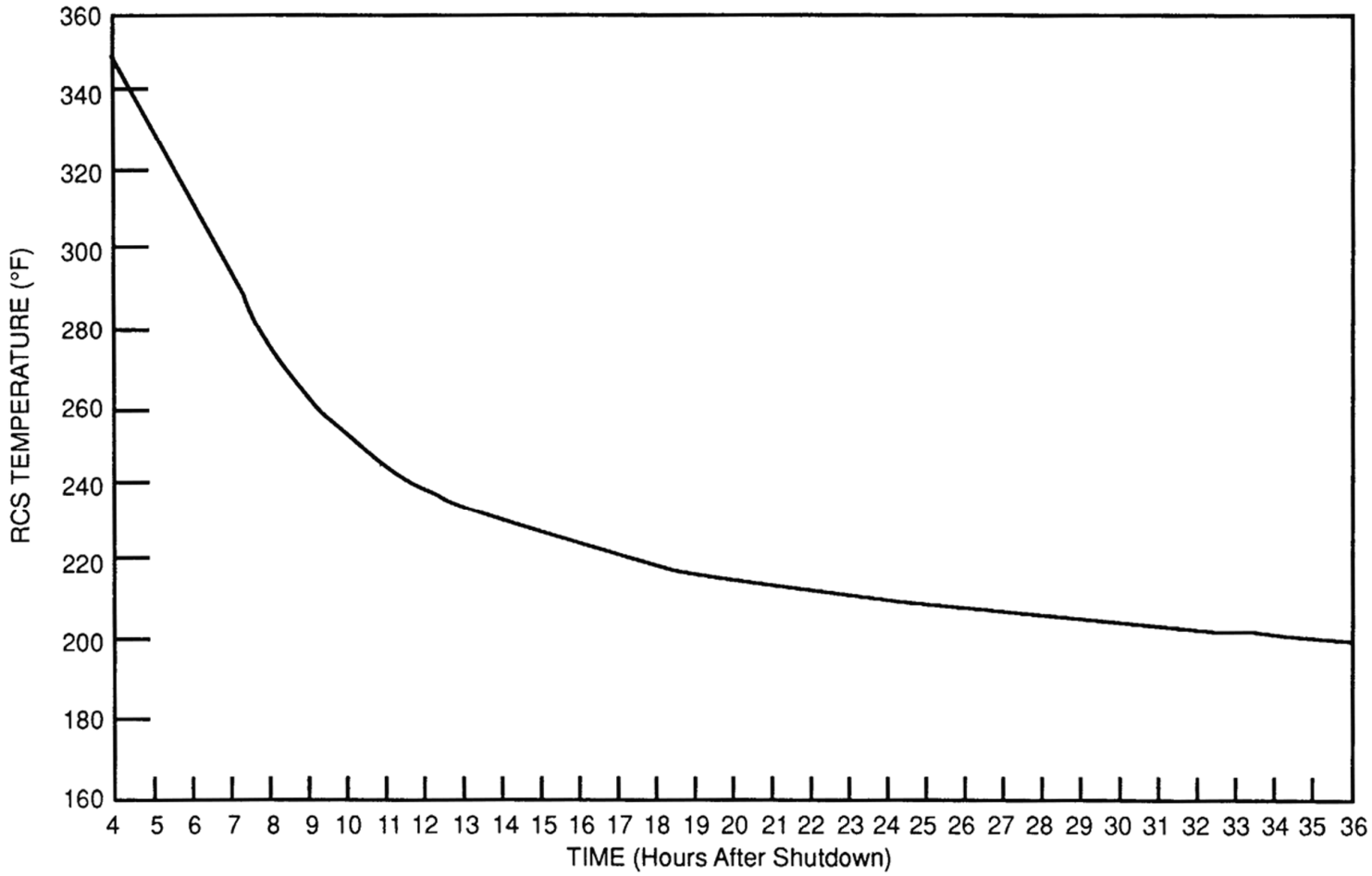
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RHRS PUMP PERFORMANCE CURVE

FIGURE 5.4.7–2



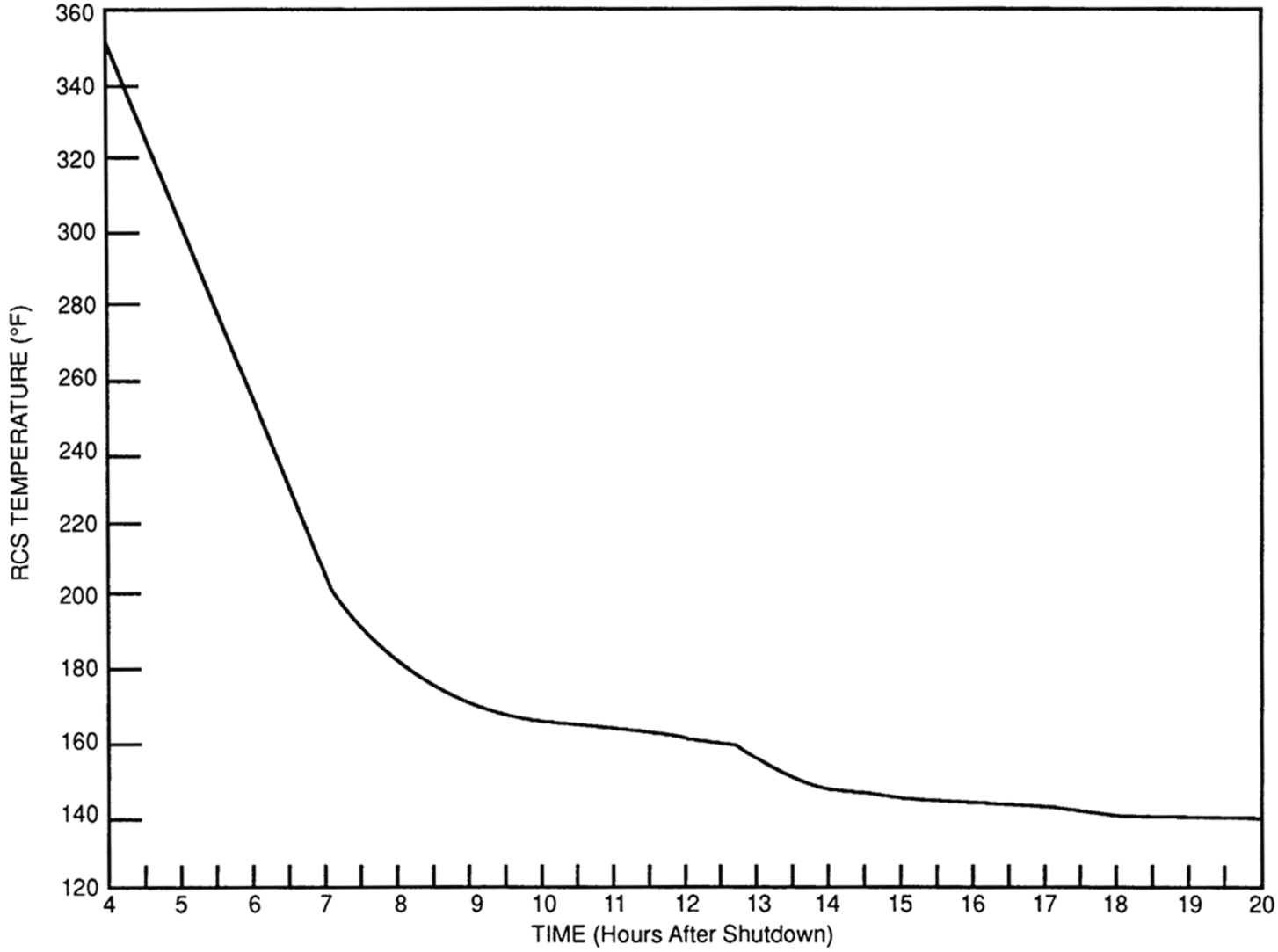
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RHRS COOLDOWN CURVE
(ONE TRAIN)

FIGURE 5.4.7–3 (SHEET 1 OF 2)



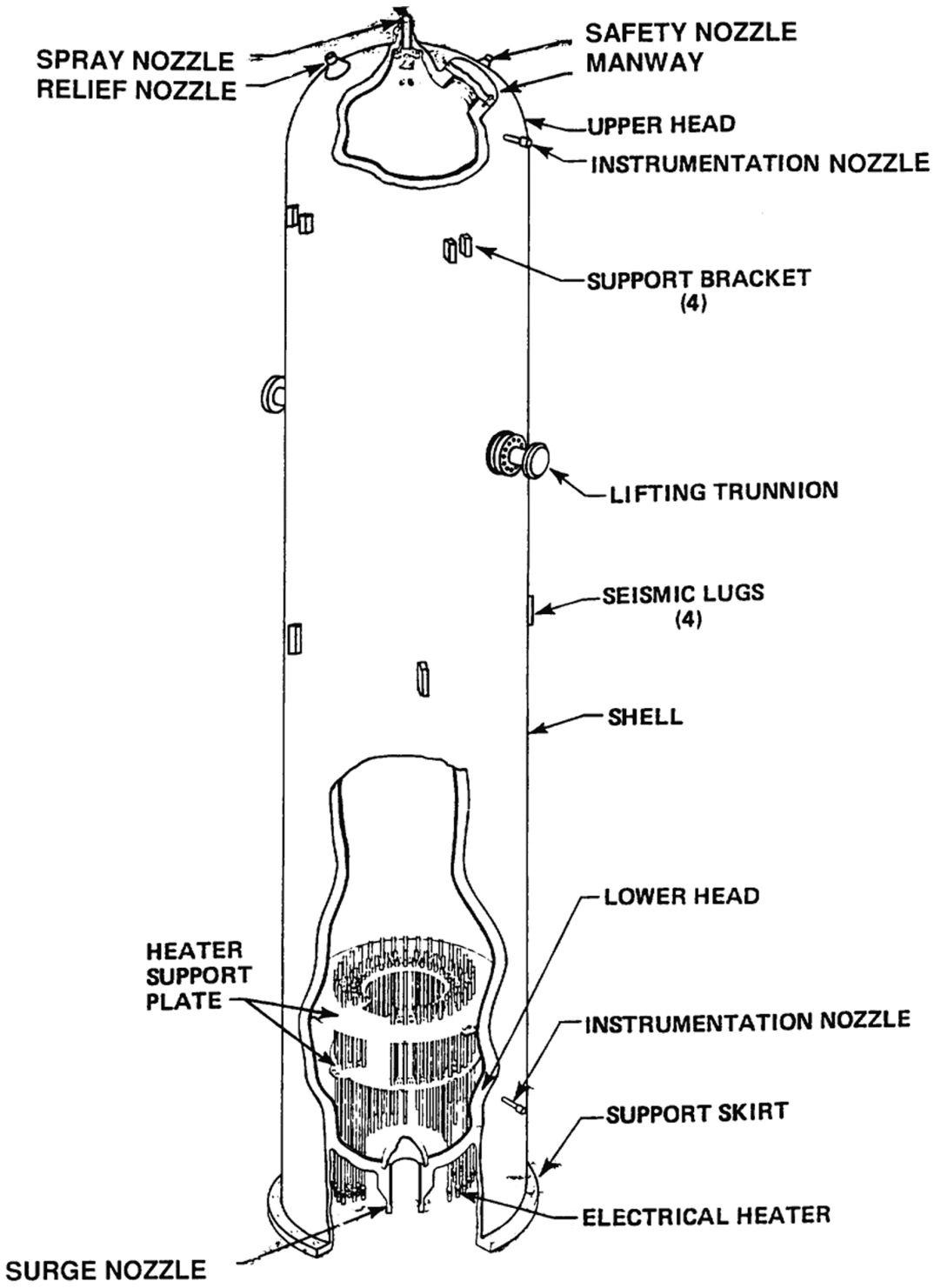
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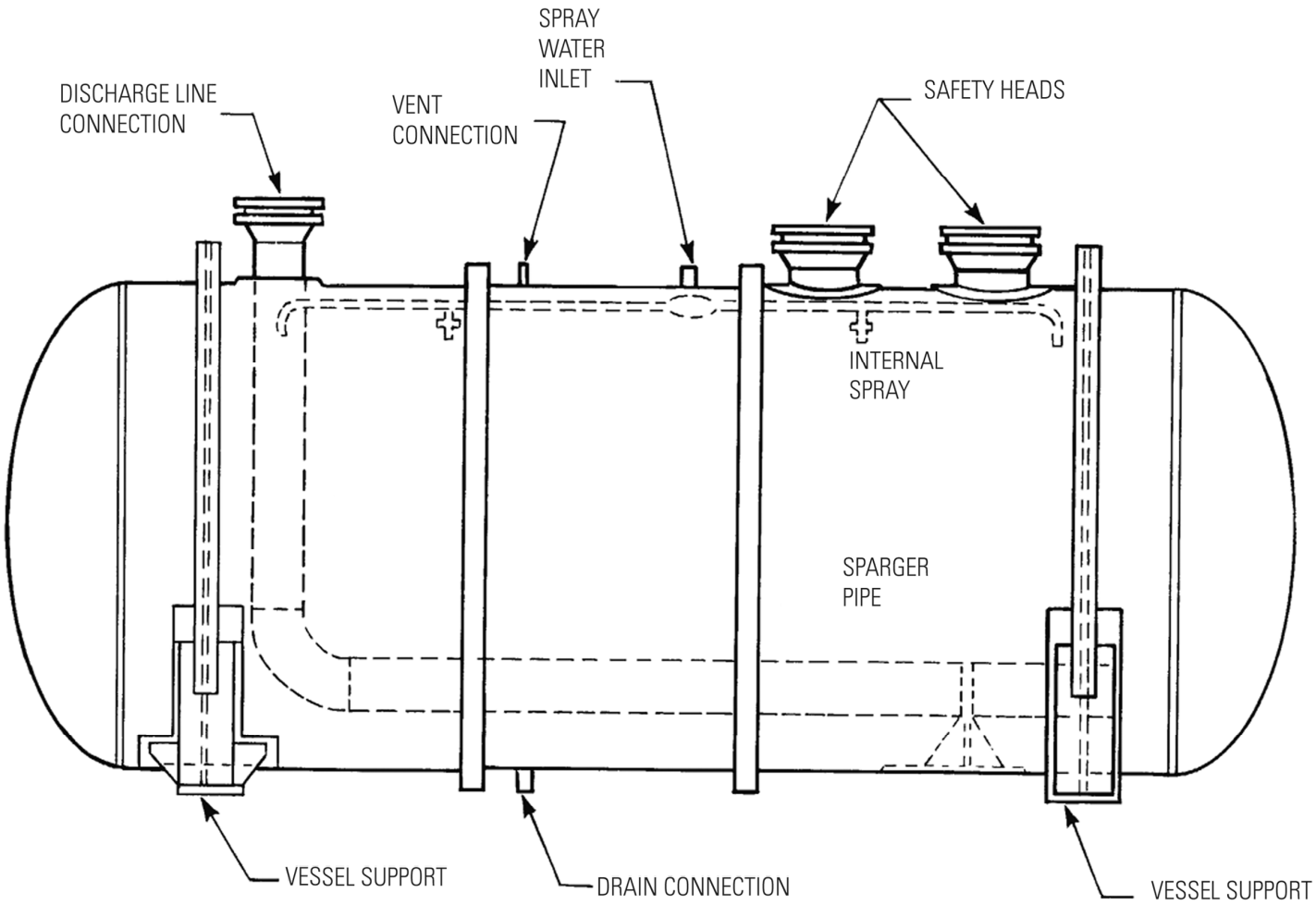
RHRS COOLDOWN CURVE
(TWO TRAIN)

FIGURE 5.4.7–3 (SHEET 2 OF 2)



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	FIGURE 5.4.10–1



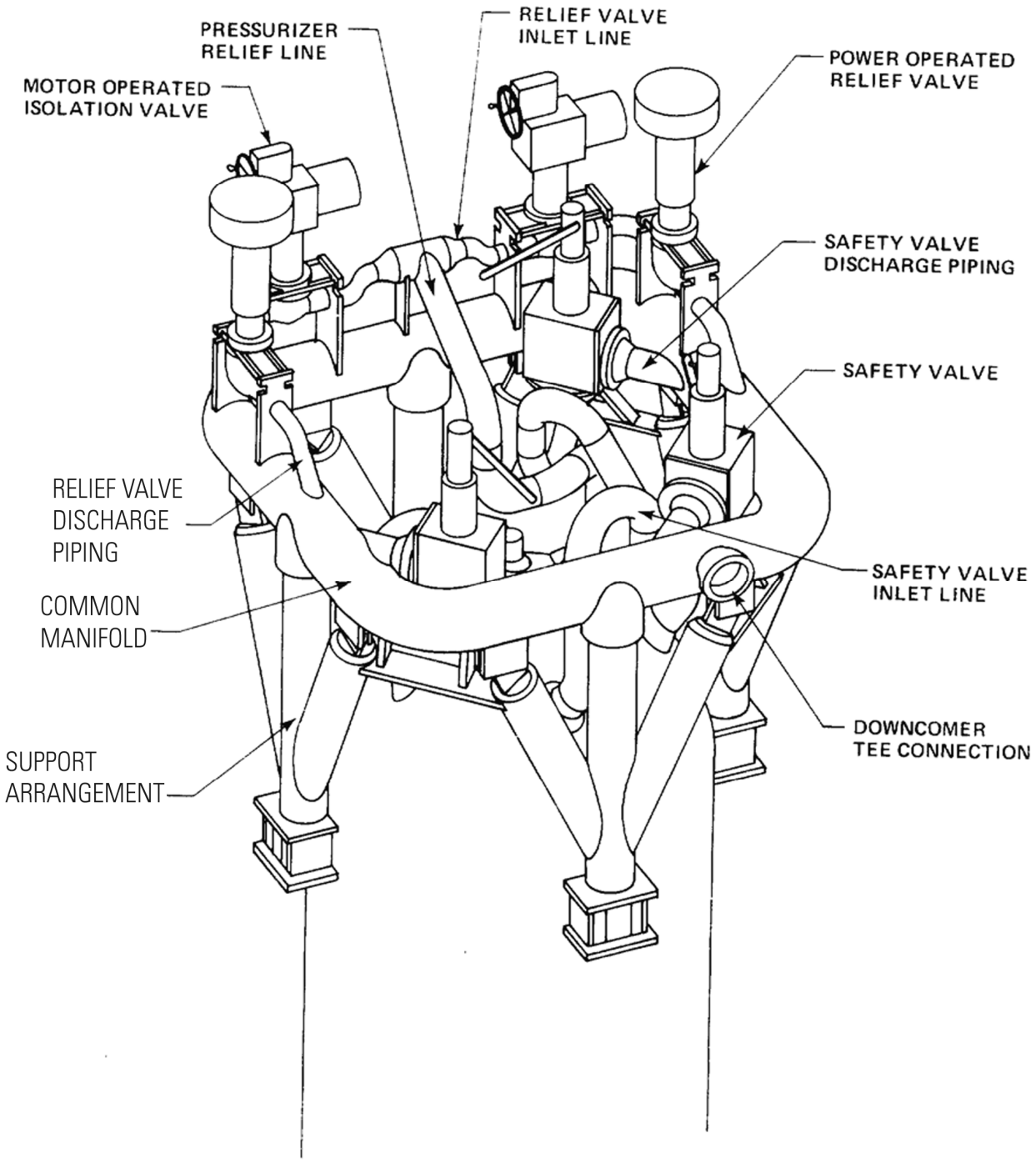
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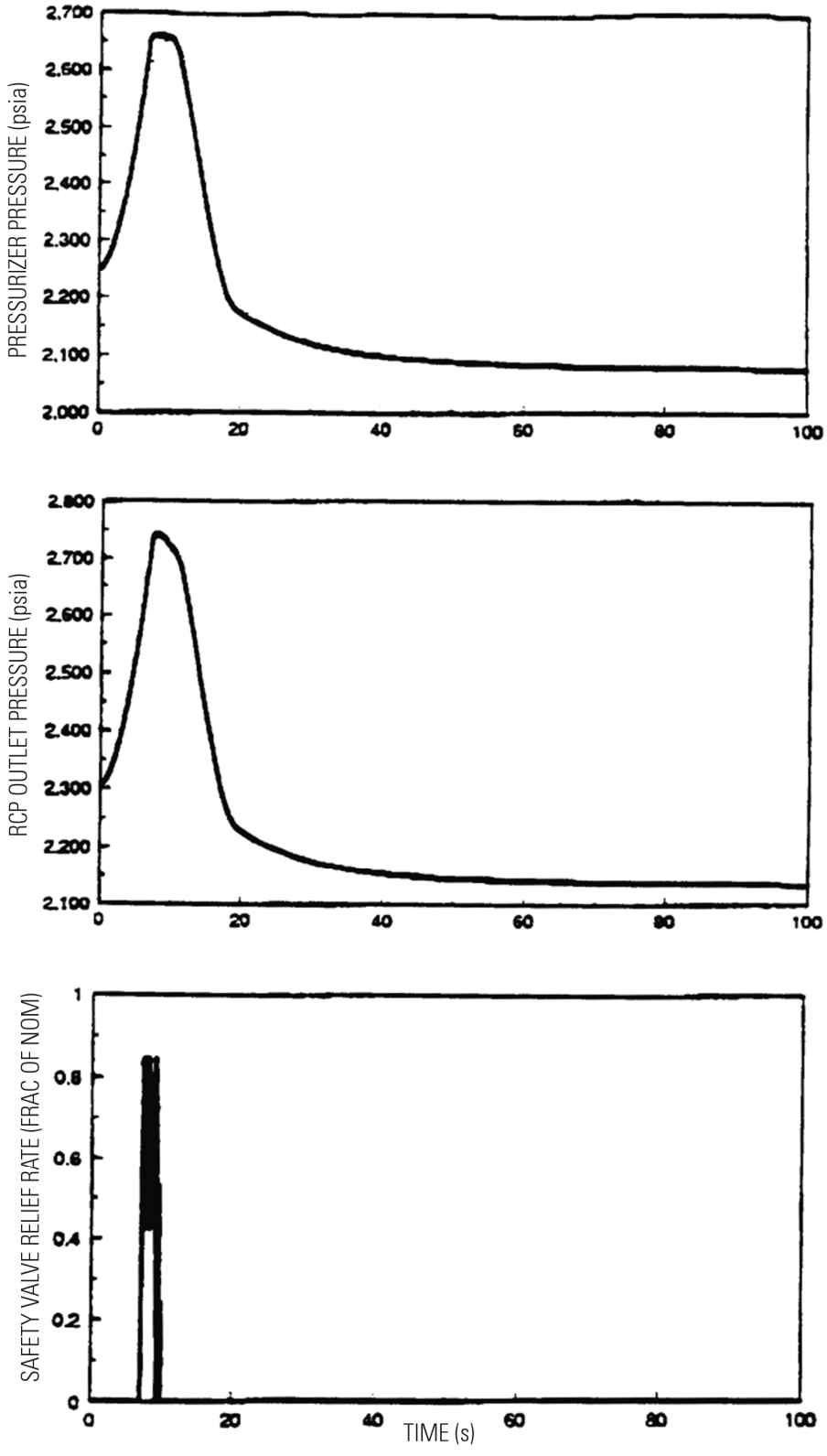
PRESSURIZER RELIEF TANK

FIGURE 5.4.11–1



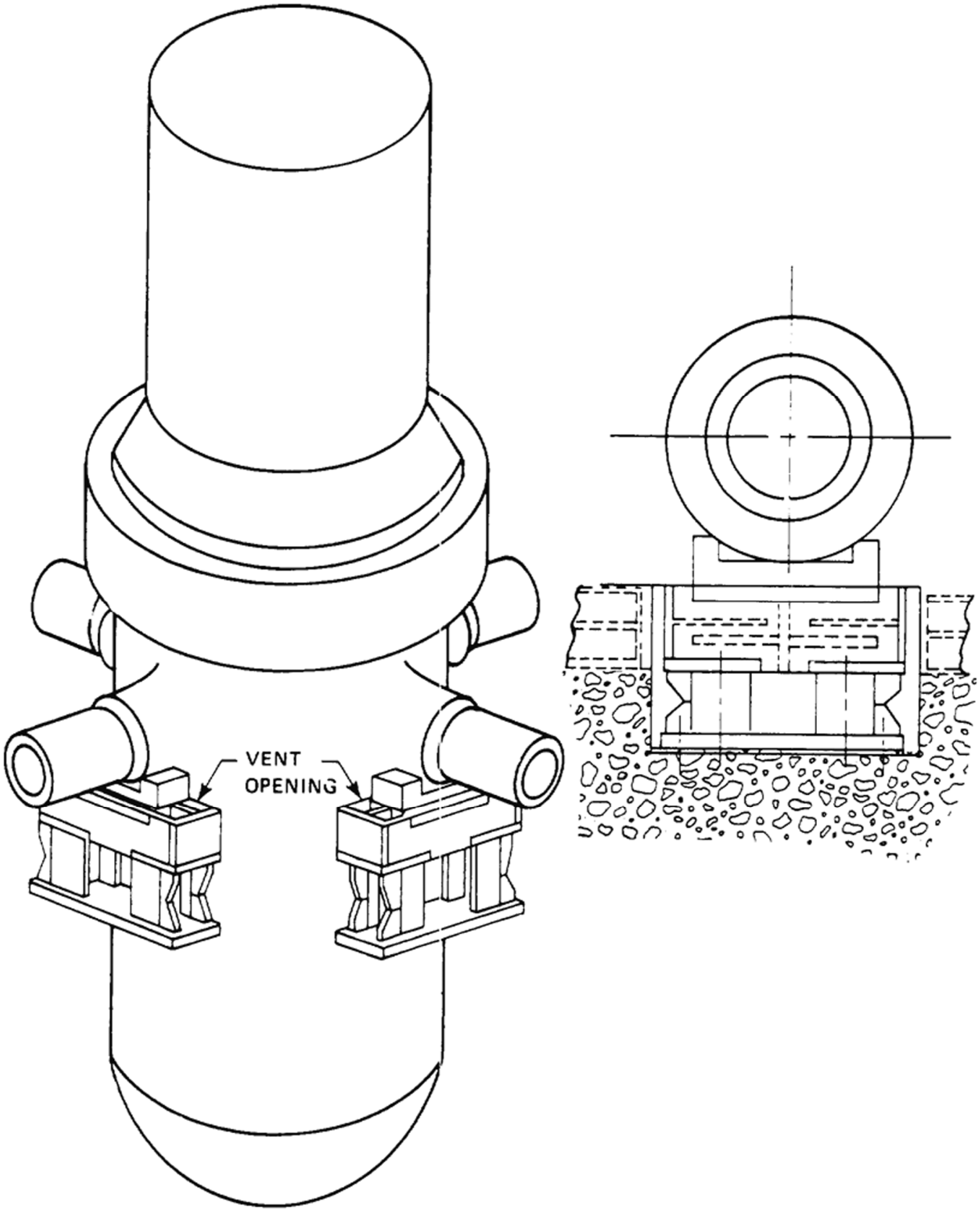
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SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	PRESSURIZER SAFETY AND RELIEF VALVE PIPING AND SUPPORT ARRANGEMENT
		FIGURE 5.4.11–2



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SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	PRESSURIZER SAFETY VALVES RELIEF RATE
		FIGURE 5.4.13–1



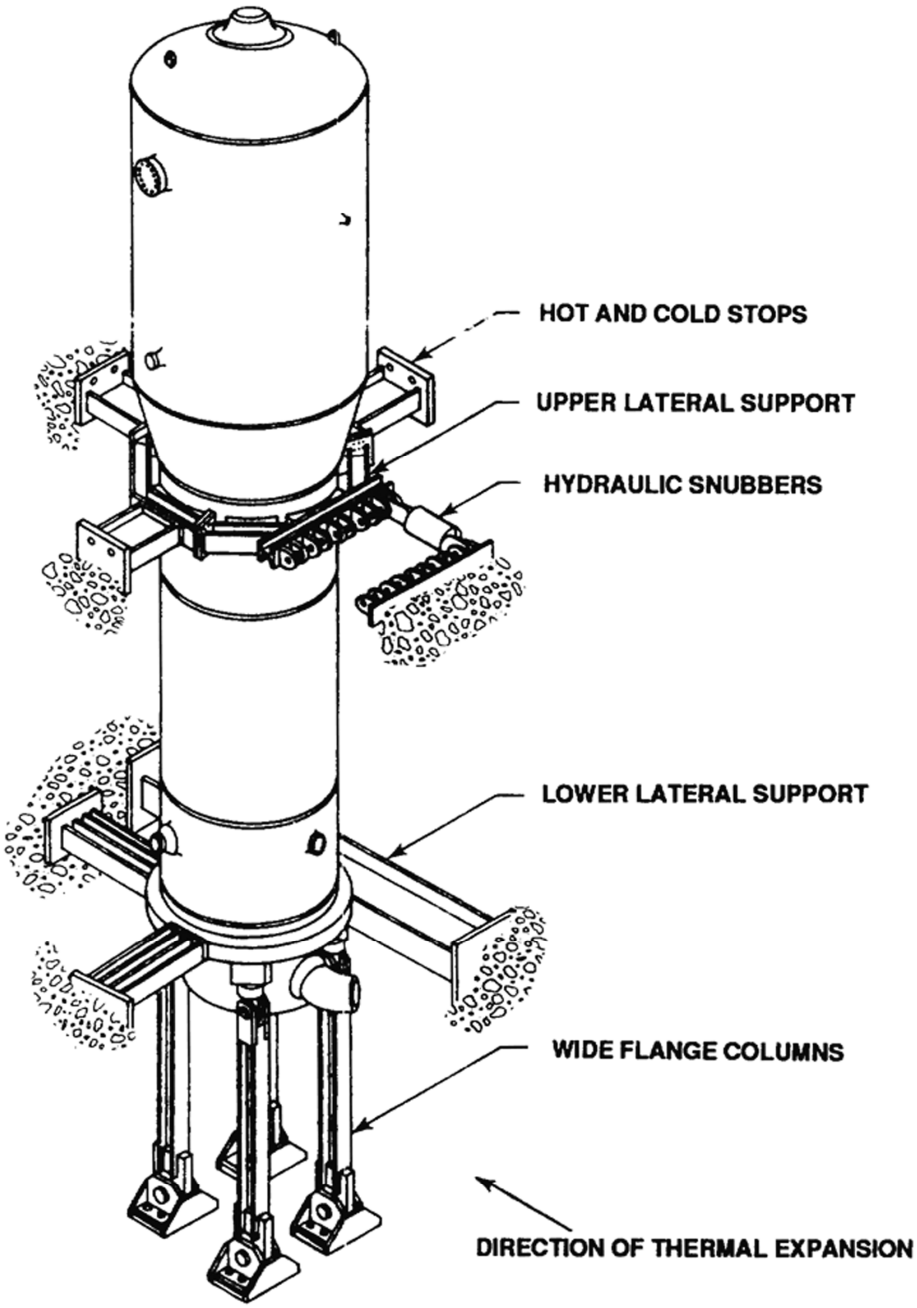
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UNIT 1 AND UNIT 2

REACTOR VESSEL SUPPORTS

FIGURE 5.4.14–1



*SYMMETRICALLY INSTALLED WITH RESPECT TO THE CENTER LINE.

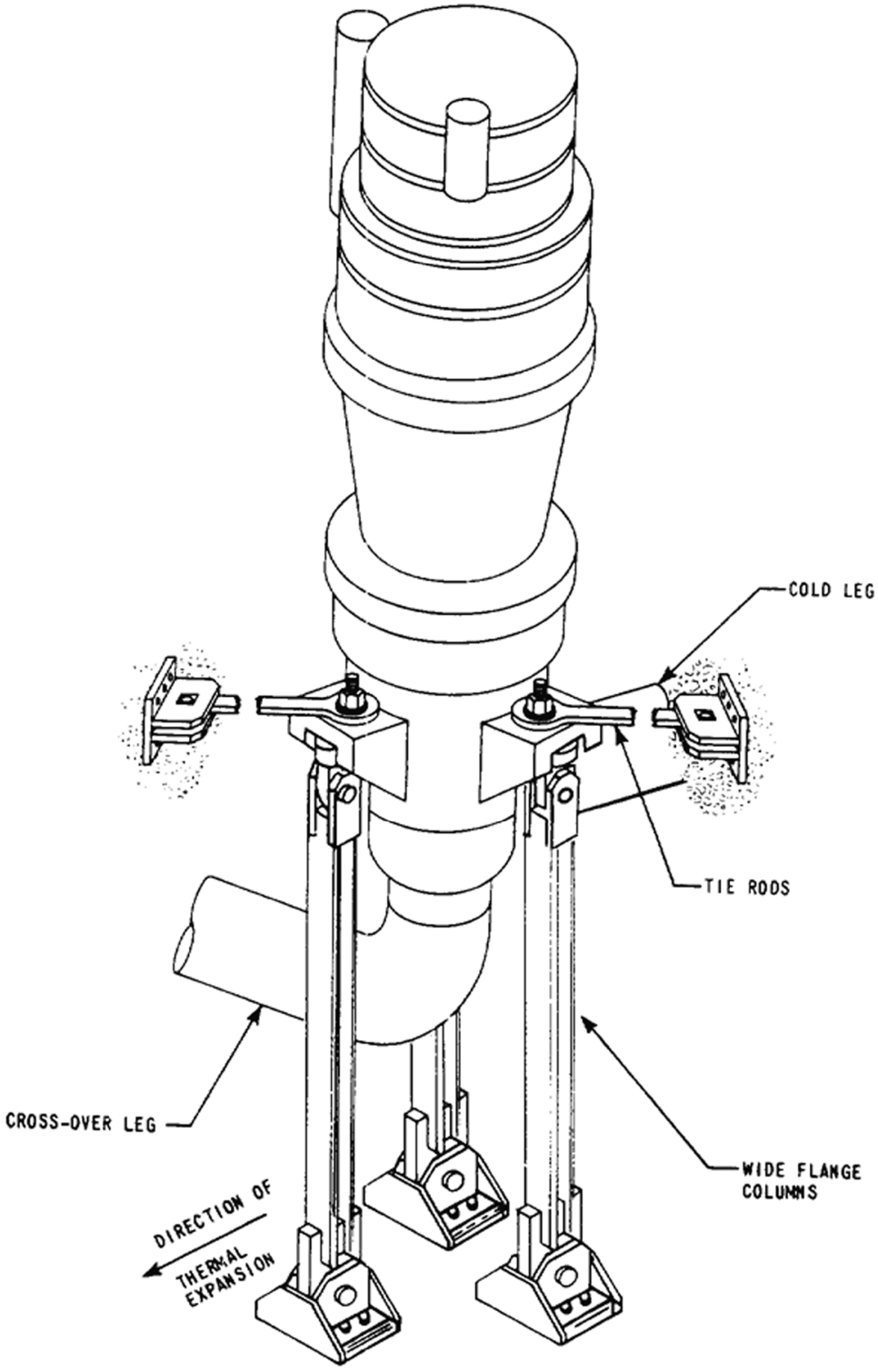
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STEAM GENERATOR SUPPORTS

FIGURE 5.4.14–2



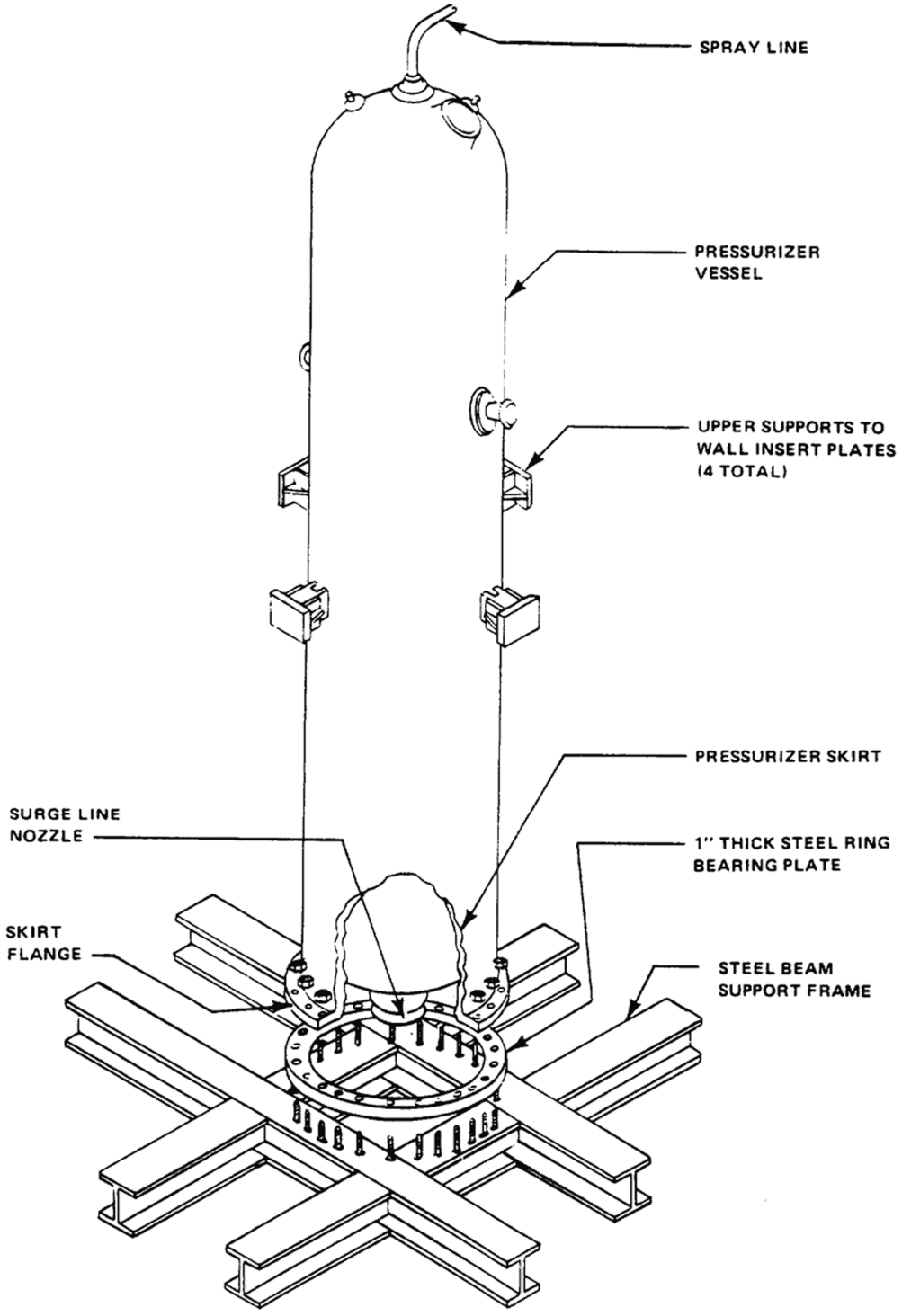
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REACTOR COOLANT PUMPS SUPPORTS

FIGURE 5.4.14–3



PRESSURIZER

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UNIT 1 AND UNIT 2

PRESSURIZER SUPPORTS

FIGURE 5.4.14–4